Evaluation of LLMs on Bloom's Taxonomy Classification

A Comparative Study on Open-Source LLMs using Monash CLO Dataset

Muhammad Daniyal

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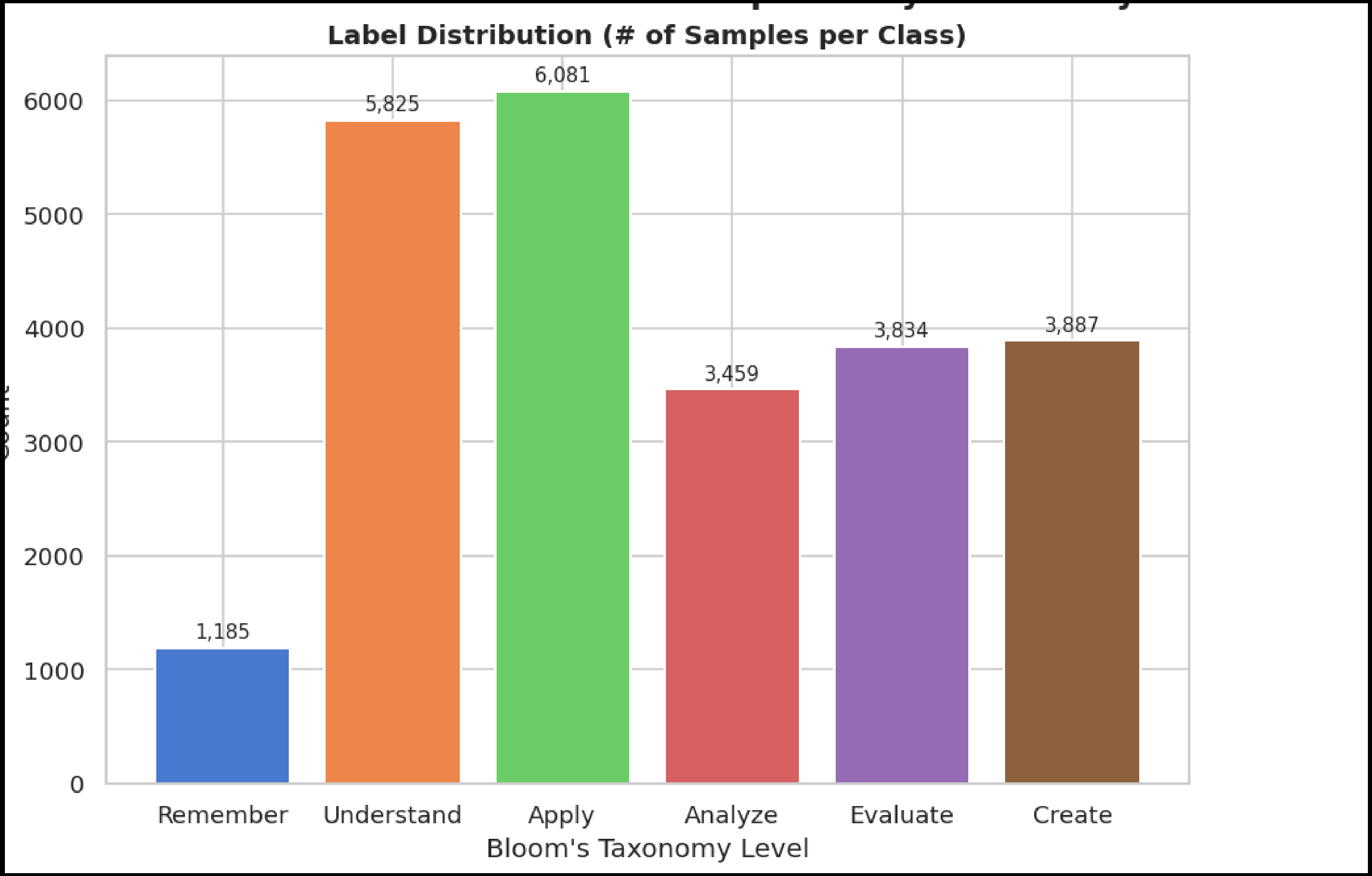
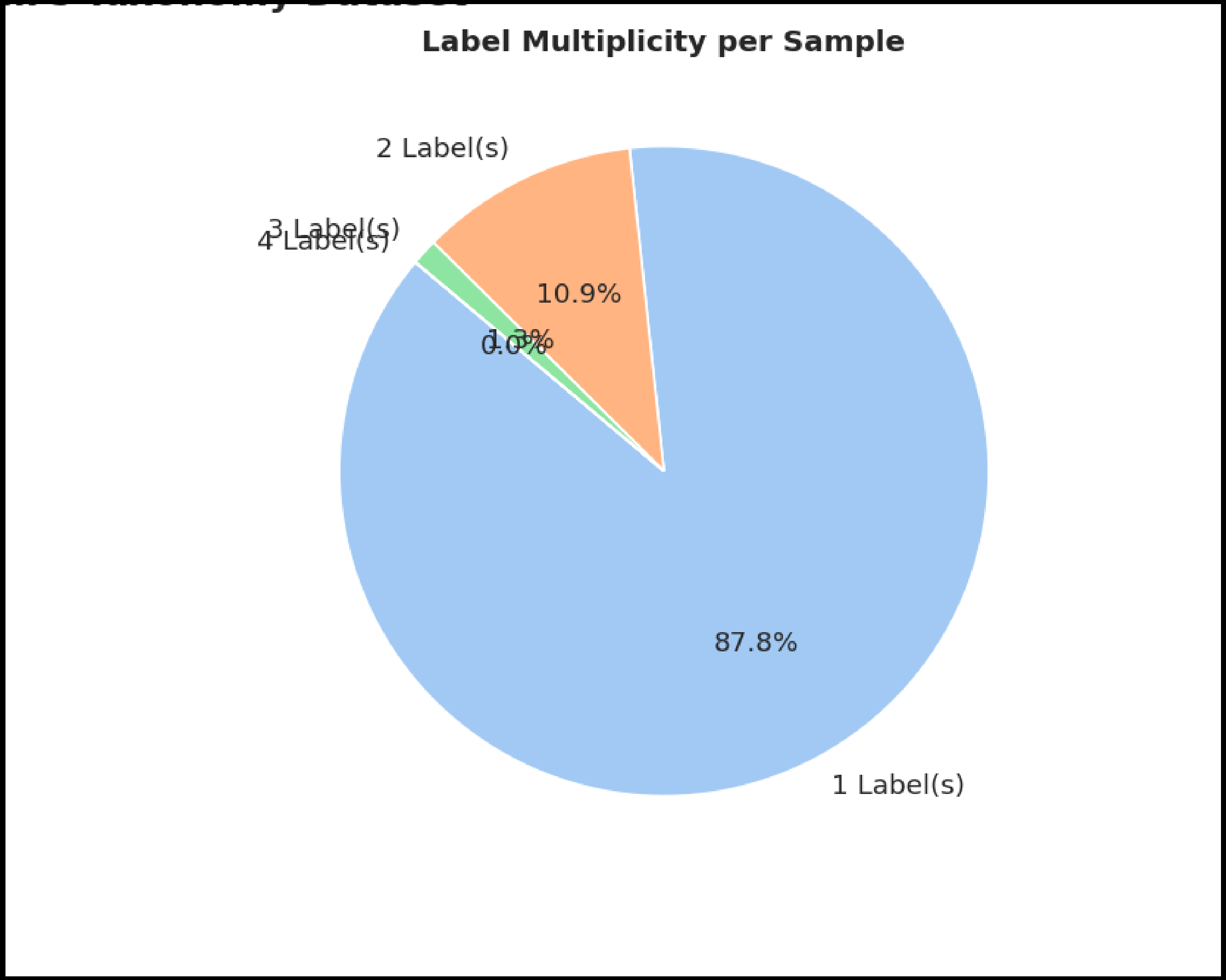
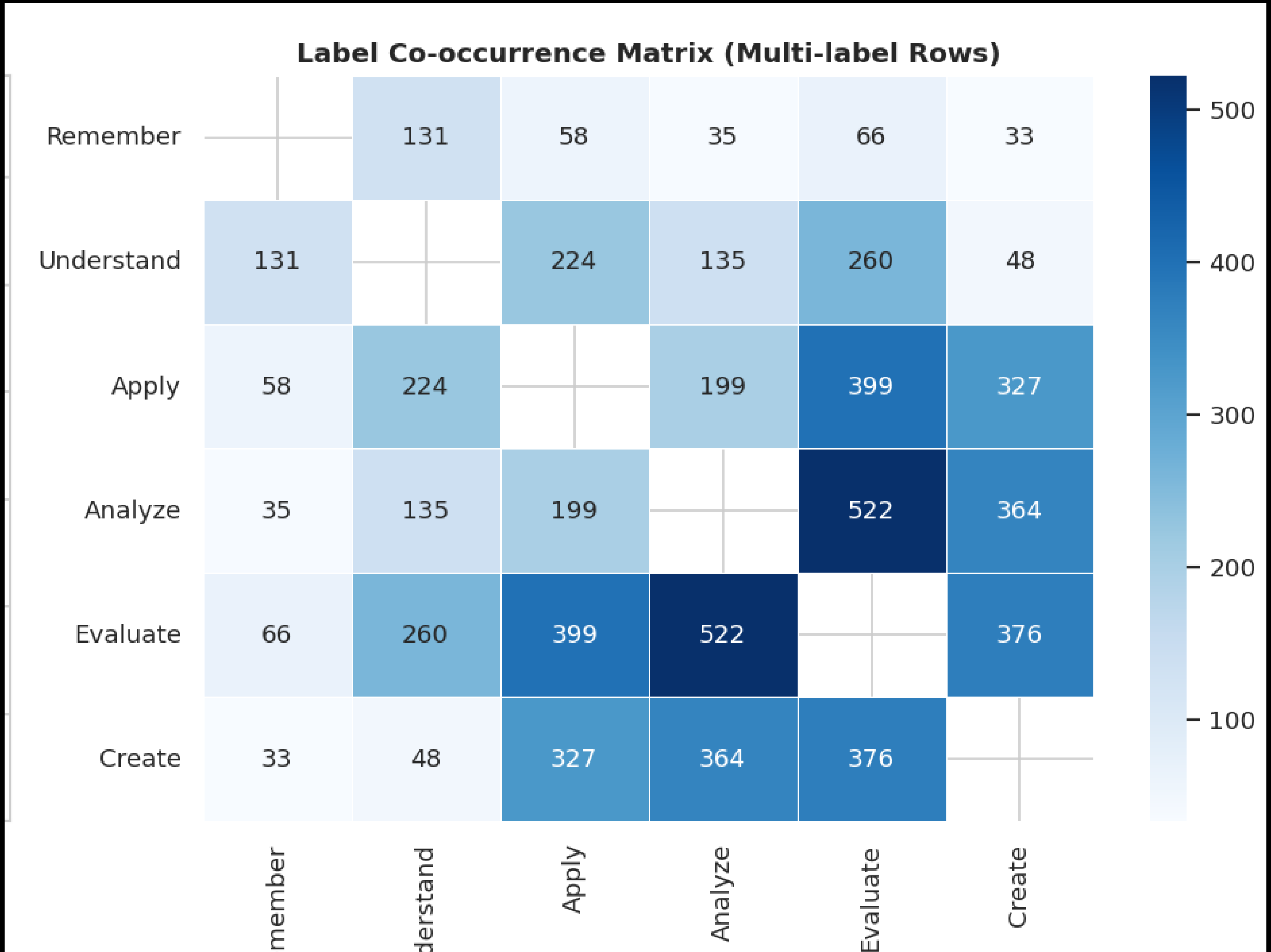
EDA – The CLO Dataset (EDM2022)



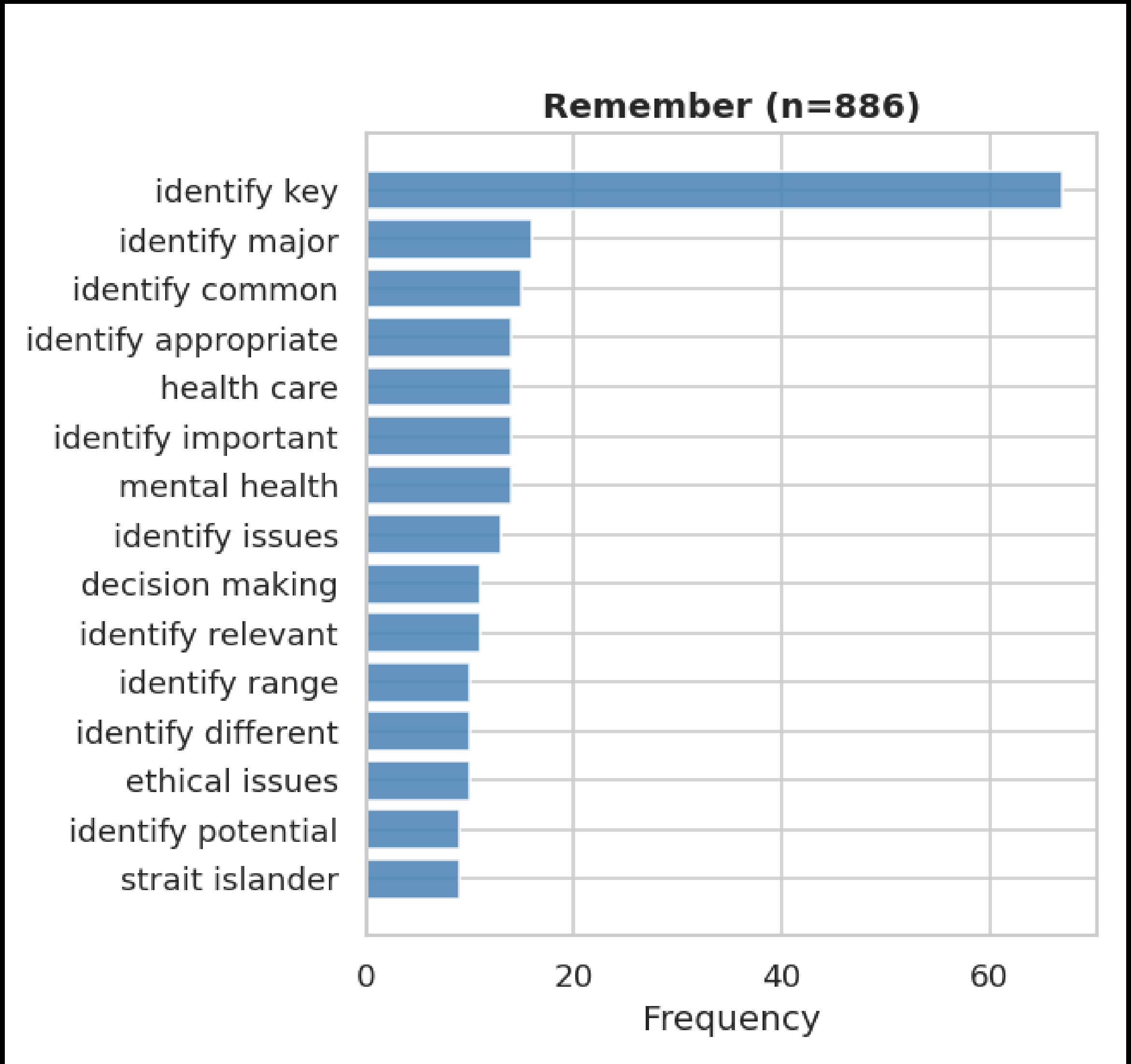
EDA – The CLO Dataset (EDM2022)



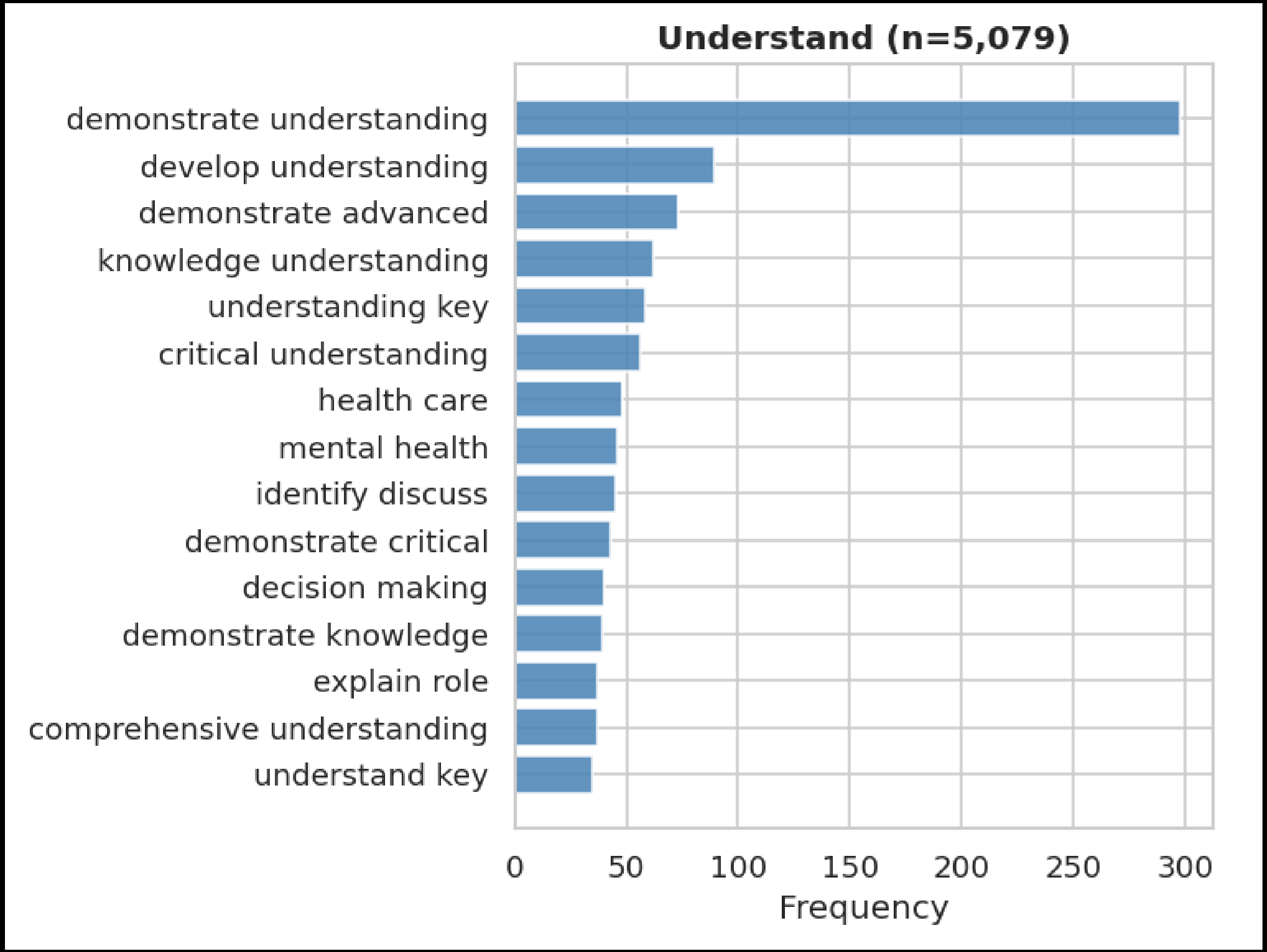
EDA – The CLO Dataset (EDM2022)



Top 15 Bigrams per level (single label)

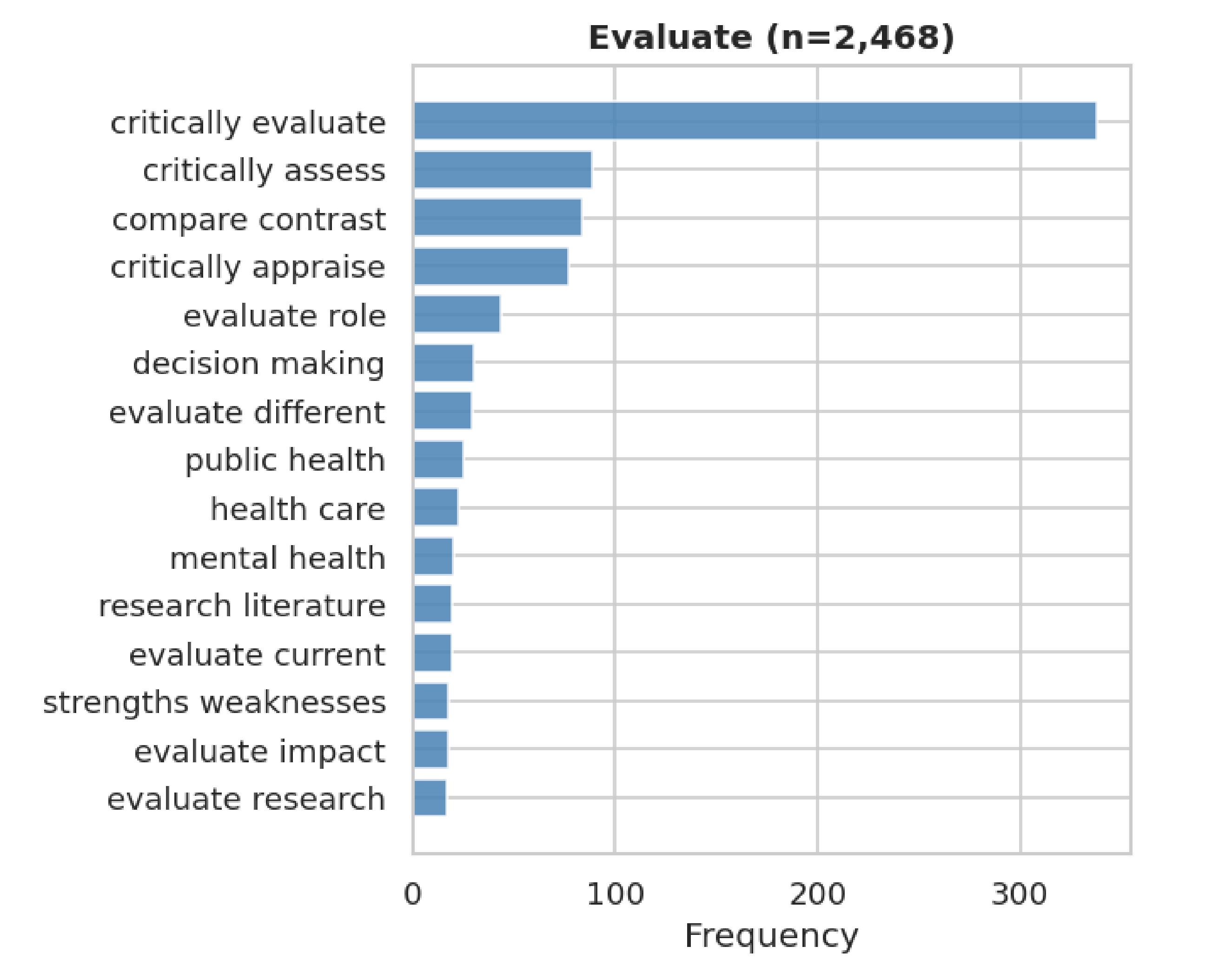


Top 15 Bigrams per level (single label)

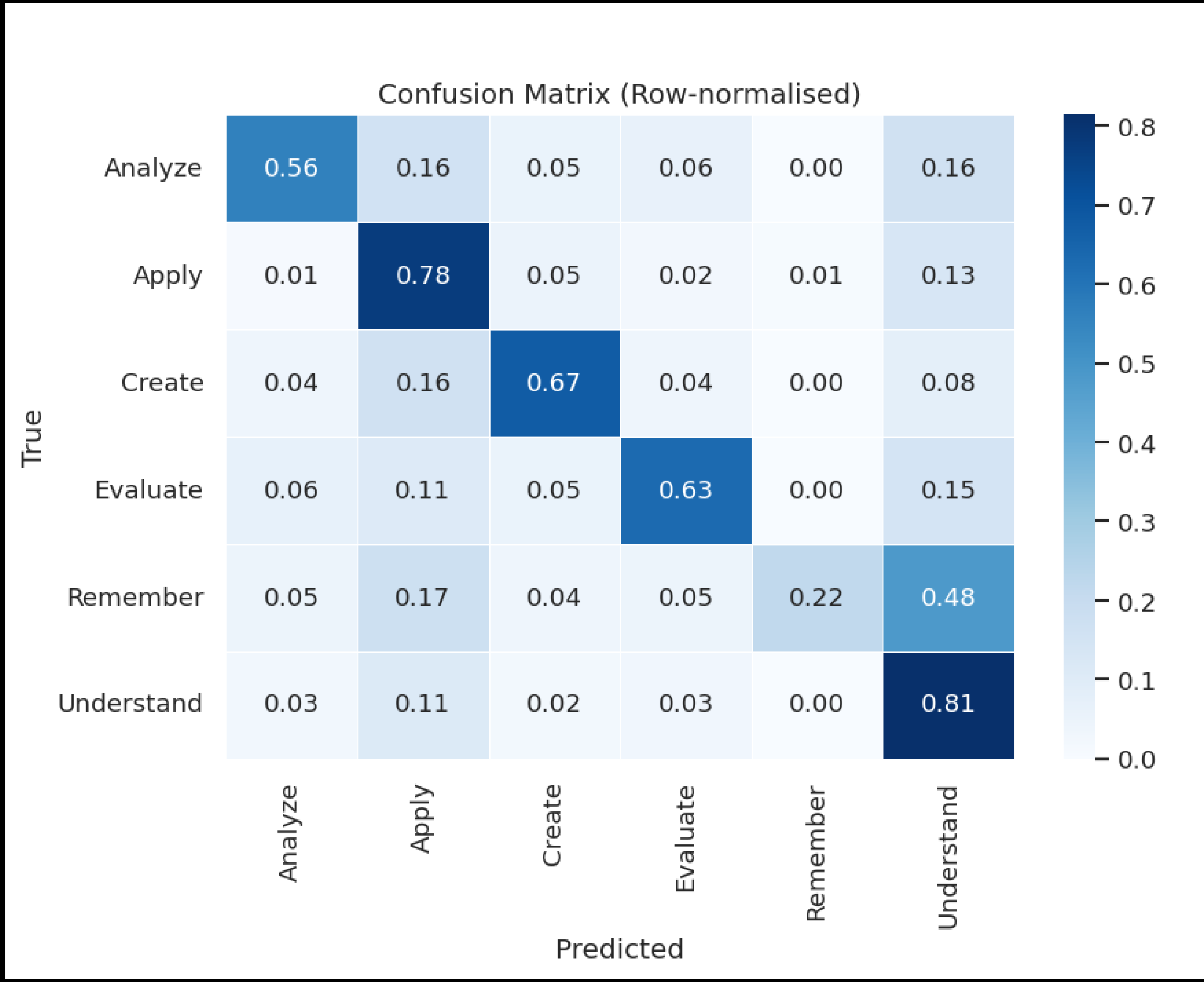


Top 15 Bigrams per level (single label)

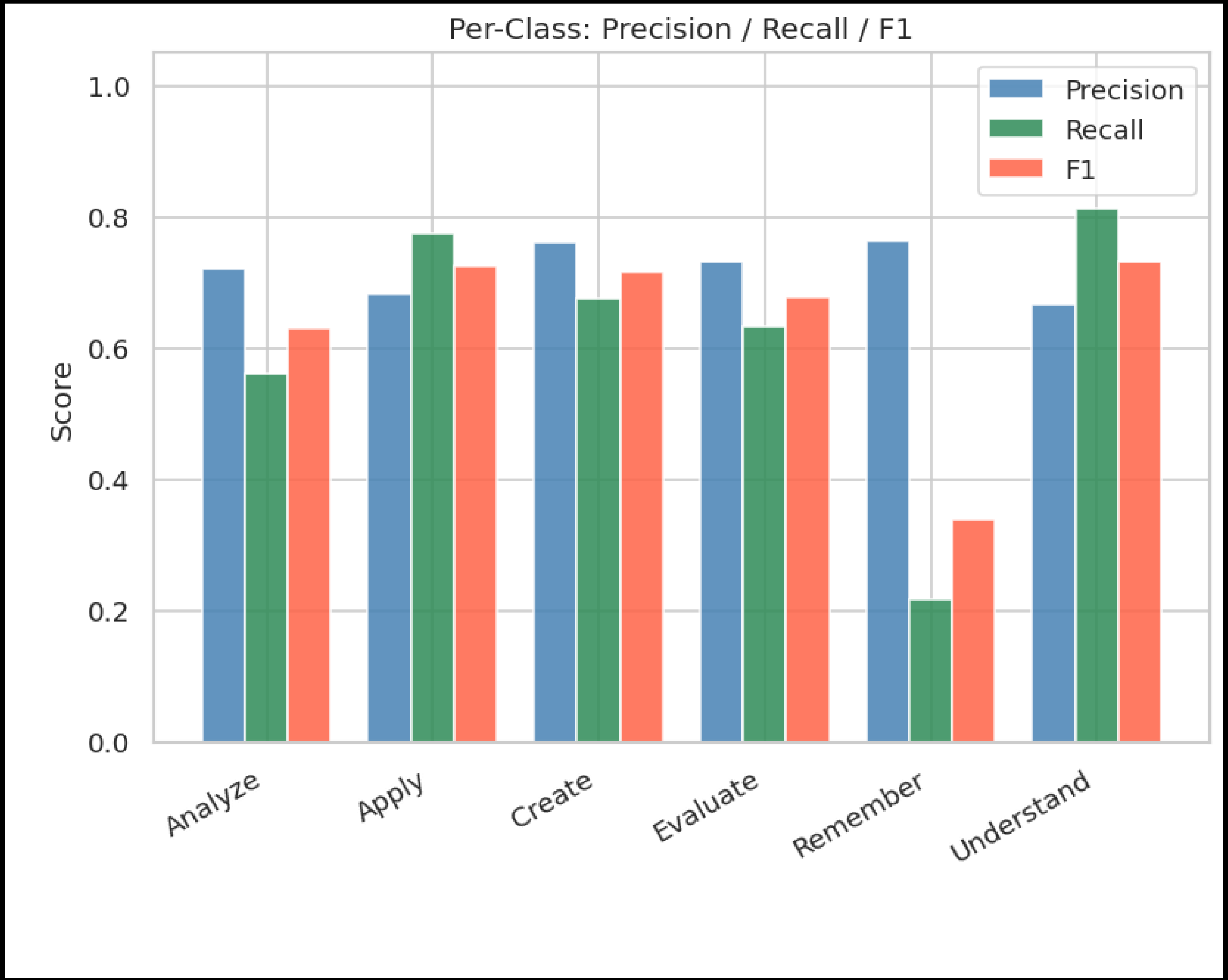
Top 15 Bigrams per level (single label)



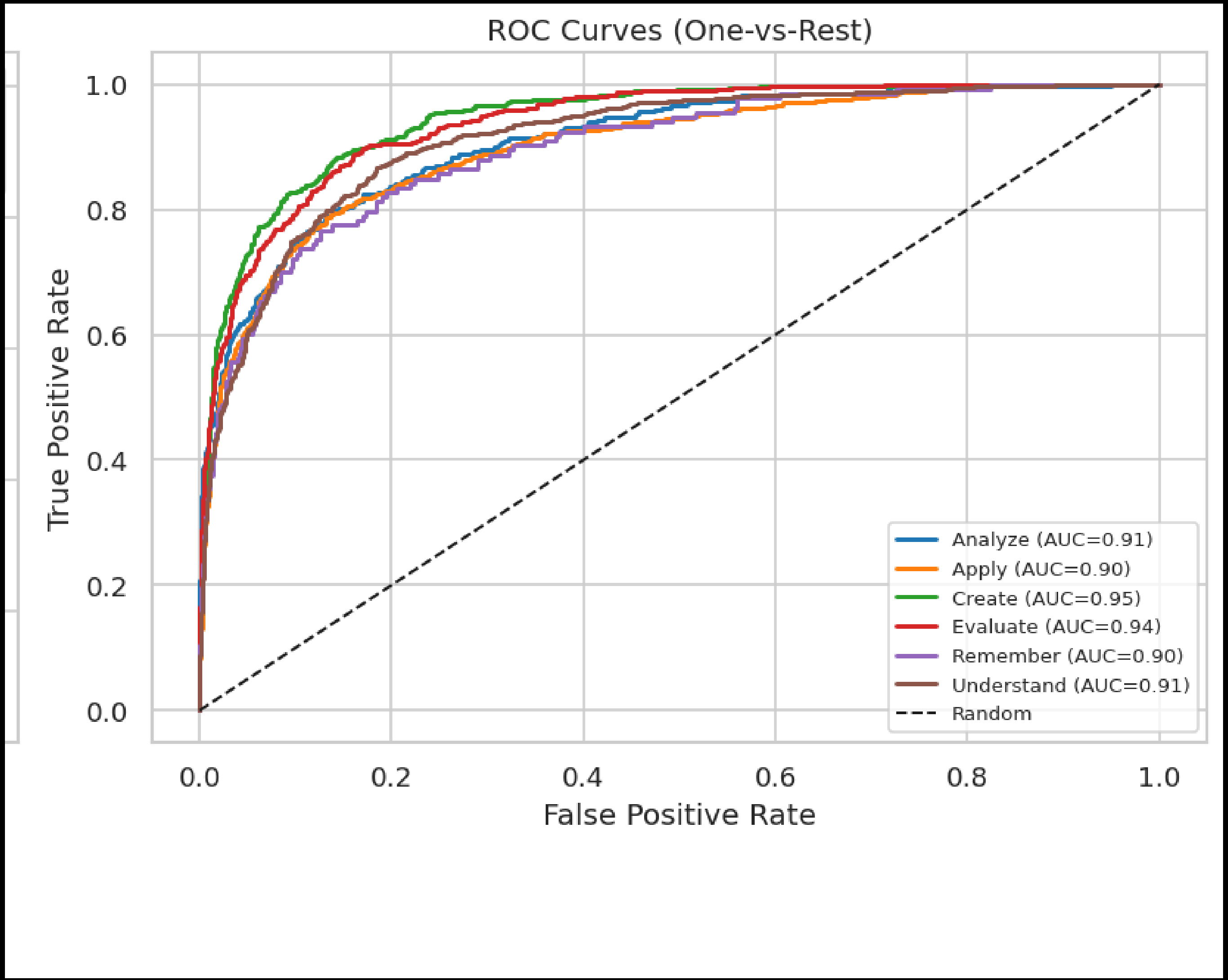
Multinomial Naïve Bayes Evaluation



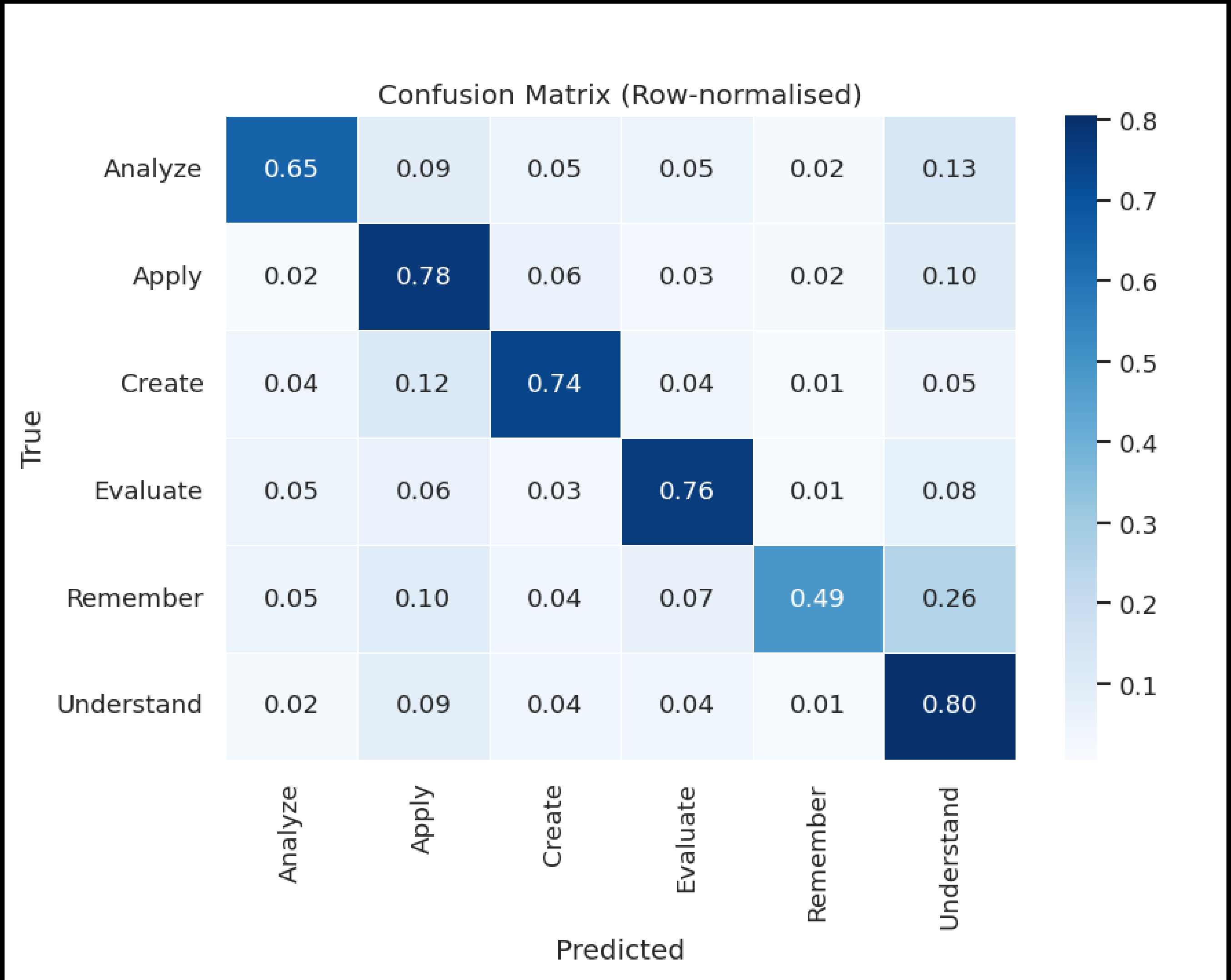
Multinomial Naïve Bayes Evaluation



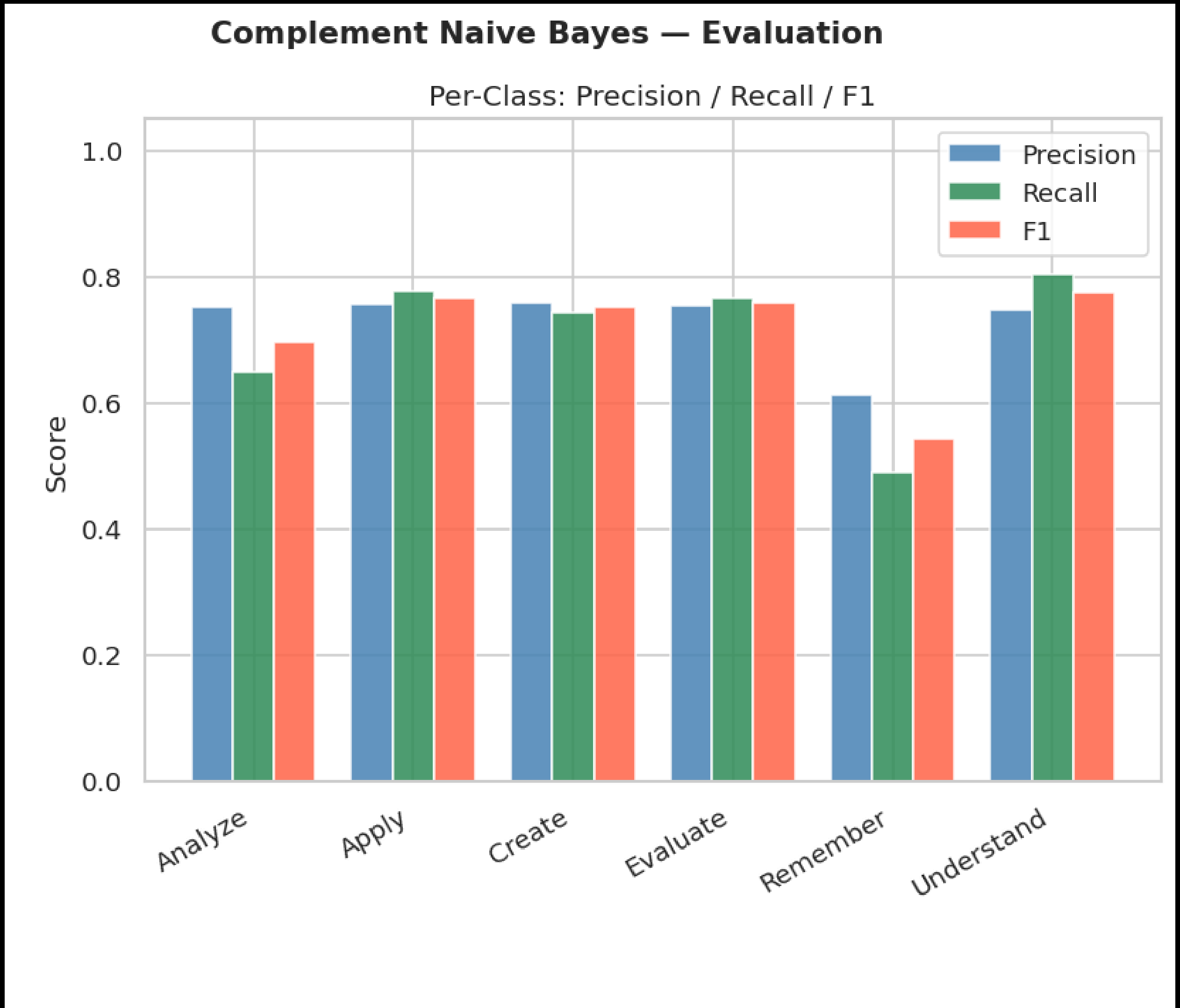
Multinomial Naïve Bayes Evaluation



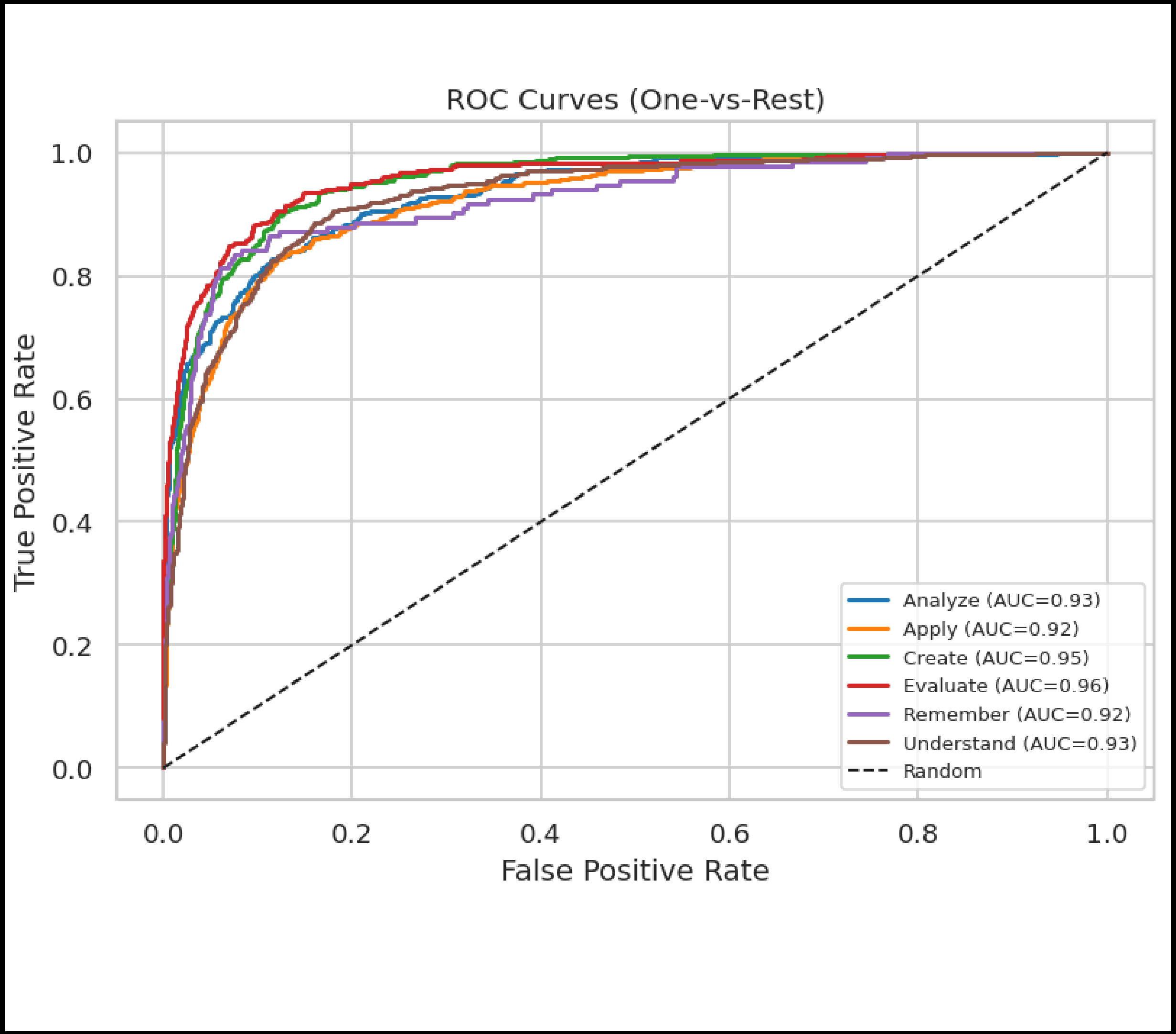
Complement Naïve Bayes Evaluation



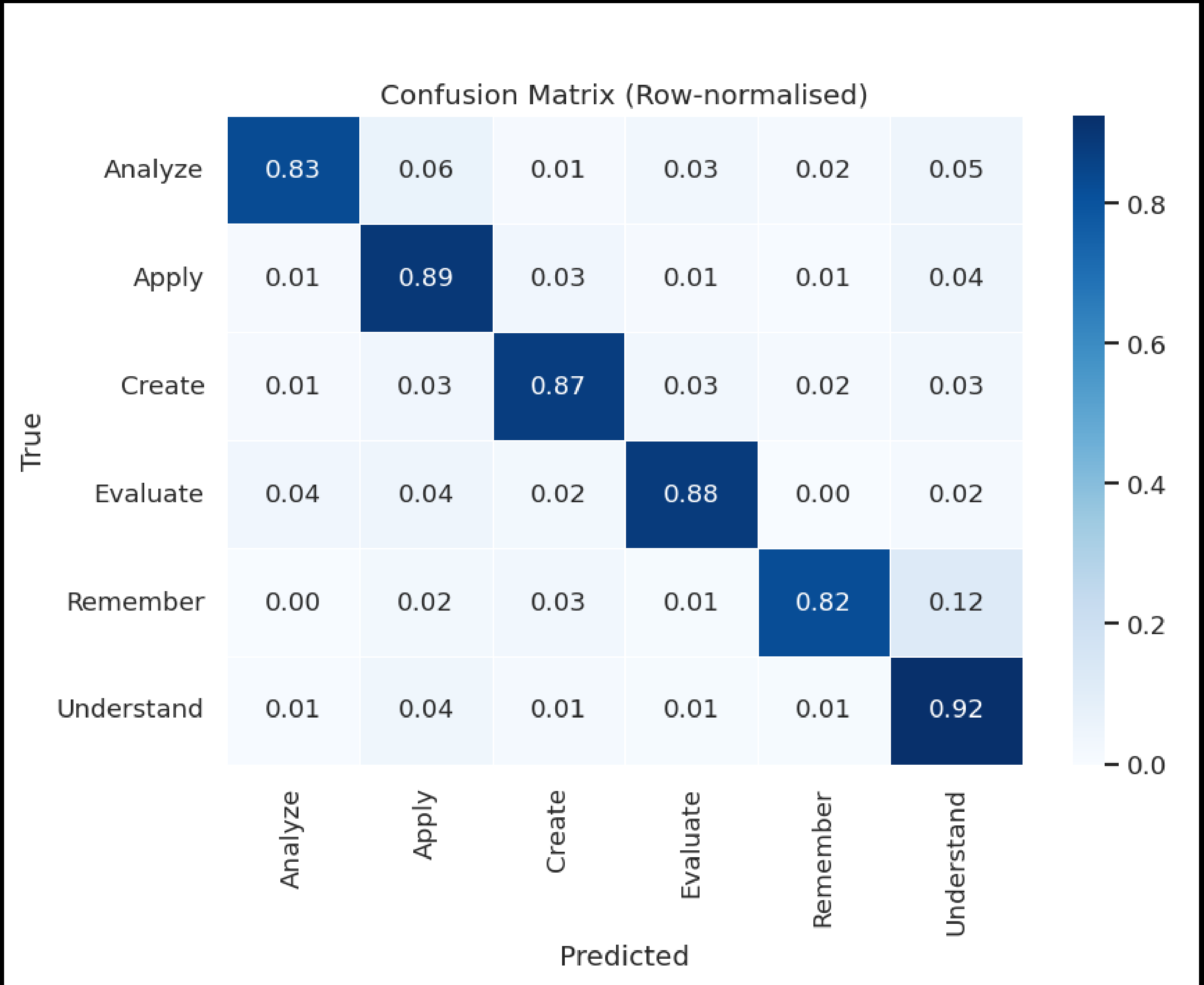
Complement Naïve Bayes Evaluation



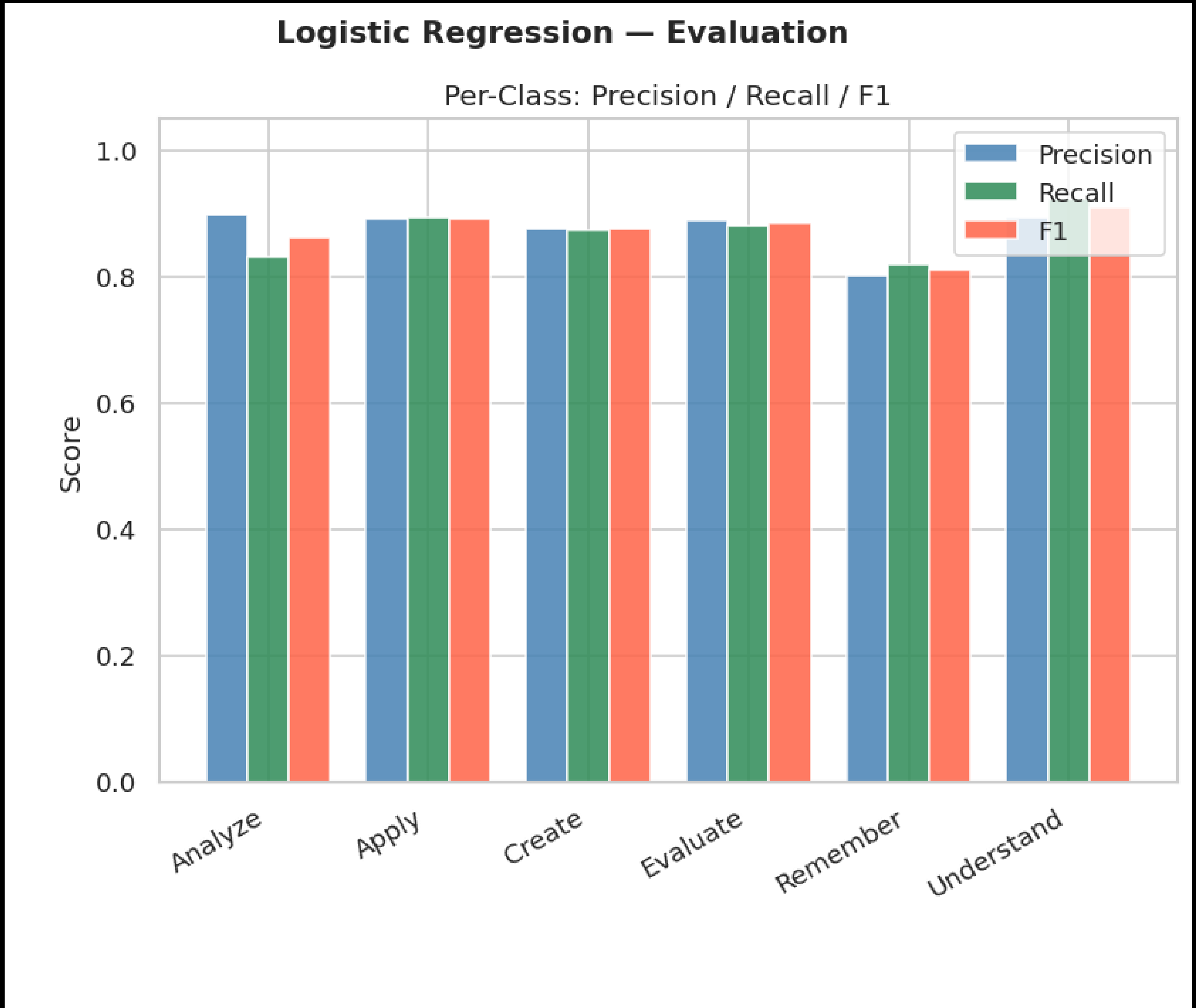
Complement Naïve Bayes Evaluation



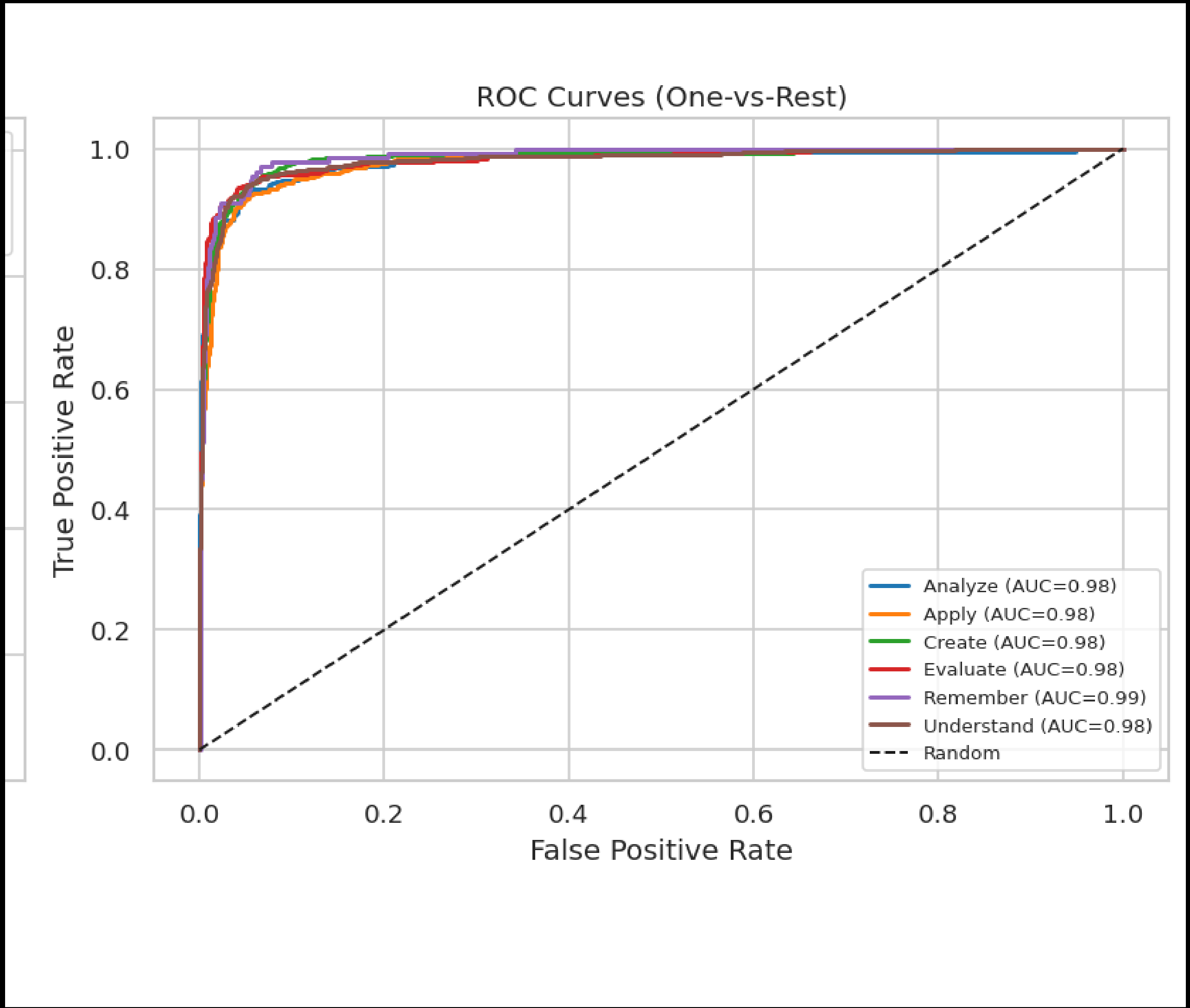
Logistic Regression - Evaluation

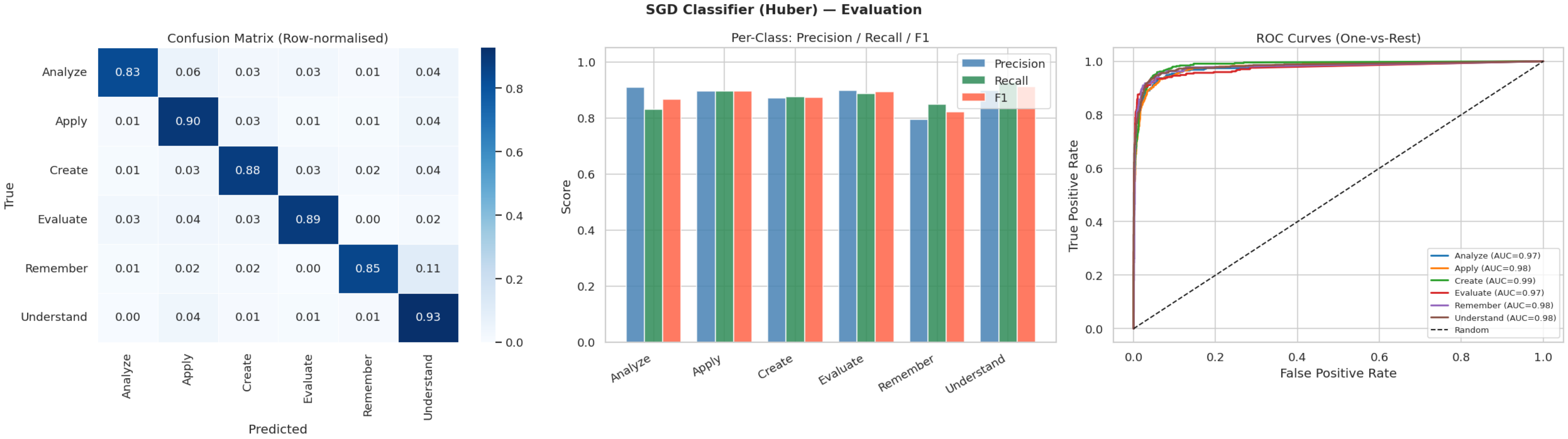


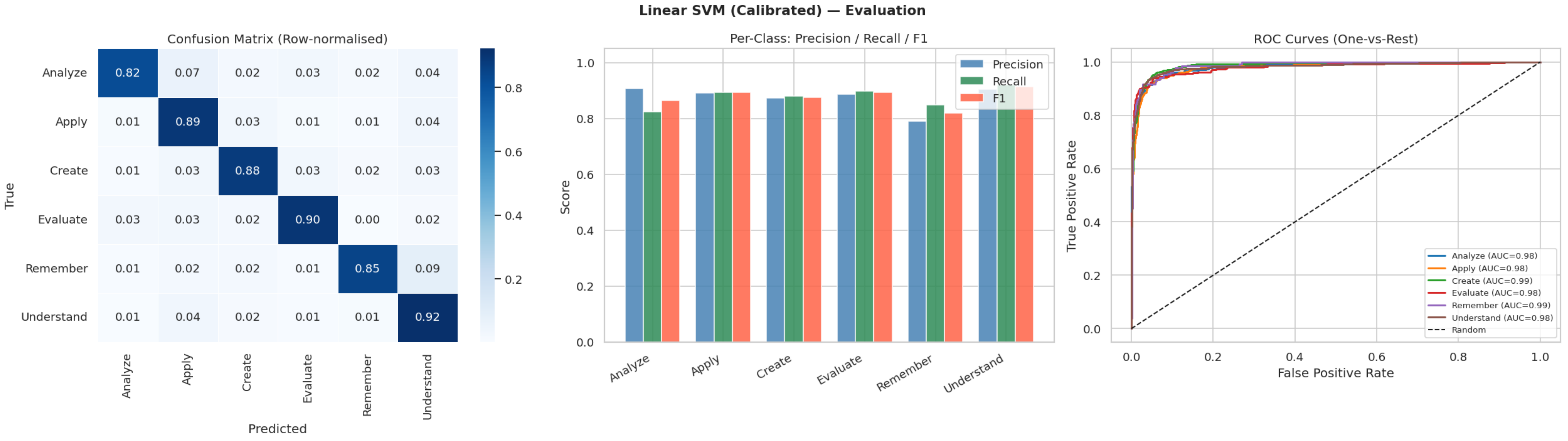
Logistic Regression - Evaluation



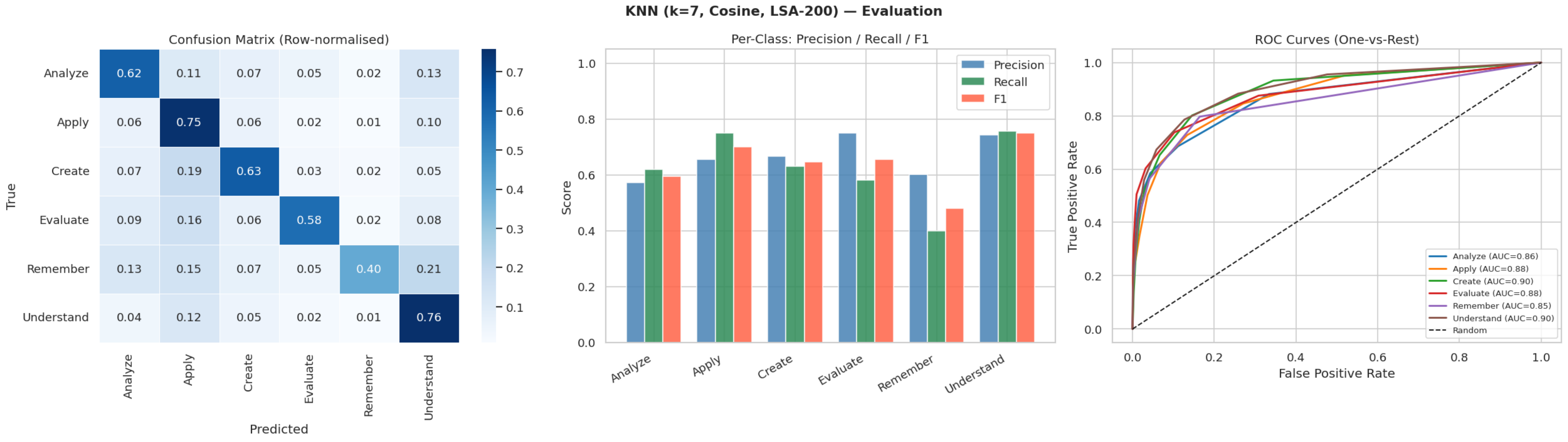
Logistic Regression - Evaluation

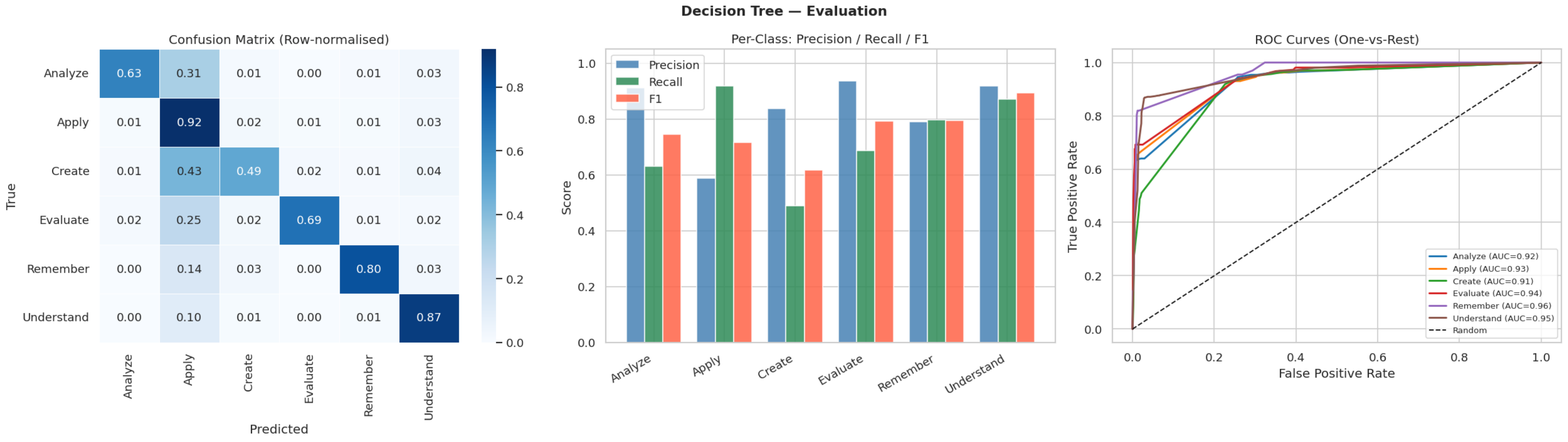


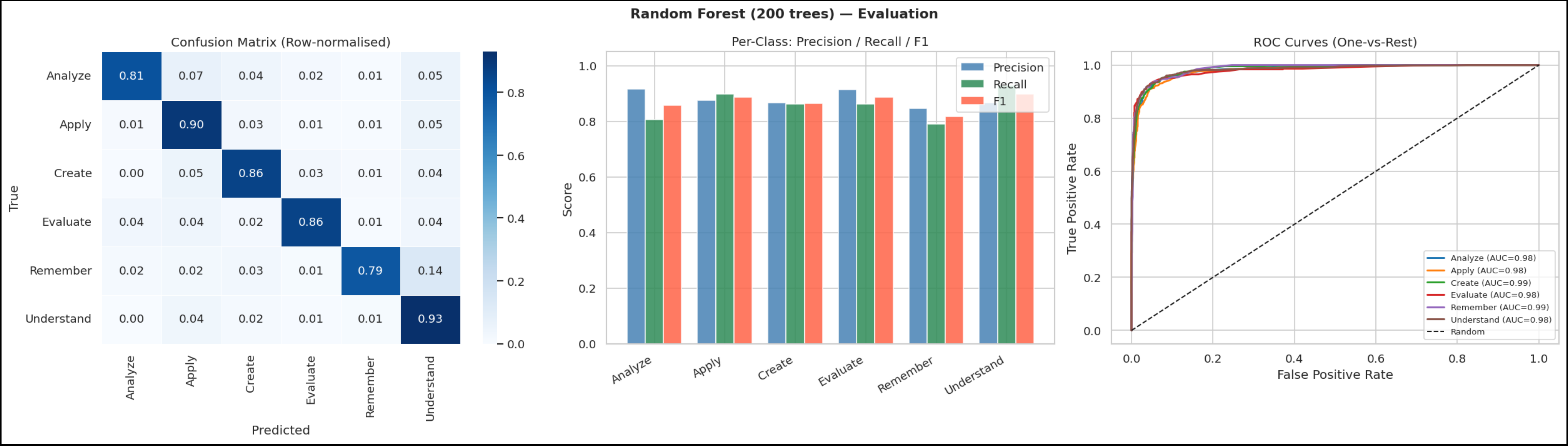

SGD Classifier (Huber) — Evaluation
Confusion Matrix (Row-normalised)
True
Predicted
Analyze
Apply
Create
Evaluate
Remember
Understand
0.83 0.06 0.03 0.03 0.01 0.04
0.01 0.90 0.03 0.01 0.01 0.04
0.01 0.03 0.88 0.03 0.02 0.04
0.03 0.04 0.03 0.89 0.00 0.02
0.01 0.02 0.02 0.00 0.85 0.11
0.00 0.04 0.01 0.01 0.01 0.93
Per-Class: Precision / Recall / F1
Score
Precision
Recall
F1
ROC Curves (One-vs-Rest)
True Positive Rate
False Positive Rate
Analyze (AUC=0.97)
Apply (AUC=0.98)
Create (AUC=0.99)
Evaluate (AUC=0.97)
Remember (AUC=0.98)
Understand (AUC=0.98)
Random

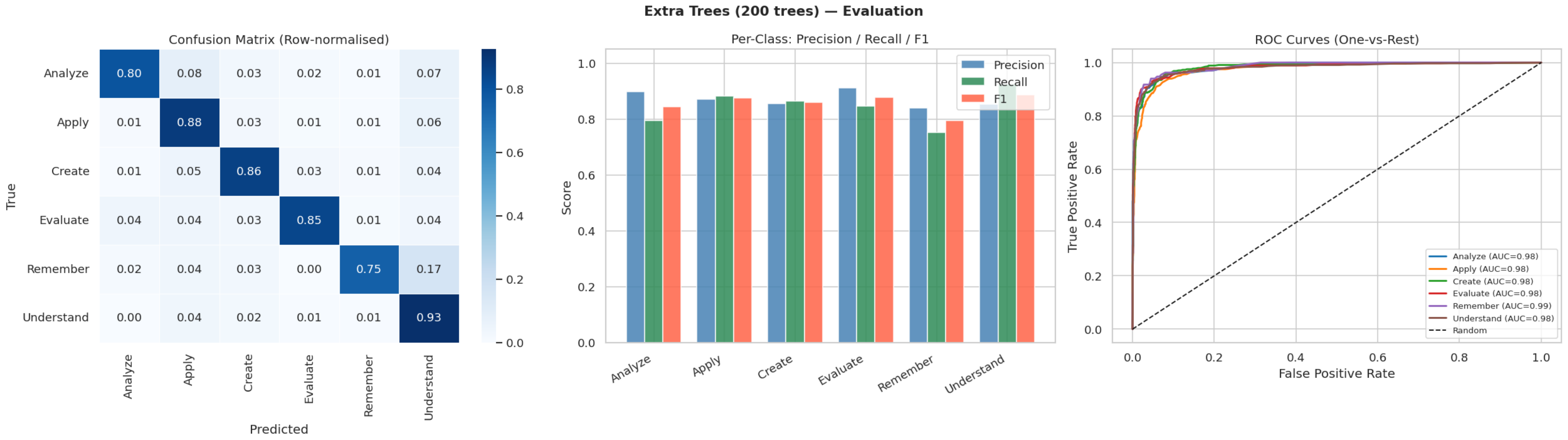

Linear SVM (Calibrated) — Evaluation
Confusion Matrix (Row-normalised)
True
Predicted
Analyze
Apply
Create
Evaluate
Remember
Understand
0.82 0.07 0.02 0.03 0.02 0.04
0.01 0.89 0.03 0.01 0.01 0.04
0.01 0.03 0.88 0.03 0.02 0.03
0.03 0.03 0.02 0.90 0.00 0.02
0.01 0.02 0.02 0.01 0.85 0.09
0.01 0.04 0.02 0.01 0.01 0.92
Per-Class: Precision / Recall / F1
Score
Precision
Recall
F1
ROC Curves (One-vs-Rest)
True Positive Rate
False Positive Rate
Analyze (AUC=0.98)
Apply (AUC=0.98)
Create (AUC=0.99)
Evaluate (AUC=0.98)
Remember (AUC=0.99)
Understand (AUC=0.98)
Random

Linear SVM (Char n-gram TF-IDF) — Evaluation

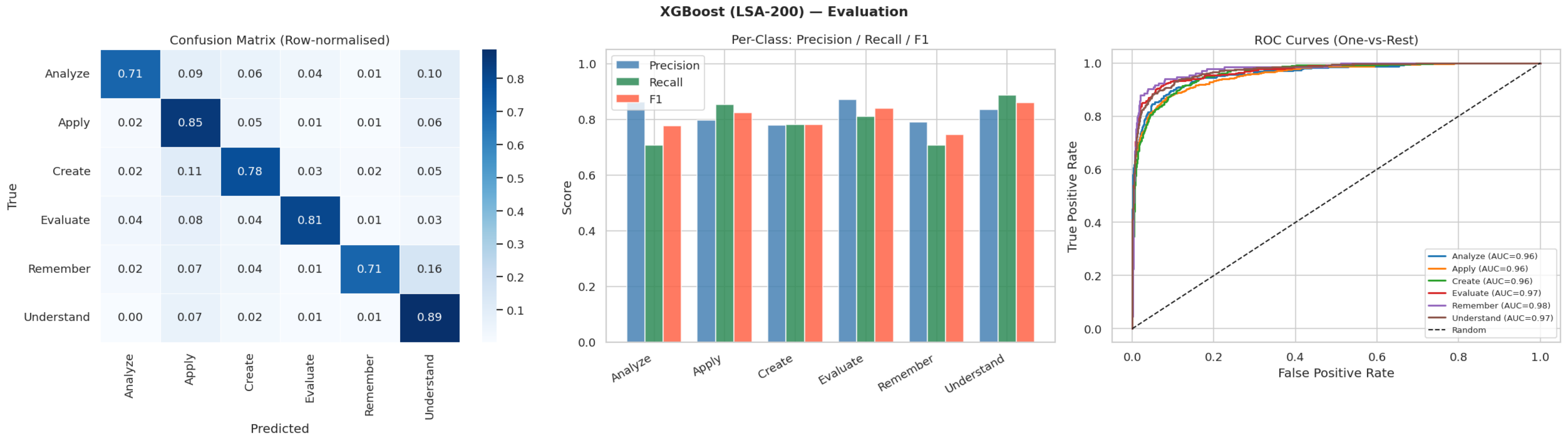

KNN (k=7, Cosine, LSA-200) — Evaluation
Confusion Matrix (Row-normalised)
True
Predicted
Analyze
Apply
Create
Evaluate
Remember
Understand
0.62 0.11 0.07 0.05 0.02 0.13
0.06 0.75 0.06 0.02 0.01 0.10
0.07 0.19 0.63 0.03 0.02 0.05
0.09 0.16 0.06 0.58 0.02 0.08
0.13 0.15 0.07 0.05 0.40 0.21
0.04 0.12 0.05 0.02 0.01 0.76
Per-Class: Precision / Recall / F1
Precision
Recall
F1
Score
ROC Curves (One-vs-Rest)
True Positive Rate
False Positive Rate
Analyze (AUC=0.86)
Apply (AUC=0.88)
Create (AUC=0.90)
Evaluate (AUC=0.88)
Remember (AUC=0.85)
Understand (AUC=0.90)
Random

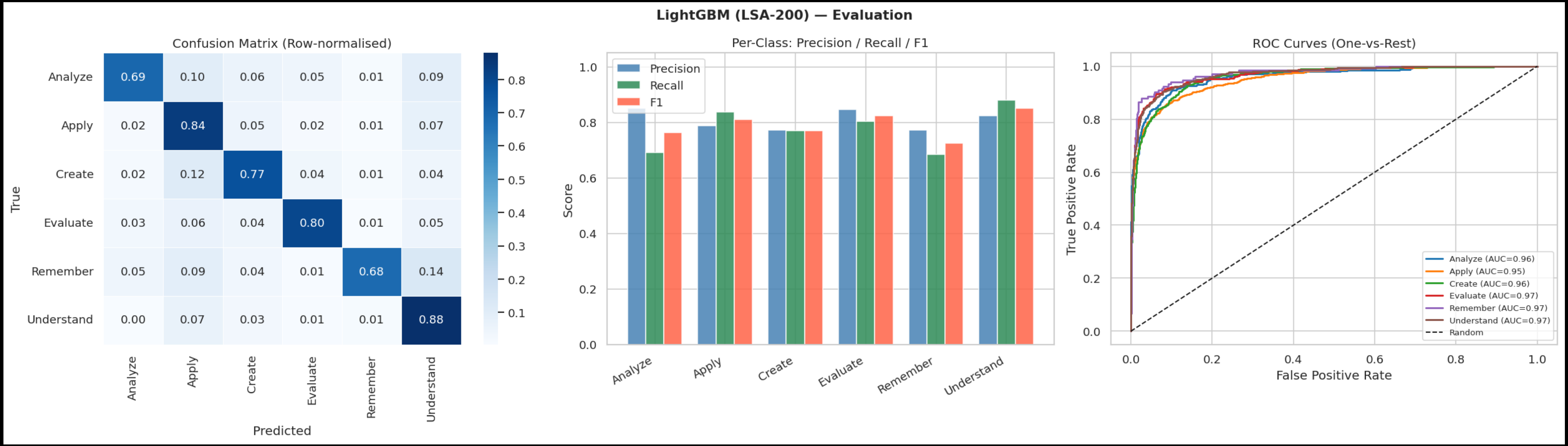

Decision Tree — Evaluation
Confusion Matrix (Row-normalised)
Analyze 0.63 0.31 0.01 0.00 0.01 0.03
Apply 0.01 0.92 0.02 0.01 0.01 0.03
Create 0.01 0.43 0.49 0.02 0.01 0.04
Evaluate 0.02 0.25 0.02 0.69 0.01 0.02
Remember 0.00 0.14 0.03 0.00 0.80 0.03
Understand 0.00 0.10 0.01 0.00 0.01 0.87
True
Predicted
Analyze
Apply
Create
Evaluate
Remember
Understand
Per-Class: Precision / Recall / F1
Precision
Recall
F1
Score
ROC Curves (One-vs-Rest)
Analyze (AUC=0.92)
Apply (AUC=0.93)
Create (AUC=0.91)
Evaluate (AUC=0.94)
Remember (AUC=0.96)
Understand (AUC=0.95)
Random
True Positive Rate
False Positive Rate

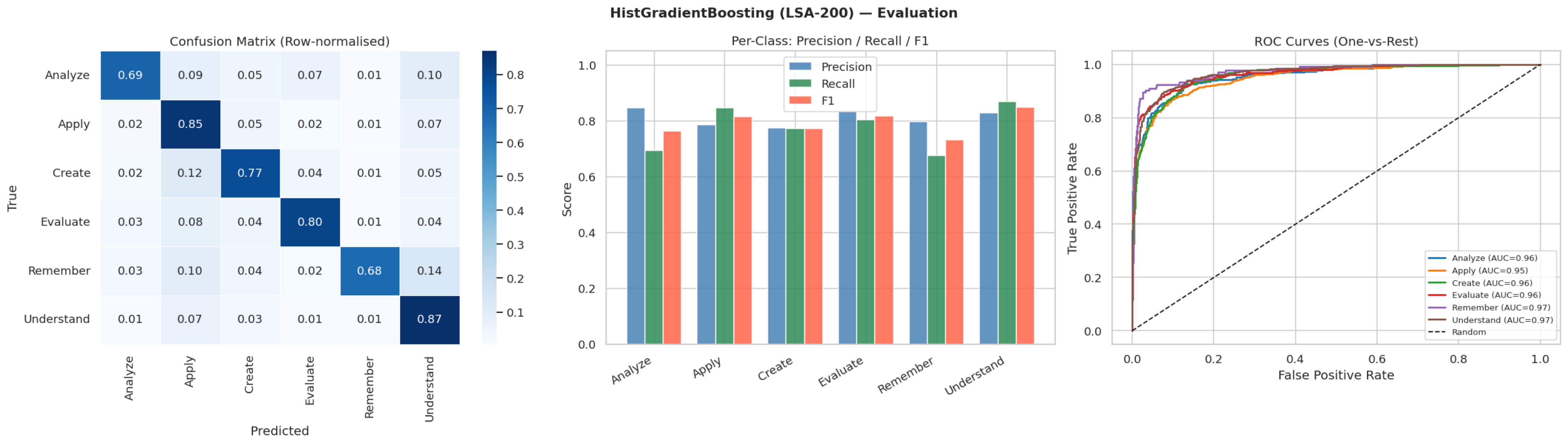

Random Forest (200 trees) — Evaluation
Confusion Matrix (Row-normalised)
True
Predicted
Analyze
Apply
Create
Evaluate
Remember
Understand
0.81 0.07 0.04 0.02 0.01 0.05
0.01 0.90 0.03 0.01 0.01 0.05
0.00 0.05 0.86 0.03 0.01 0.04
0.04 0.04 0.02 0.86 0.01 0.04
0.02 0.02 0.03 0.01 0.79 0.14
0.00 0.04 0.02 0.01 0.01 0.93
Per-Class: Precision / Recall / F1
Score
Precision
Recall
F1
ROC Curves (One-vs-Rest)
True Positive Rate
False Positive Rate
Analyze (AUC=0.98)
Apply (AUC=0.98)
Create (AUC=0.99)
Evaluate (AUC=0.98)
Remember (AUC=0.99)
Understand (AUC=0.98)
Random

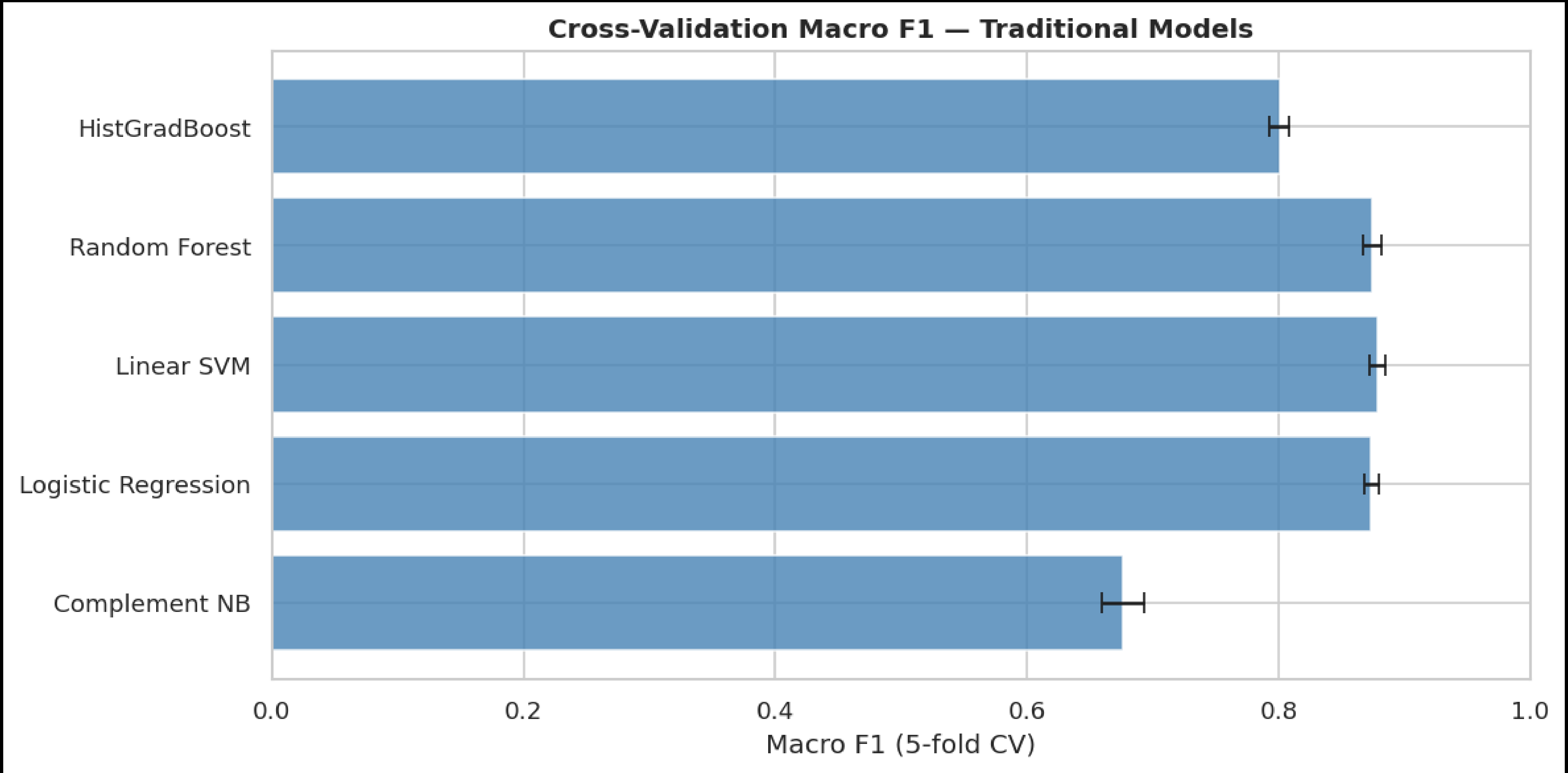


Extra Trees (200 trees) — Evaluation
Confusion Matrix (Row-normalised)
True
Predicted
Analyze
Apply
Create
Evaluate
Remember
Understand
0.80 0.08 0.03 0.02 0.01 0.07
0.01 0.88 0.03 0.01 0.01 0.06
0.01 0.05 0.86 0.03 0.01 0.04
0.04 0.04 0.03 0.85 0.01 0.04
0.02 0.04 0.03 0.00 0.75 0.17
0.00 0.04 0.02 0.01 0.01 0.93
Per-Class: Precision / Recall / F1
Precision
Recall
F1
Score
ROC Curves (One-vs-Rest)
True Positive Rate
False Positive Rate
Analyze (AUC=0.98)
Apply (AUC=0.98)
Create (AUC=0.98)
Evaluate (AUC=0.98)
Remember (AUC=0.99)
Understand (AUC=0.98)
Random


XGBoost (LSA-200) — Evaluation
Confusion Matrix (Row-normalised)
True
Predicted
Analyze
Apply
Create
Evaluate
Remember
Understand
0.71 0.09 0.06 0.04 0.01 0.10
0.02 0.85 0.05 0.01 0.01 0.06
0.02 0.11 0.78 0.03 0.02 0.05
0.04 0.08 0.04 0.81 0.01 0.03
0.02 0.07 0.04 0.01 0.71 0.16
0.00 0.07 0.02 0.01 0.01 0.89
Per-Class: Precision / Recall / F1
Precision
Recall
F1
Score
ROC Curves (One-vs-Rest)
True Positive Rate
False Positive Rate
Analyze (AUC=0.96)
Apply (AUC=0.96)
Create (AUC=0.96)
Evaluate (AUC=0.97)
Remember (AUC=0.98)
Understand (AUC=0.97)
Random

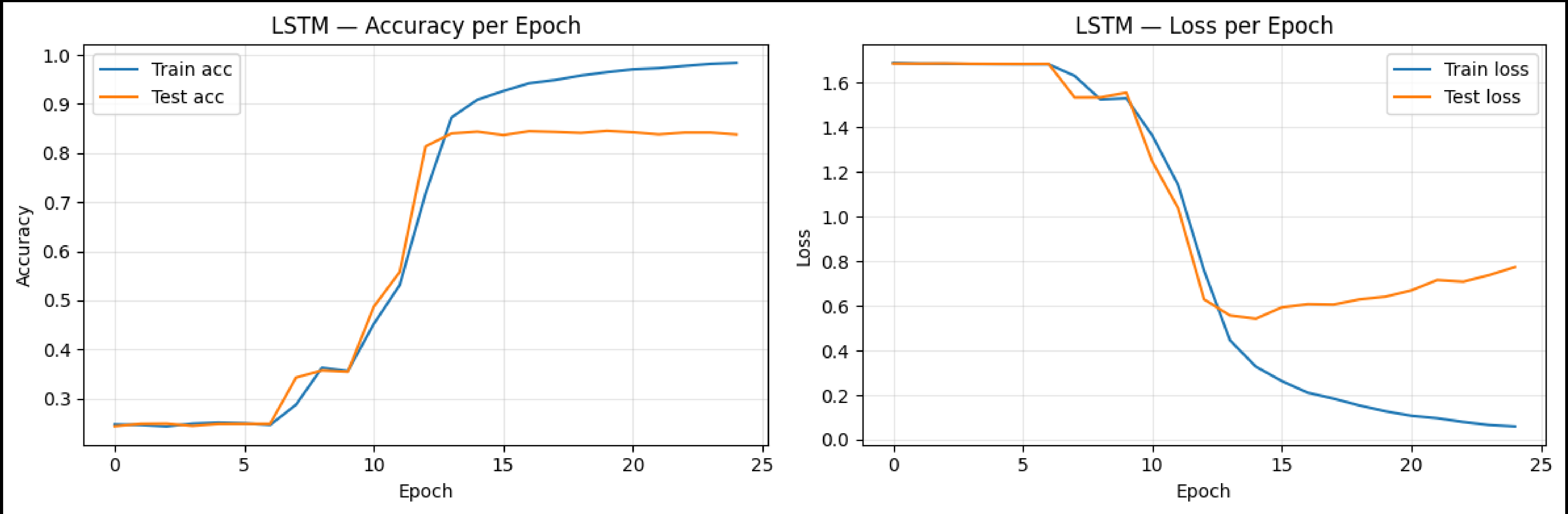

LightGBM (LSA-200) — Evaluation
Confusion Matrix (Row-normalised)
True
Predicted
Analyze
Apply
Create
Evaluate
Remember
Understand
0.69 0.10 0.06 0.05 0.01 0.09
0.02 0.84 0.05 0.02 0.01 0.07
0.02 0.12 0.77 0.04 0.01 0.04
0.03 0.06 0.04 0.80 0.01 0.05
0.05 0.09 0.04 0.01 0.68 0.14
0.00 0.07 0.03 0.01 0.01 0.88
Per-Class: Precision / Recall / F1
Precision
Recall
F1
Score
ROC Curves (One-vs-Rest)
True Positive Rate
False Positive Rate
Analyze (AUC=0.96)
Apply (AUC=0.95)
Create (AUC=0.96)
Evaluate (AUC=0.97)
Remember (AUC=0.97)
Understand (AUC=0.97)
Random


HistGradientBoosting (LSA-200) — Evaluation
Confusion Matrix (Row-normalised)
Analyze 0.69 0.09 0.05 0.07 0.01 0.10
Apply 0.02 0.85 0.05 0.02 0.01 0.07
Create 0.02 0.12 0.77 0.04 0.01 0.05
Evaluate 0.03 0.08 0.04 0.80 0.01 0.04
Remember 0.03 0.10 0.04 0.02 0.68 0.14
Understand 0.01 0.07 0.03 0.01 0.01 0.87
True
Predicted
Analyze Apply Create Evaluate Remember Understand
Per-Class: Precision / Recall / F1
Precision
Recall
F1
Score
ROC Curves (One-vs-Rest)
Analyze (AUC=0.96)
Apply (AUC=0.95)
Create (AUC=0.96)
Evaluate (AUC=0.96)
Remember (AUC=0.97)
Understand (AUC=0.97)
Random
True Positive Rate
False Positive Rate

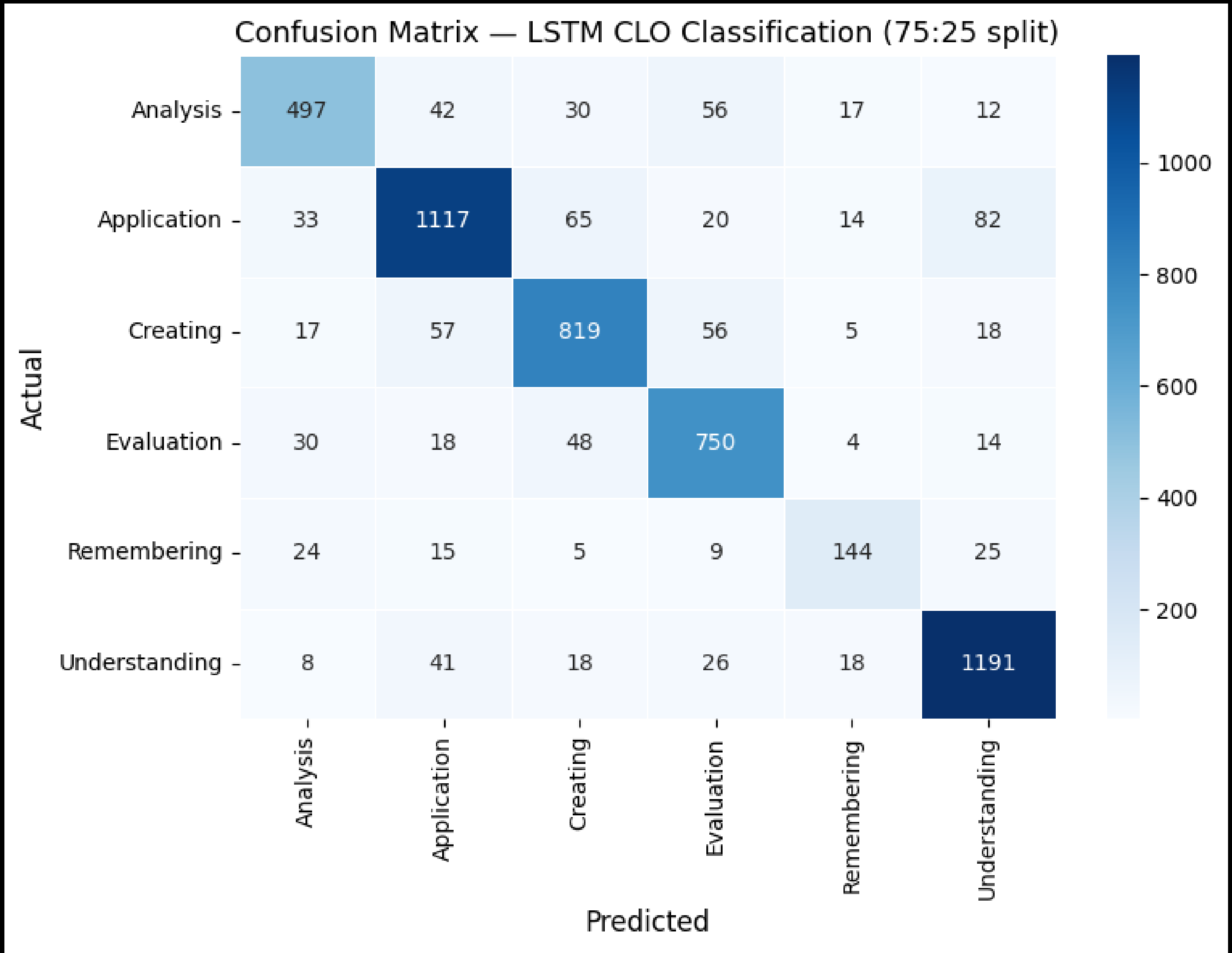


Cross-Validation Macro F1 — Traditional Models
HistGradBoost
Random Forest
Linear SVM
Logistic Regression
Complement NB
0.0
0.2
0.4
0.6
0.8
1.0
Macro F1 (5-fold CV)

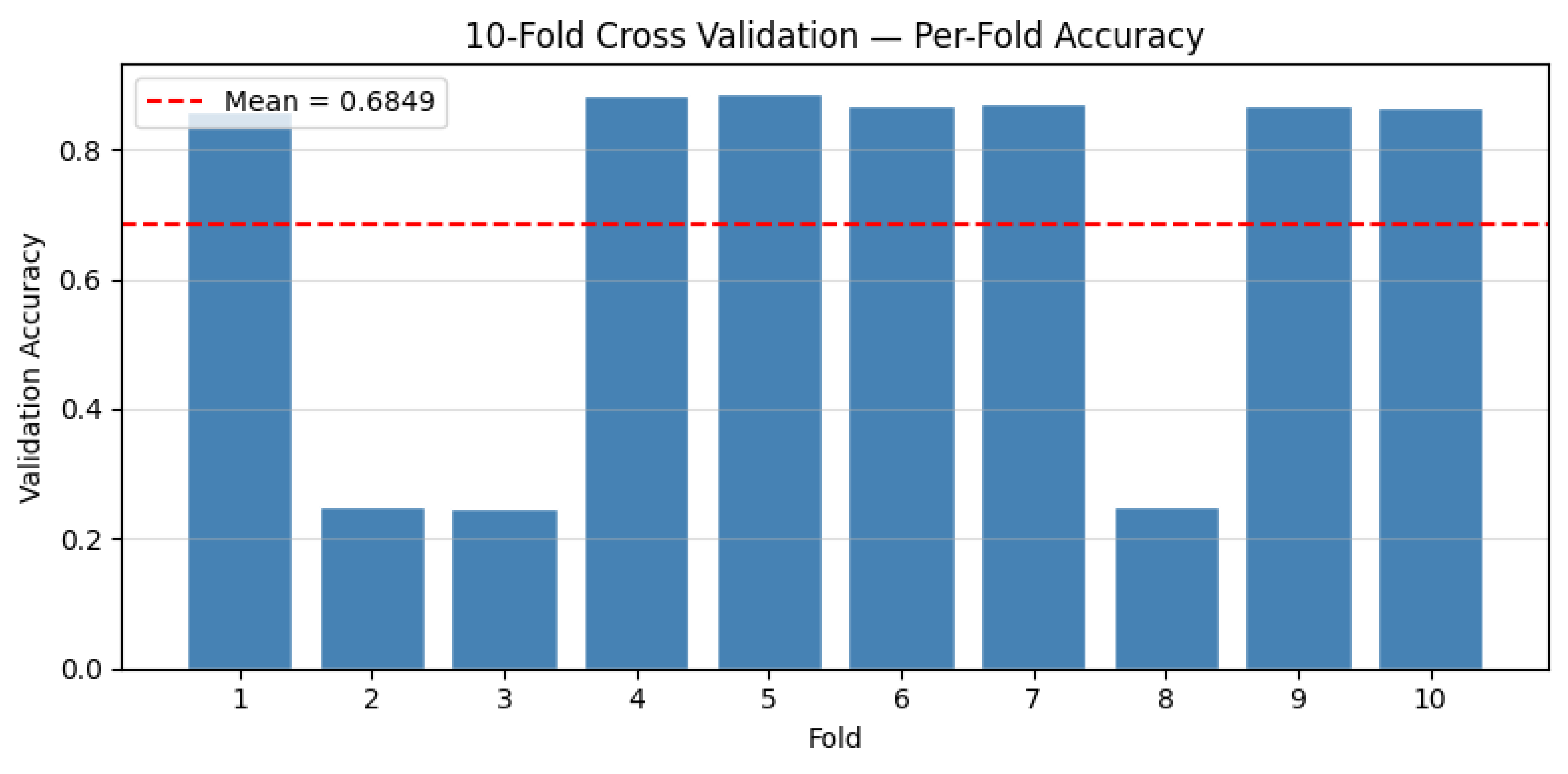
LSTM


LSTM — Accuracy per Epoch
Train acc
Test acc
Accuracy
Epoch
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0
5
10
15
20
25
LSTM — Loss per Epoch
Train loss
Test loss
Loss
Epoch
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
25

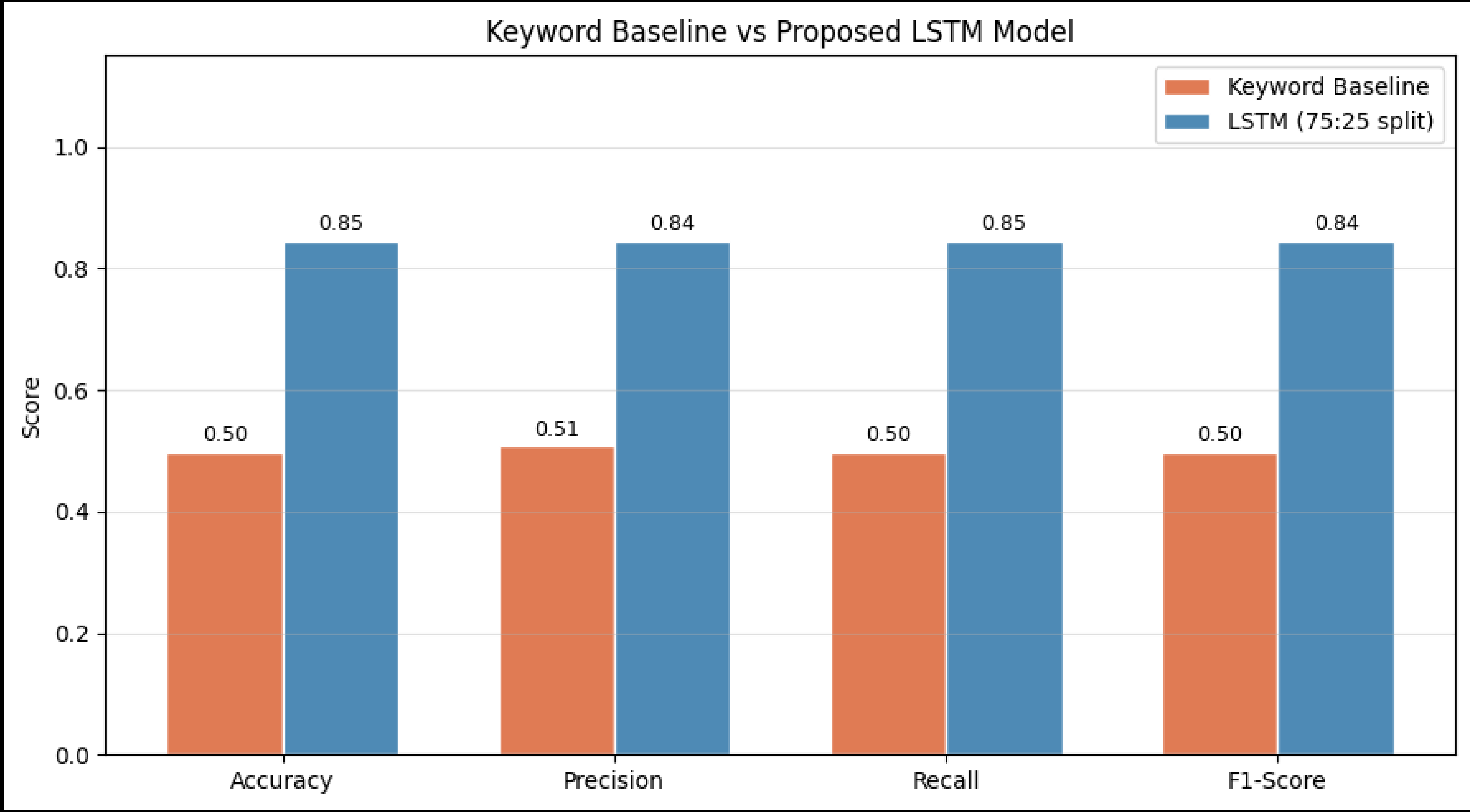
LSTM


Confusion Matrix — LSTM CLO Classification (75:25 split)
Actual
Predicted
Analysis
Application
Creating
Evaluation
Remembering
Understanding
497 42 30 56 17 12
33 1117 65 20 14 82
17 57 819 56 5 18
30 18 48 750 4 14
24 15 5 9 144 25
8 41 18 26 18 1191
1000
800
600
400
200

LSTM

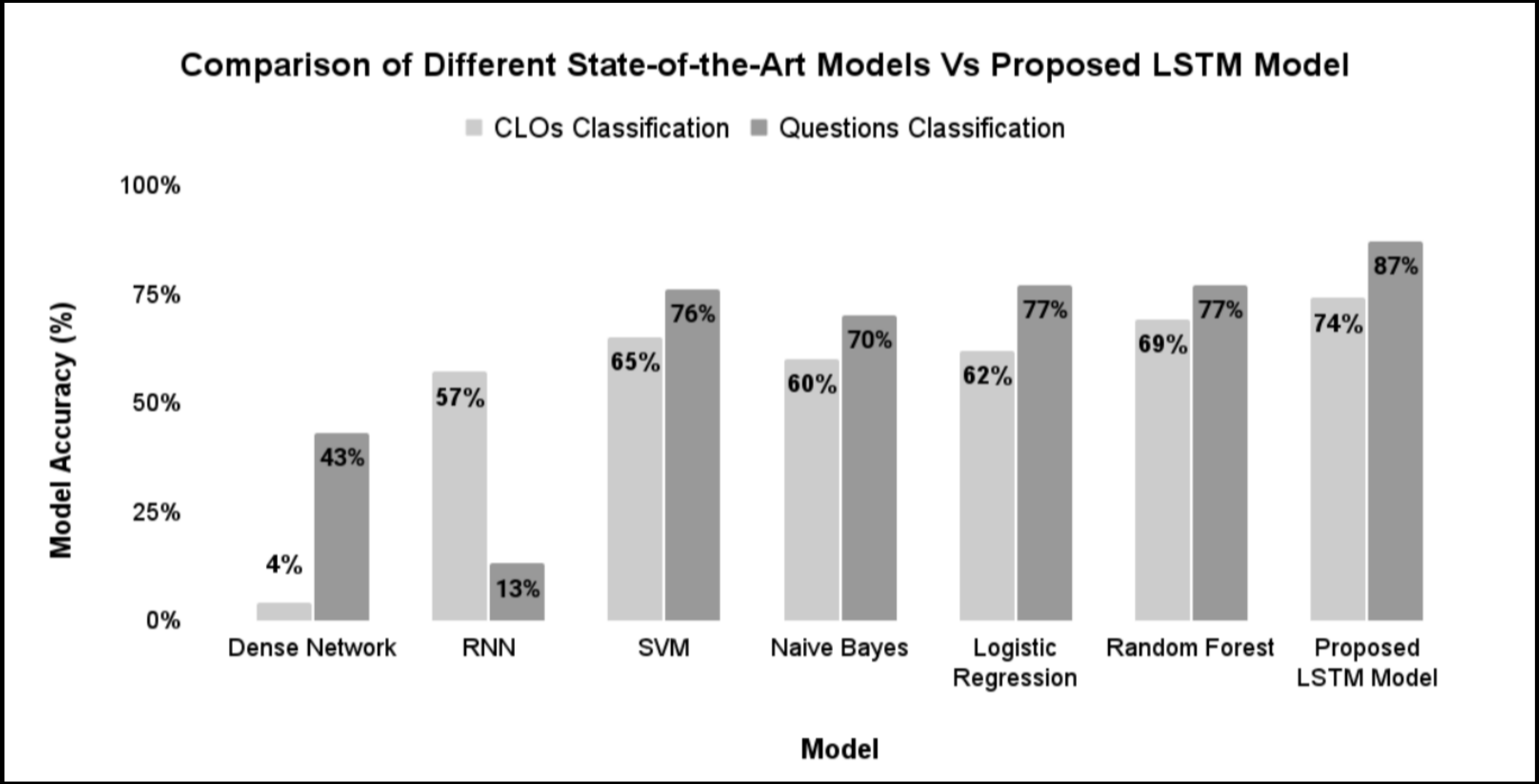

10-Fold Cross Validation — Per-Fold Accuracy
Mean = 0.6849
Validation Accuracy
0.0
0.2
0.4
0.6
0.8
1
2
3
4
5
6
7
8
9
10
Fold

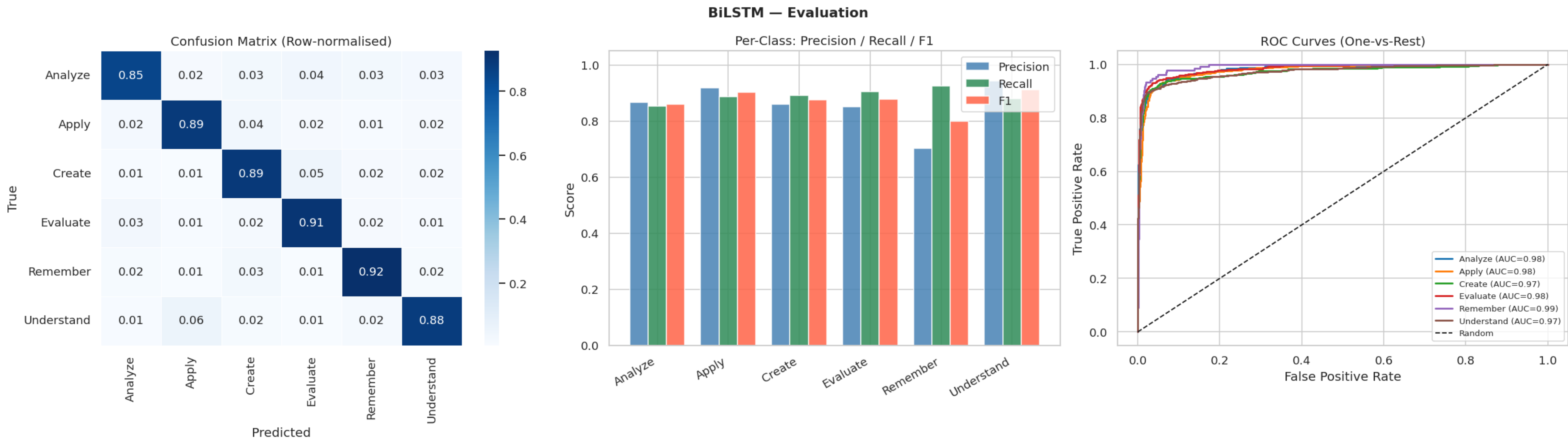
LSTM

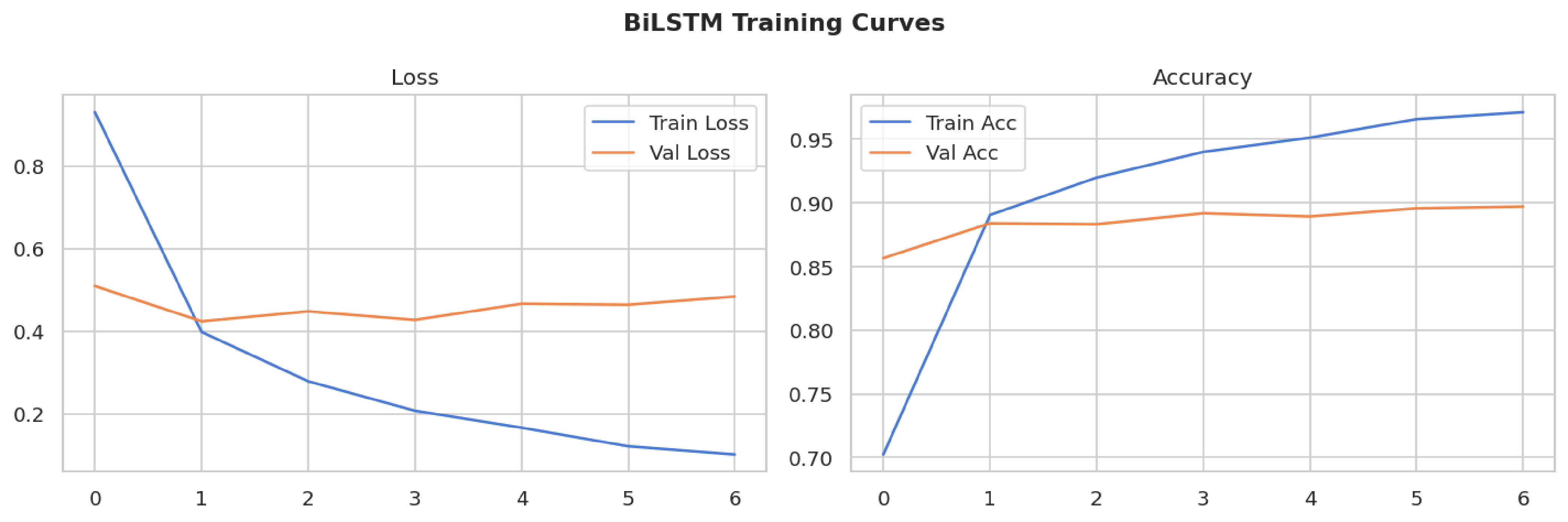


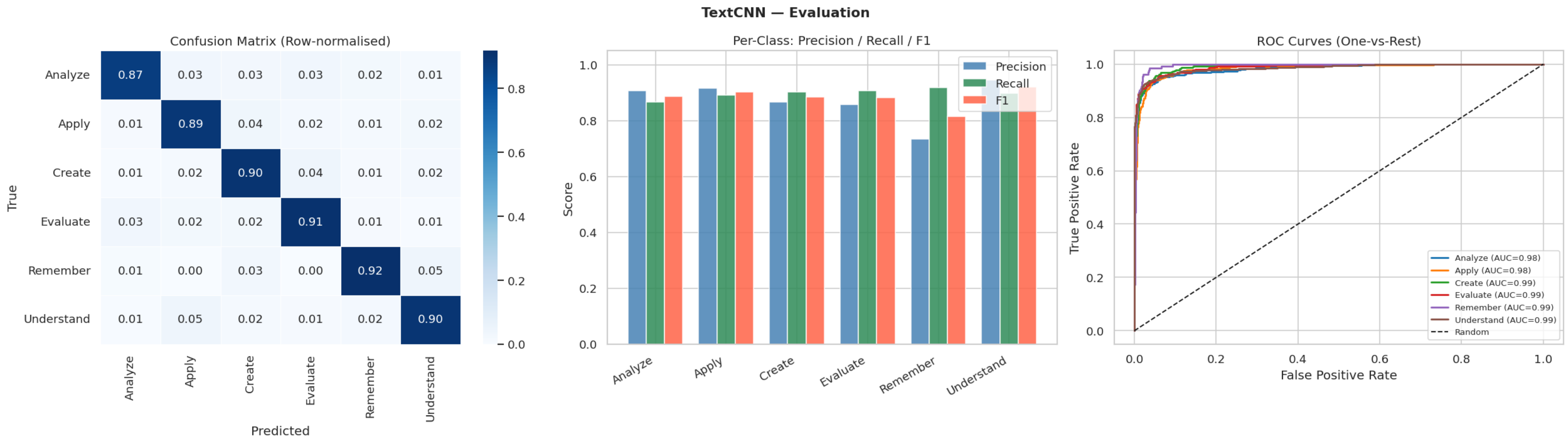
Keyword Baseline vs Proposed LSTM Model
Keyword Baseline
LSTM (75:25 split)
Score
1.0
0.8
0.6
0.4
0.2
0.0
0.85
0.84
0.85
0.84
0.50
0.51
0.50
0.50
Accuracy
Precision
Recall
F1-Score

LSTM (Shaikh et al. 2021)

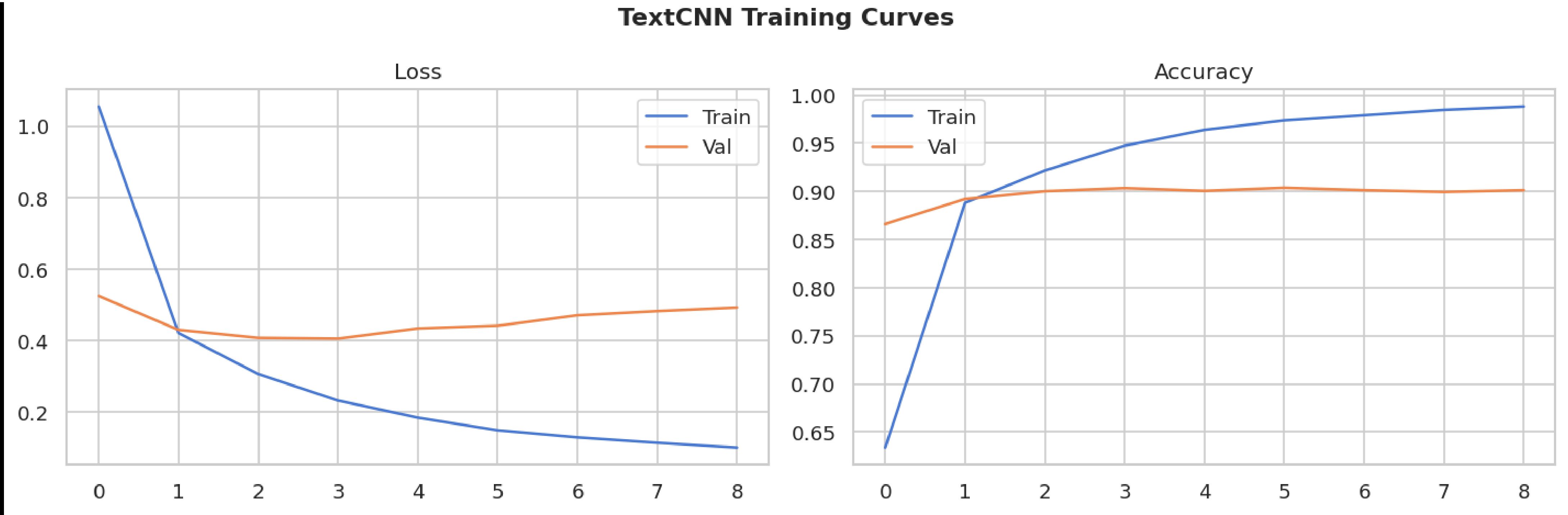



BiLSTM — Evaluation
Confusion Matrix (Row-normalised)
True
Predicted
Analyze
Apply
Create
Evaluate
Remember
Understand
0.85 0.02 0.03 0.04 0.03 0.03
0.02 0.89 0.04 0.02 0.01 0.02
0.01 0.01 0.89 0.05 0.02 0.02
0.03 0.01 0.02 0.91 0.02 0.01
0.02 0.01 0.03 0.01 0.92 0.02
0.01 0.06 0.02 0.01 0.02 0.88
Per-Class: Precision / Recall / F1
Score
Precision
Recall
F1
ROC Curves (One-vs-Rest)
True Positive Rate
False Positive Rate
Analyze (AUC=0.98)
Apply (AUC=0.98)
Create (AUC=0.97)
Evaluate (AUC=0.98)
Remember (AUC=0.99)
Understand (AUC=0.97)
Random

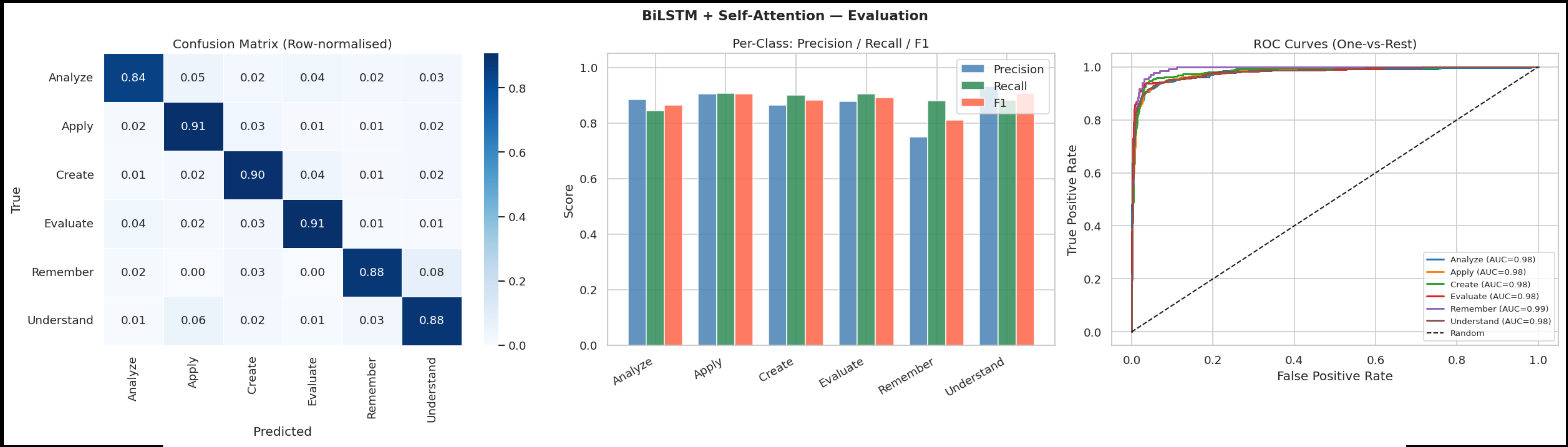

BiLSTM Training Curves
Loss
Train Loss
Val Loss
Accuracy
Train Acc
Val Acc

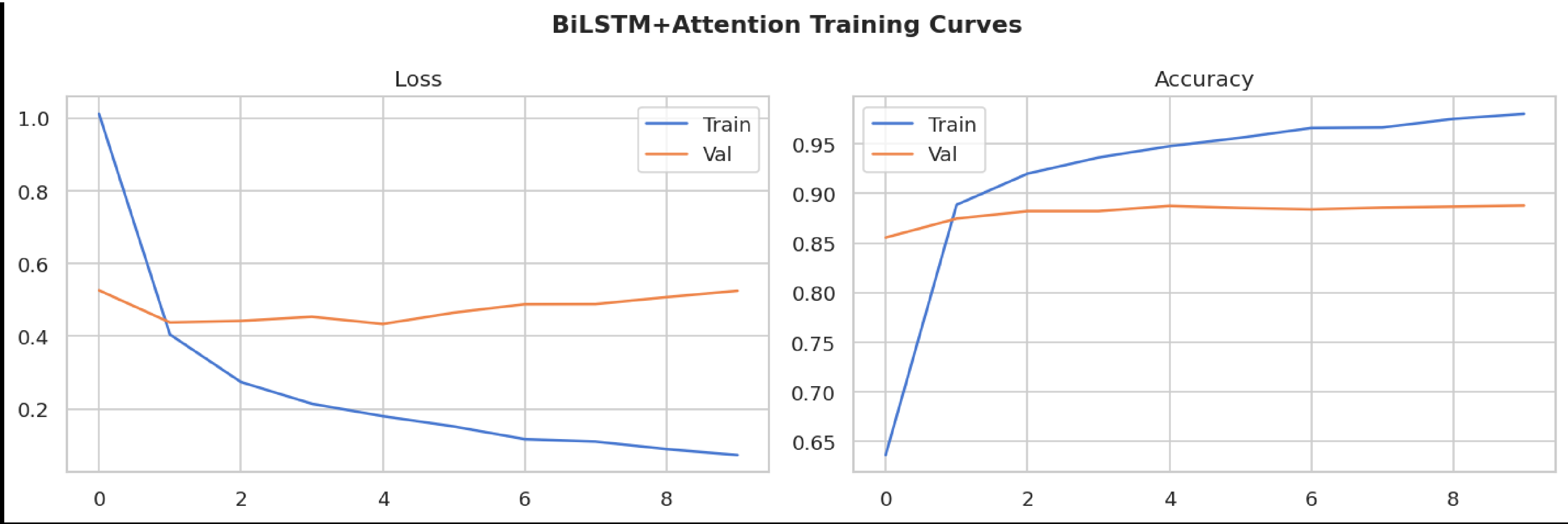


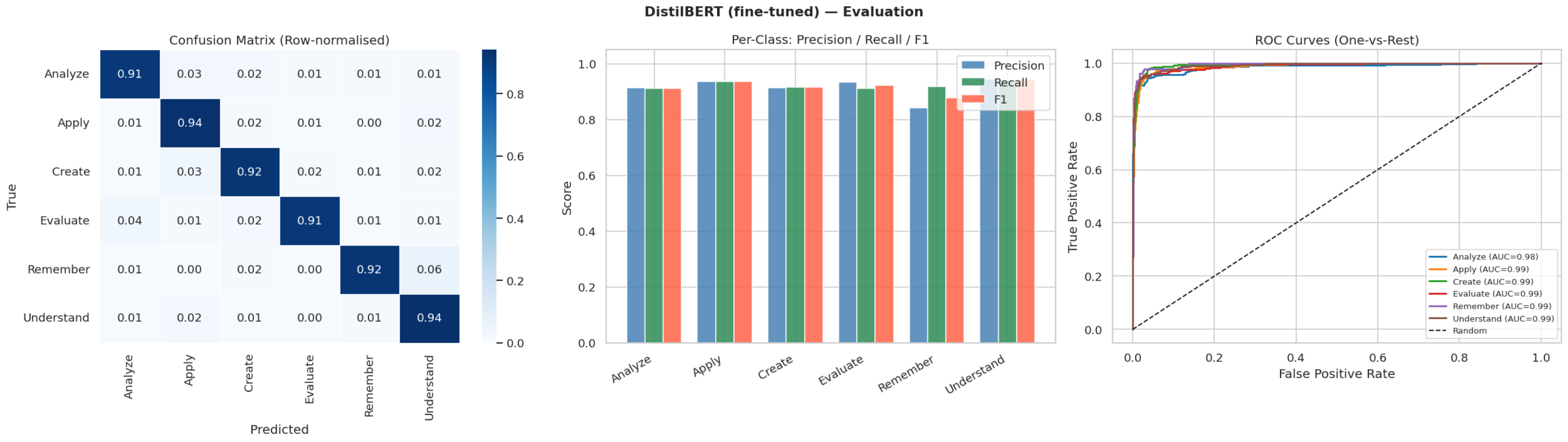
TextCNN — Evaluation
Confusion Matrix (Row-normalised)
Per-Class: Precision / Recall / F1
ROC Curves (One-vs-Rest)
Precision
Recall
F1
Analyze (AUC=0.98)
Apply (AUC=0.98)
Create (AUC=0.99)
Evaluate (AUC=0.99)
Remember (AUC=0.99)
Understand (AUC=0.99)
Random
True
Predicted
Score
True Positive Rate
False Positive Rate
Analyze
Apply
Create
Evaluate
Remember
Understand



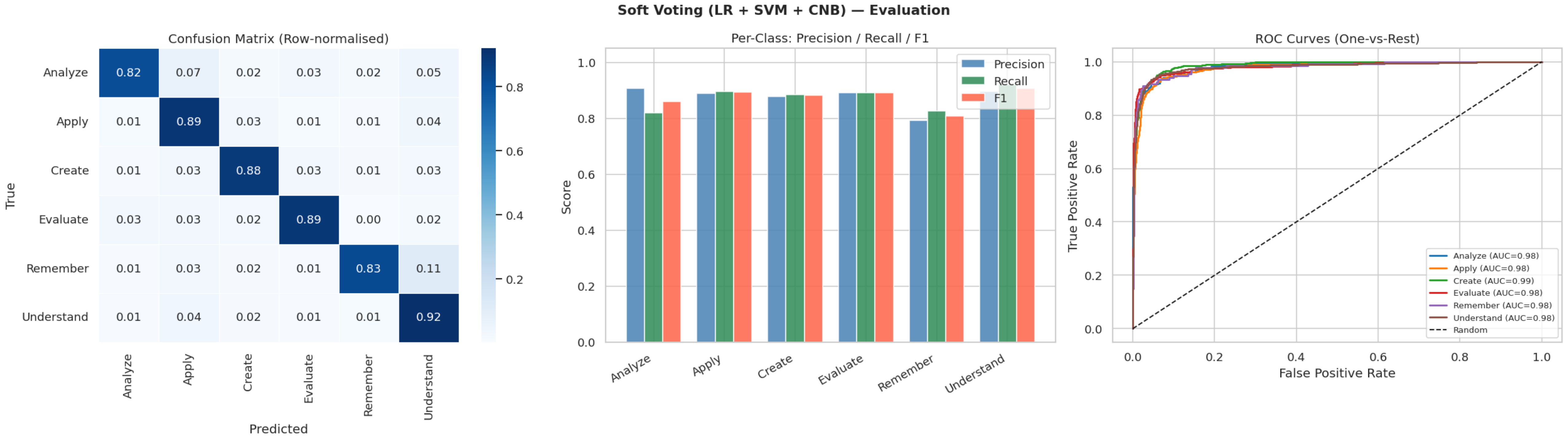
TextCNN Training Curves
Loss
Accuracy
Train
Val


BiLSTM + Self-Attention — Evaluation
Confusion Matrix (Row-normalised)
True
Predicted
Analyze
Apply
Create
Evaluate
Remember
Understand
0.84 0.05 0.02 0.04 0.02 0.03
0.02 0.91 0.03 0.01 0.01 0.02
0.01 0.02 0.90 0.04 0.01 0.02
0.04 0.02 0.03 0.91 0.01 0.01
0.02 0.00 0.03 0.00 0.88 0.08
0.01 0.06 0.02 0.01 0.03 0.88
Per-Class: Precision / Recall / F1
Score
Precision
Recall
F1
ROC Curves (One-vs-Rest)
True Positive Rate
False Positive Rate
Analyze (AUC=0.98)
Apply (AUC=0.98)
Create (AUC=0.98)
Evaluate (AUC=0.98)
Remember (AUC=0.99)
Understand (AUC=0.98)
Random


BiLSTM+Attention Training Curves
Loss
Accuracy
Train
Val

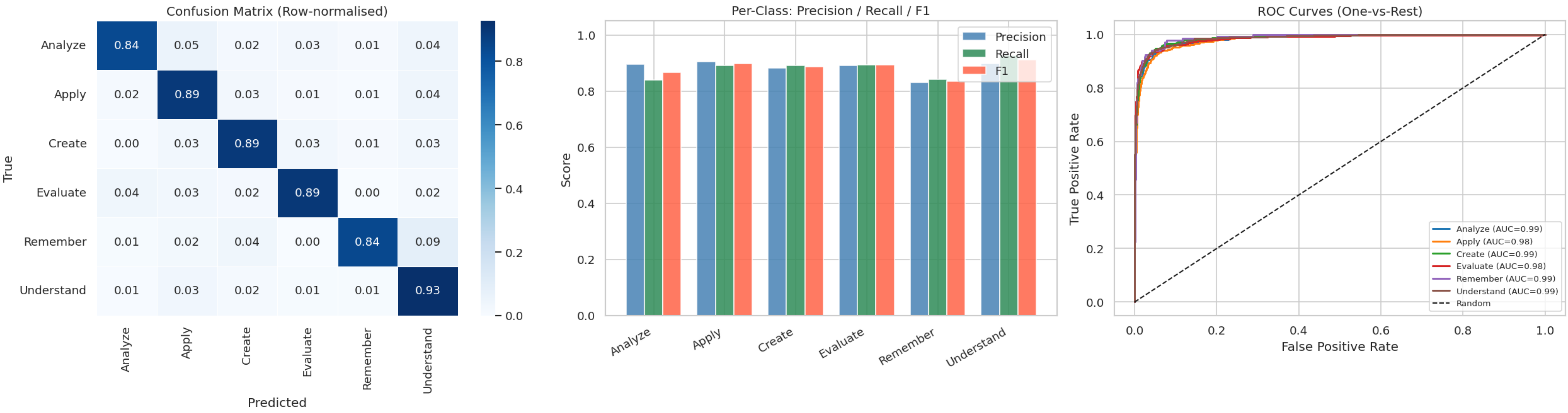


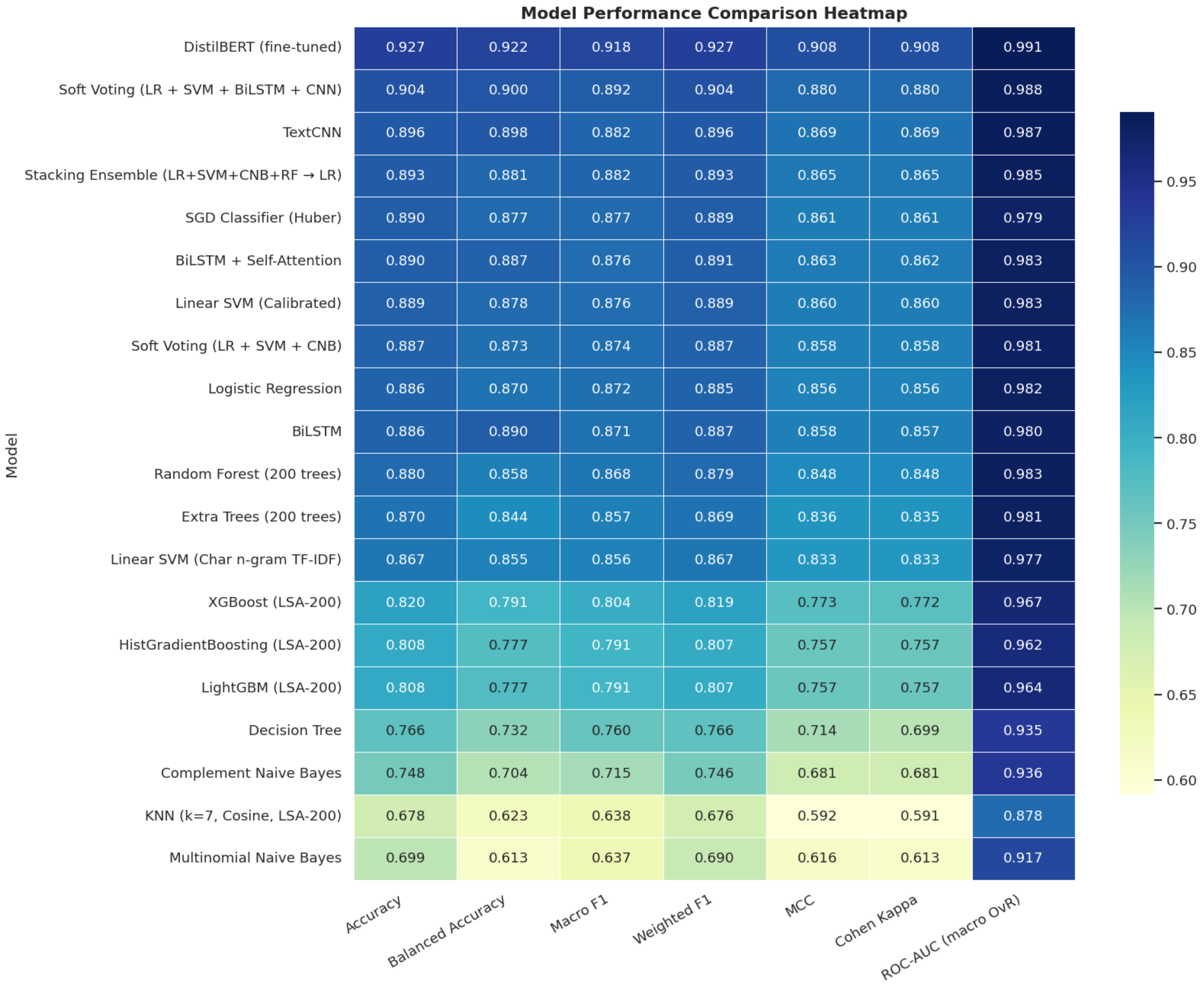
DistilBERT (fine-tuned) — Evaluation
Confusion Matrix (Row-normalised)
True
Predicted
Analyze
Apply
Create
Evaluate
Remember
Understand
0.91 0.03 0.02 0.01 0.01 0.01
0.01 0.94 0.02 0.01 0.00 0.02
0.01 0.03 0.92 0.02 0.01 0.02
0.04 0.01 0.02 0.91 0.01 0.01
0.01 0.00 0.02 0.00 0.92 0.06
0.01 0.02 0.01 0.00 0.01 0.94
Per-Class: Precision / Recall / F1
Score
Precision
Recall
F1
ROC Curves (One-vs-Rest)
True Positive Rate
False Positive Rate
Analyze (AUC=0.98)
Apply (AUC=0.99)
Create (AUC=0.99)
Evaluate (AUC=0.99)
Remember (AUC=0.99)
Understand (AUC=0.99)
Random

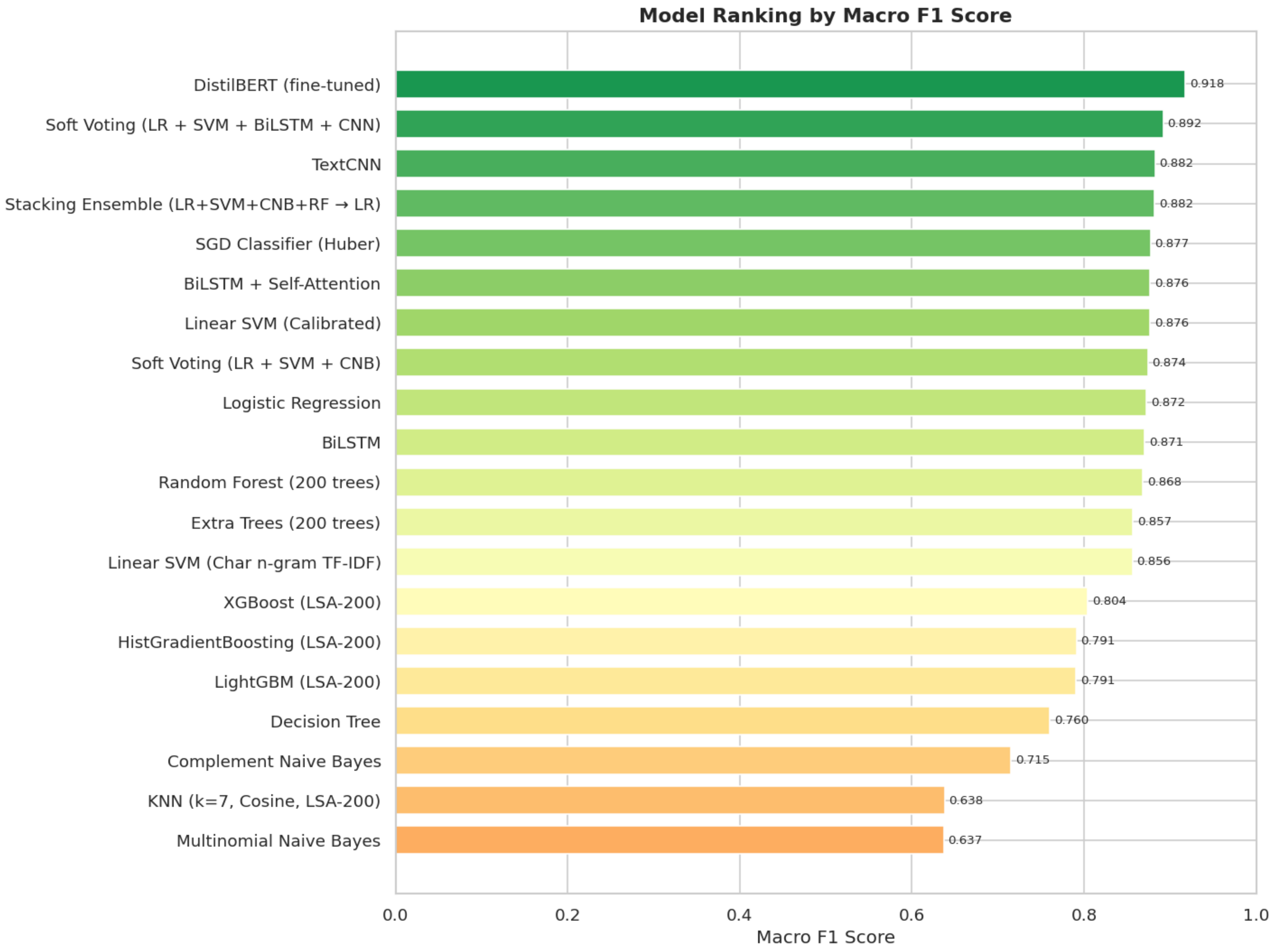


Soft Voting (LR + SVM + CNB) — Evaluation
Confusion Matrix (Row-normalised)
True
Predicted
Analyze
Apply
Create
Evaluate
Remember
Understand
0.82 0.07 0.02 0.03 0.02 0.05
0.01 0.89 0.03 0.01 0.01 0.04
0.01 0.03 0.88 0.03 0.01 0.03
0.03 0.03 0.02 0.89 0.00 0.02
0.01 0.03 0.02 0.01 0.83 0.11
0.01 0.04 0.02 0.01 0.01 0.92
Per-Class: Precision / Recall / F1
Precision
Recall
F1
Score
ROC Curves (One-vs-Rest)
True Positive Rate
False Positive Rate
Analyze (AUC=0.98)
Apply (AUC=0.98)
Create (AUC=0.99)
Evaluate (AUC=0.98)
Remember (AUC=0.98)
Understand (AUC=0.98)
Random

Stacking Ensemble (LR+SVM+CNB+RF → LR) — Evaluation




Model Performance Comparison Heatmap
Accuracy
Balanced Accuracy
Macro F1
Weighted F1
MCC
Cohen Kappa
ROC-AUC (macro OvR)
Model
DistilBERT (fine-tuned) 0.927 0.922 0.918 0.927 0.908 0.908 0.991
Soft Voting (LR + SVM + BiLSTM + CNN) 0.904 0.900 0.892 0.904 0.880 0.880 0.988
TextCNN 0.896 0.898 0.882 0.896 0.869 0.869 0.987
Stacking Ensemble (LR+SVM+CNB+RF → LR) 0.893 0.881 0.882 0.893 0.865 0.865 0.985
SGD Classifier (Huber) 0.890 0.877 0.877 0.889 0.861 0.861 0.979
BiLSTM + Self-Attention 0.890 0.887 0.876 0.891 0.863 0.862 0.983
Linear SVM (Calibrated) 0.889 0.878 0.876 0.889 0.860 0.860 0.983
Soft Voting (LR + SVM + CNB) 0.887 0.873 0.874 0.887 0.858 0.858 0.981
Logistic Regression 0.886 0.870 0.872 0.885 0.856 0.856 0.982
BiLSTM 0.886 0.890 0.871 0.887 0.858 0.857 0.980
Random Forest (200 trees) 0.880 0.858 0.868 0.879 0.848 0.848 0.983
Extra Trees (200 trees) 0.870 0.844 0.857 0.869 0.836 0.835 0.981
Linear SVM (Char n-gram TF-IDF) 0.867 0.855 0.856 0.867 0.833 0.833 0.977
XGBoost (LSA-200) 0.820 0.791 0.804 0.819 0.773 0.772 0.967
HistGradientBoosting (LSA-200) 0.808 0.777 0.791 0.807 0.757 0.757 0.962
LightGBM (LSA-200) 0.808 0.777 0.791 0.807 0.757 0.757 0.964
Decision Tree 0.766 0.732 0.760 0.766 0.714 0.699 0.935
Complement Naive Bayes 0.748 0.704 0.715 0.746 0.681 0.681 0.936
KNN (k=7, Cosine, LSA-200) 0.678 0.623 0.638 0.676 0.592 0.591 0.878
Multinomial Naive Bayes 0.699 0.613 0.637 0.690 0.616 0.613 0.917
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60


Model Ranking by Macro F1 Score
DistilBERT (fine-tuned)
0.918
Soft Voting (LR + SVM + BiLSTM + CNN)
0.892
TextCNN
0.882
Stacking Ensemble (LR+SVM+CNB+RF → LR)
0.882
SGD Classifier (Huber)
0.877
BiLSTM + Self-Attention
0.876
Linear SVM (Calibrated)
0.876
Soft Voting (LR + SVM + CNB)
0.874
Logistic Regression
0.872
BiLSTM
0.871
Random Forest (200 trees)
0.868
Extra Trees (200 trees)
0.857
Linear SVM (Char n-gram TF-IDF)
0.856
XGBoost (LSA-200)
0.804
HistGradientBoosting (LSA-200)
0.791
LightGBM (LSA-200)
0.791
Decision Tree
0.760
Complement Naive Bayes
0.715
KNN (k=7, Cosine, LSA-200)
0.638
Multinomial Naive Bayes
0.637
0.0
0.2
0.4
0.6
0.8
1.0
Macro F1 Score

13 Open Source LLMs

1. mistral_7b_instruct_v02
2. llama3_8b_instruct
3. phi35_mini_4k_instruct
4. qwen2_7b_instruct
5. llama31_8b_instruct
6. qwen25_7b_instruct
7. qwen25_3b_instruct
8. qwen25_1_5b_instruct
9. smollm2_1_7b_instruct
10. gemma2_2b_instruct
11. gemma2_9b_instruct
12. phi35_mini_Instruct
13. mistral_nemo_instruct

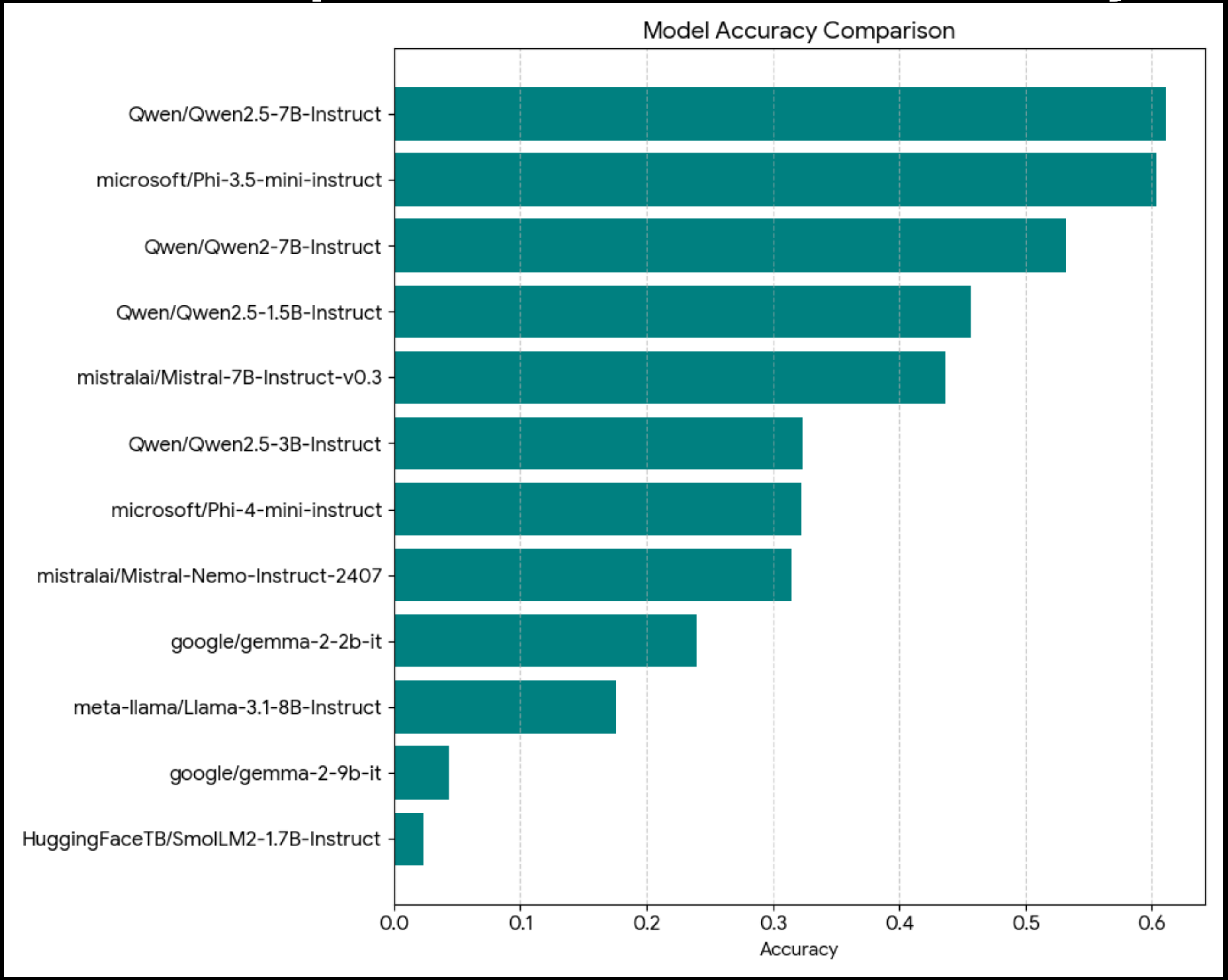
Metrics for Evaluation

- ACCURACY: Did the model guess right overall?
- MICRO F1: How good is the model when treating all predictions as one big group?
- MACRO F1: How good is the model on average for each category separately?
- WEIGHTED F1: How good is the model for each category, adjusted by how many items are in each?
- MACRO PRECISION: On average, when the model says "yes," is it actually correct for each category?
- MACRO RECALL: On average, does the model catch all the correct items for each category?
- BALANCED ACCURACY: Does the model do equally well at recognizing each category?
- MCC: How strongly does the model's prediction match reality (from perfect to completely wrong)?
- COHEN'S KAPPA: Does the model agree with the real answer more than just lucky guessing?

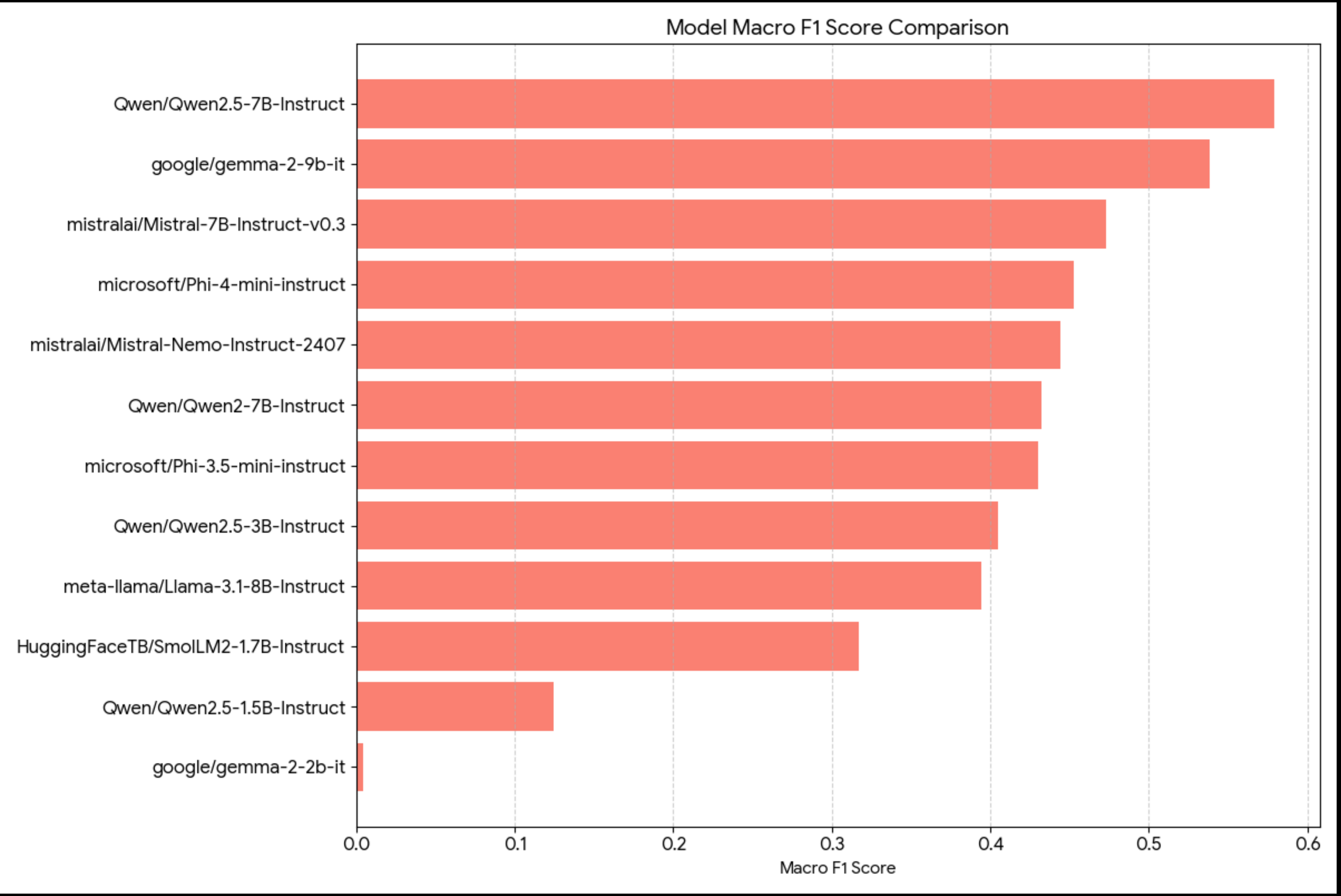
Metrics for Evaluation

- ACCURACY: The fraction of all predictions that were correct.
- MICRO F1: Overall F1 score treating every prediction equally across all classes.
- MACRO F1: Average of each class’s F1 score (treats all classes equally).
- WEIGHTED F1: Average of class F1 scores weighted by each class’s support (frequency).
- MACRO PRECISION: Average of each class’s precision (how correct positive predictions are), averaged across classes.
- MACRO RECALL: Average of each class’s recall (how many true items are found), averaged across classes.
- BALANCED ACCURACY: Average of per-class recall (equal weight per class), i.e., mean recall across classes.
- MCC (Matthews Correlation Coefficient): Single-number correlation between predictions and true labels, ranging from -1 (worst) to +1 (best), 0 = random.
- COHEN'S KAPPA: Agreement between predictions and truth adjusted for chance agreement (1 = perfect, 0 = chance).

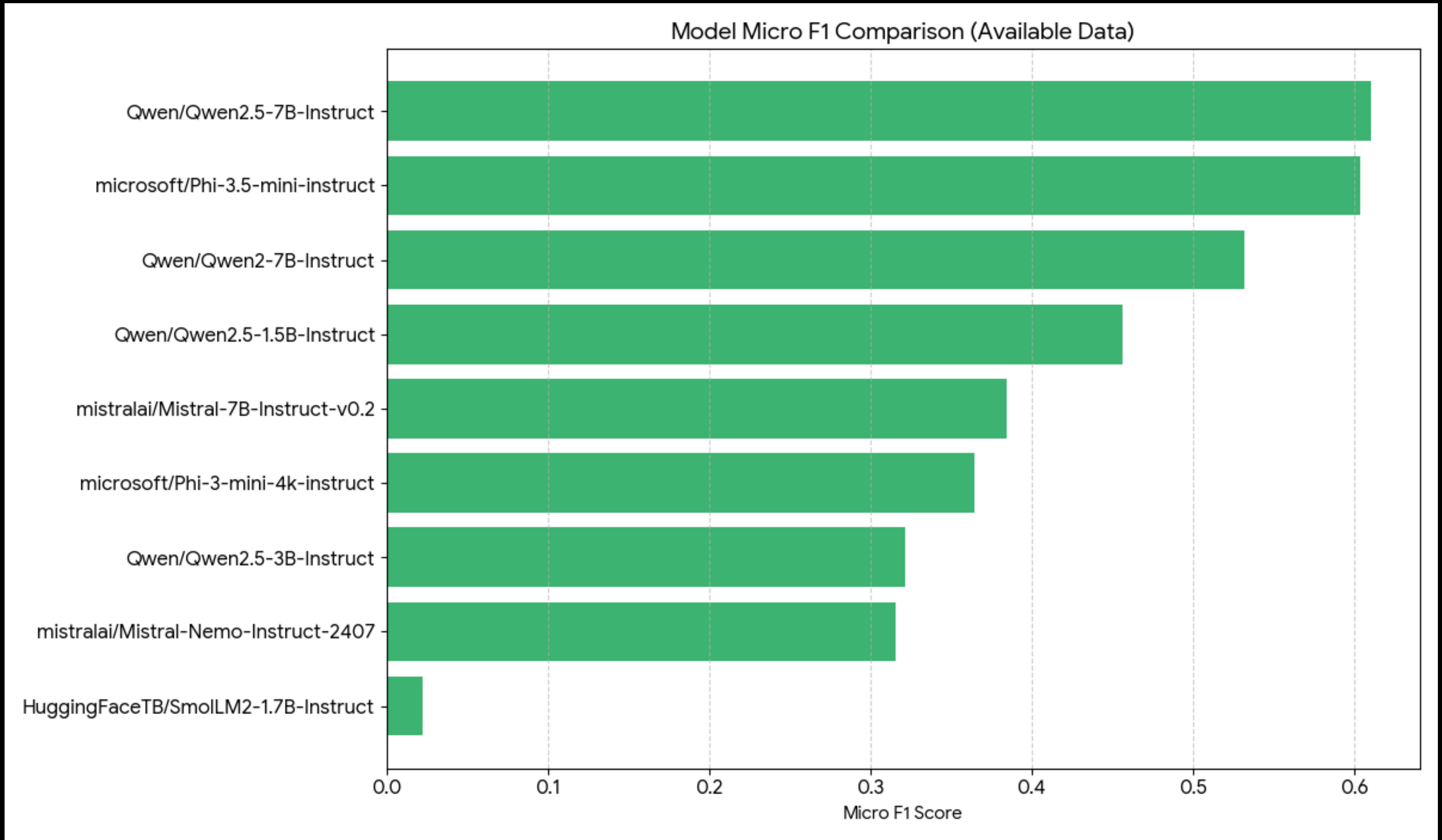
Zero-shot comparison via Accuracy



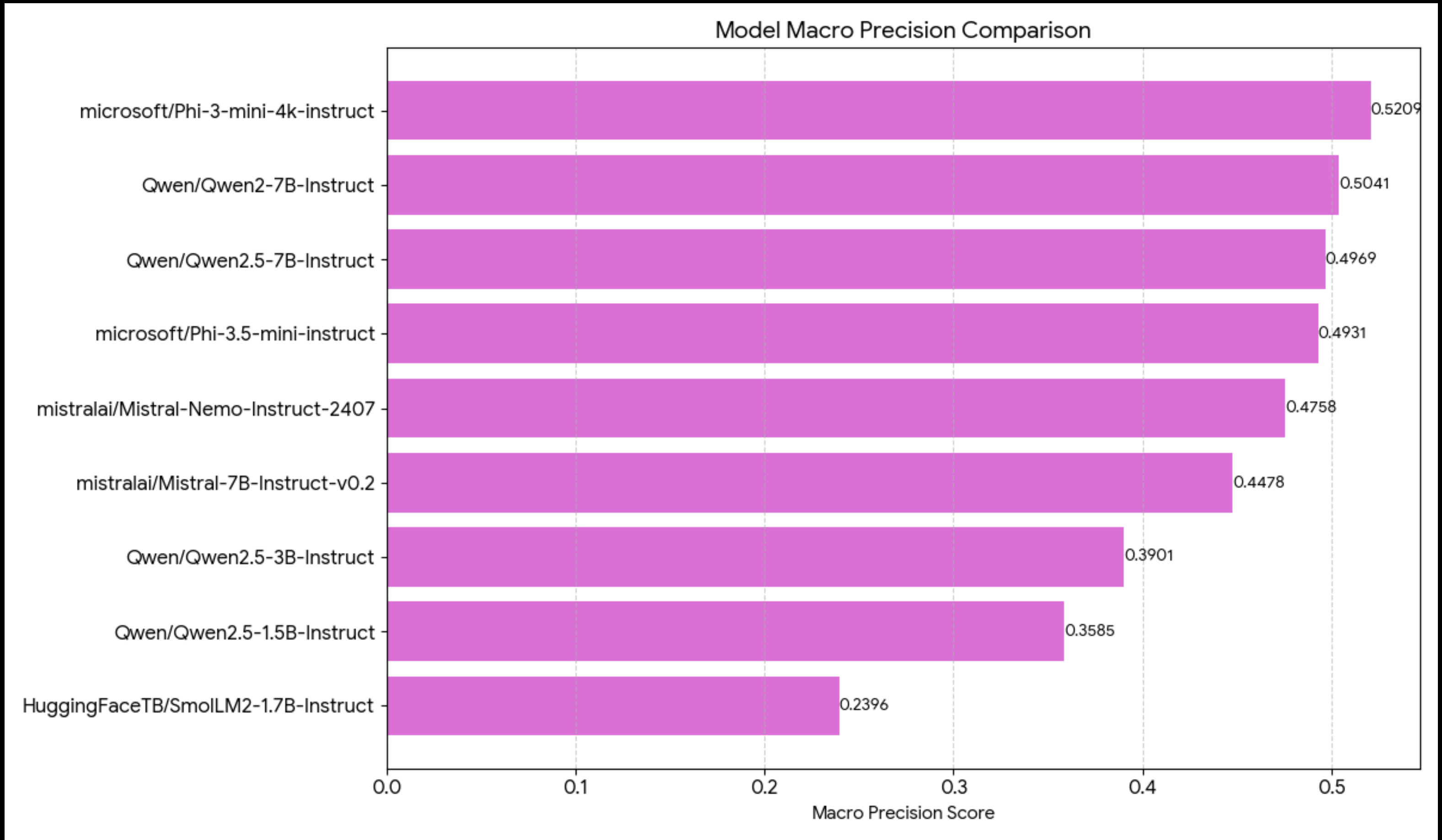
Zero-shot comparison via Macro F1



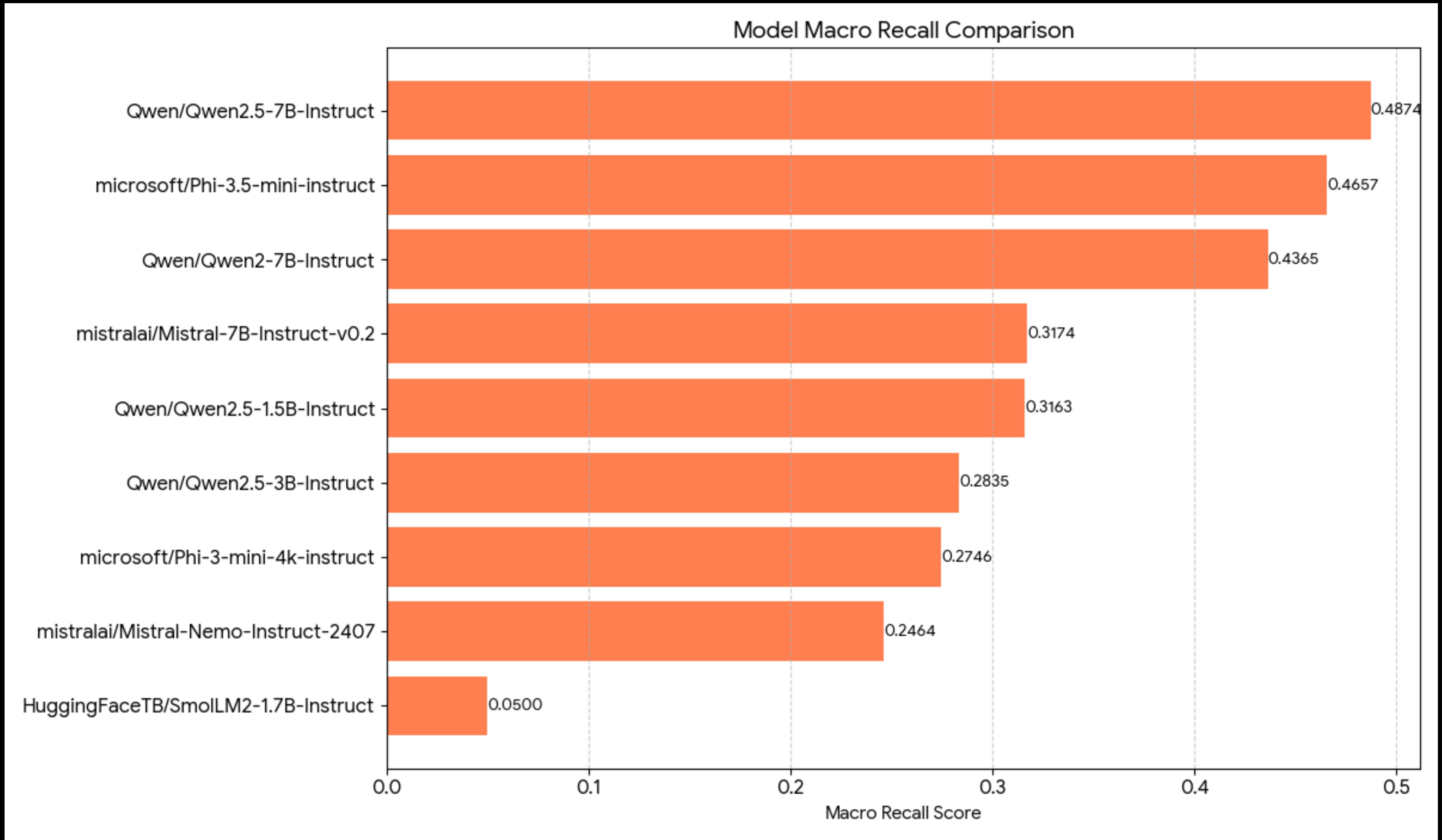
Zero Shot Comparison with Micro F1



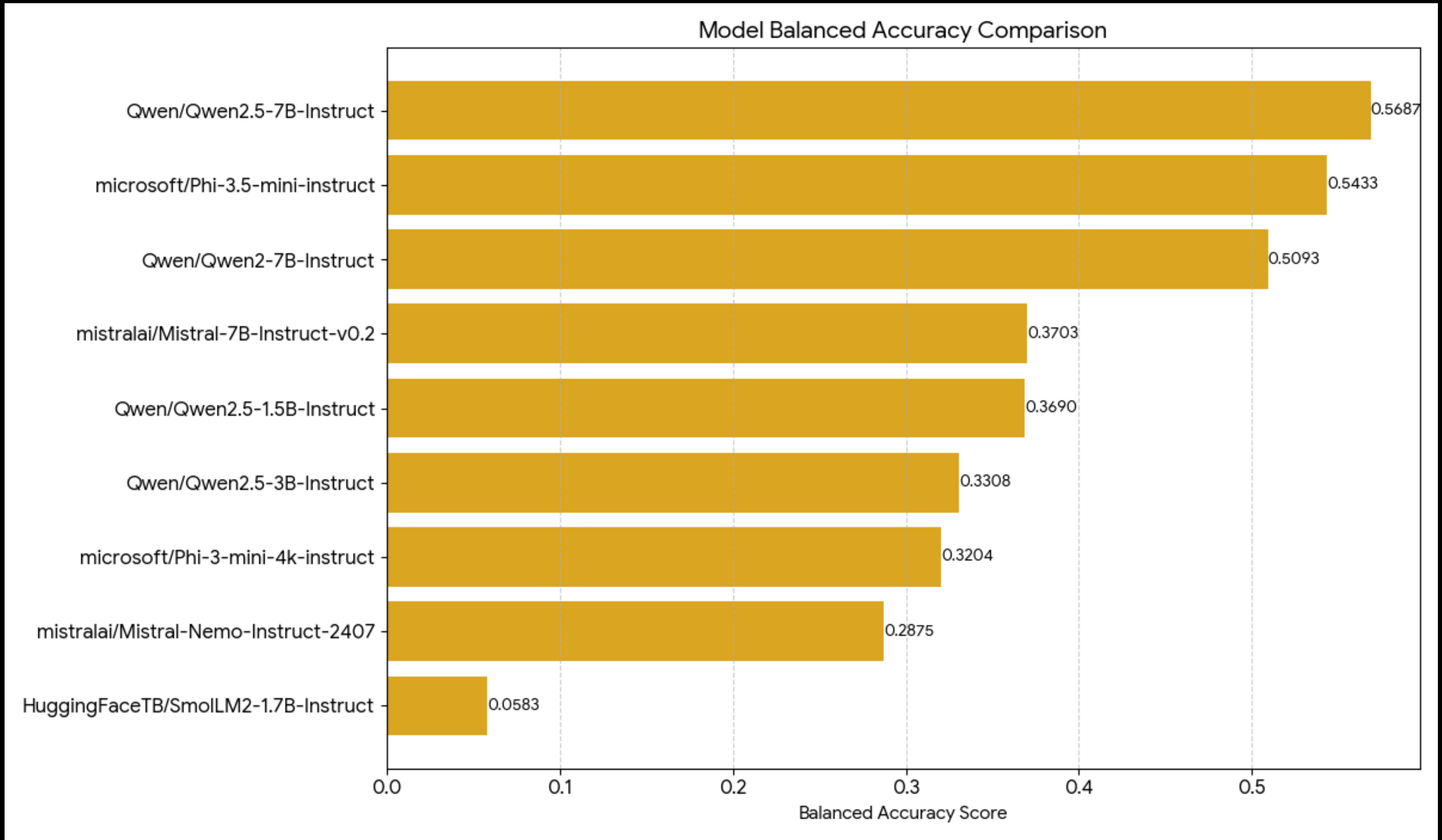
Zero Shot Comparison with Macro Precision



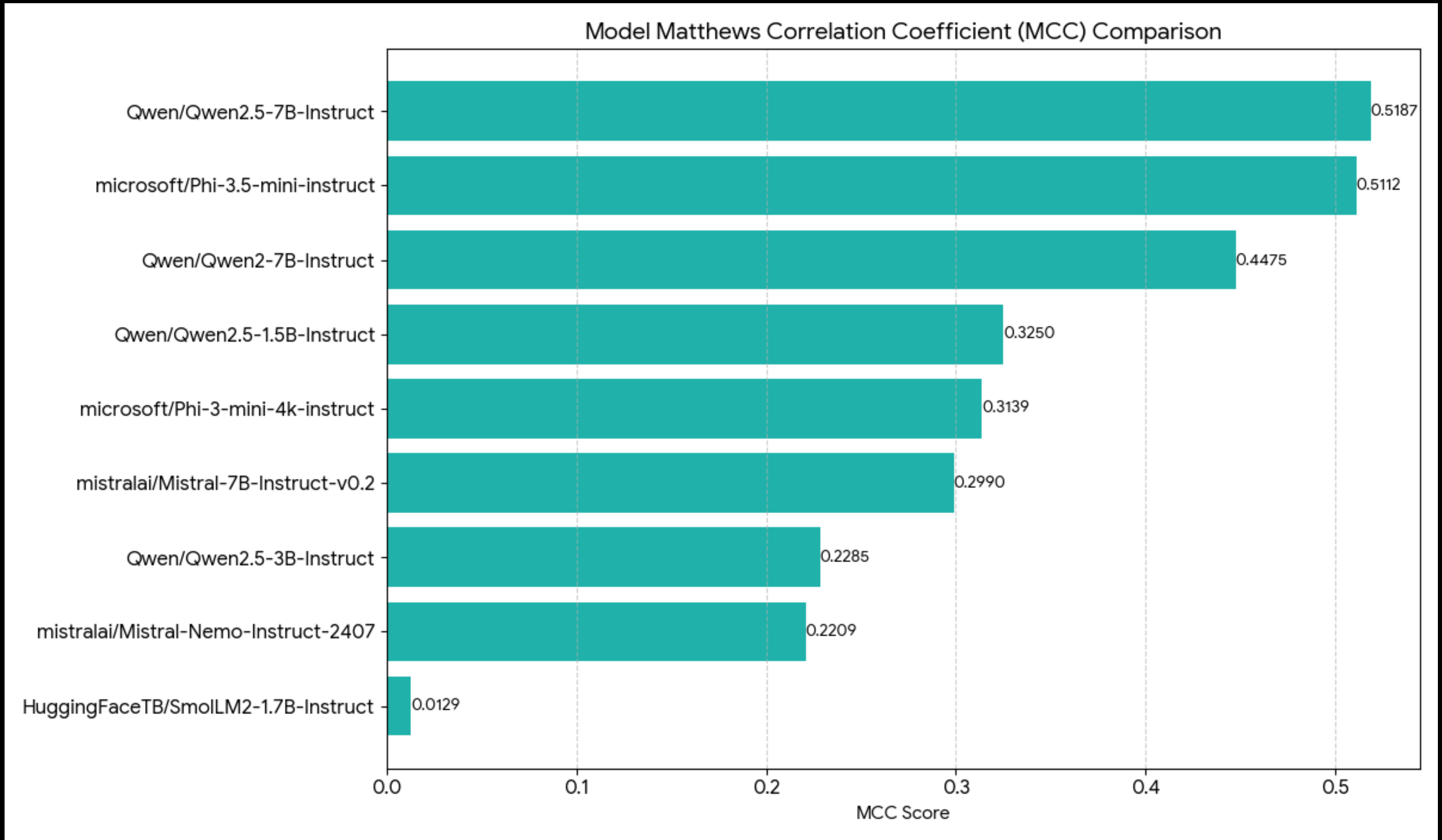
Zero Shot Comparison with Macro Recall



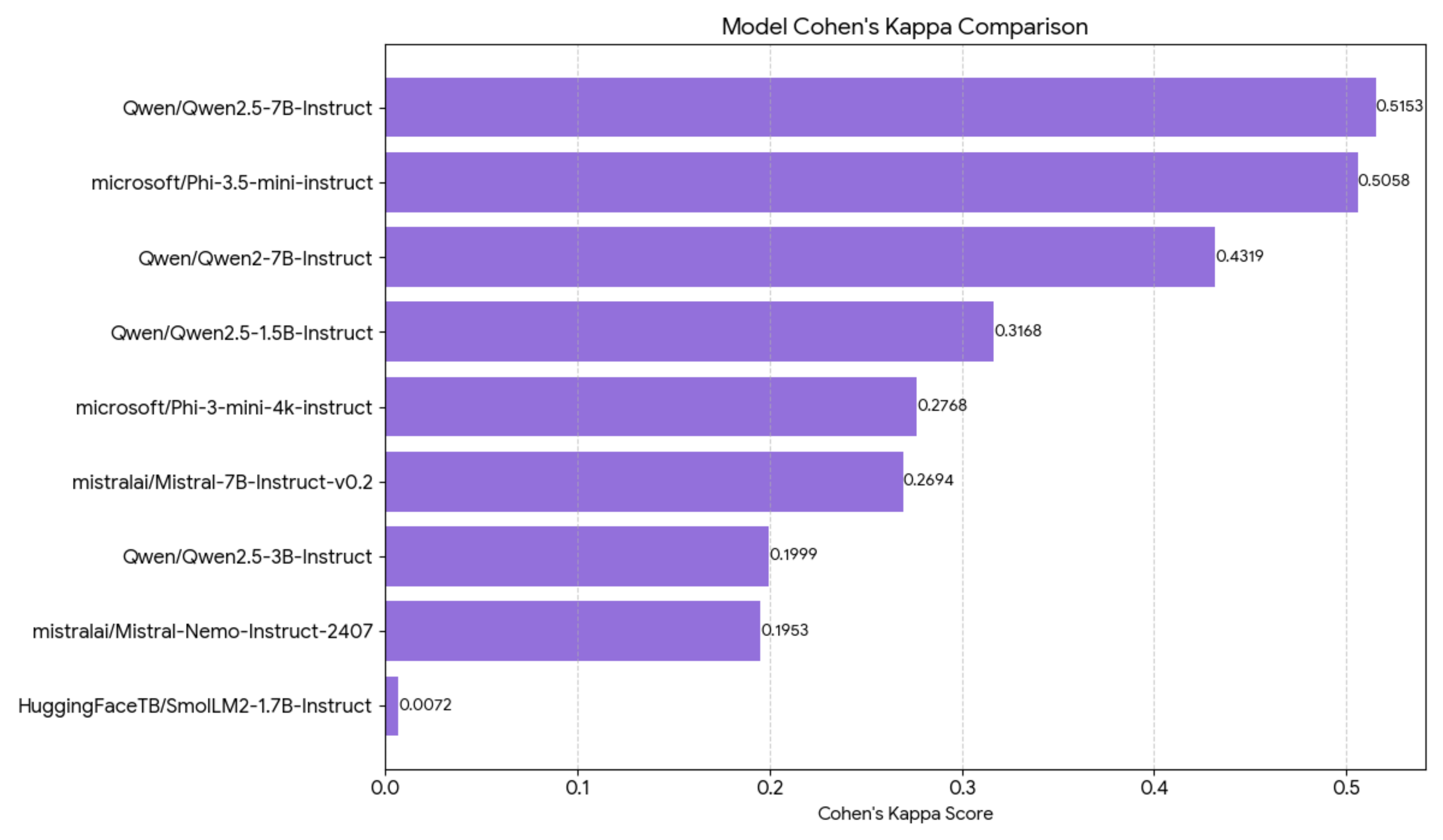
Zero Shot Comparison with Balanced Accuracy



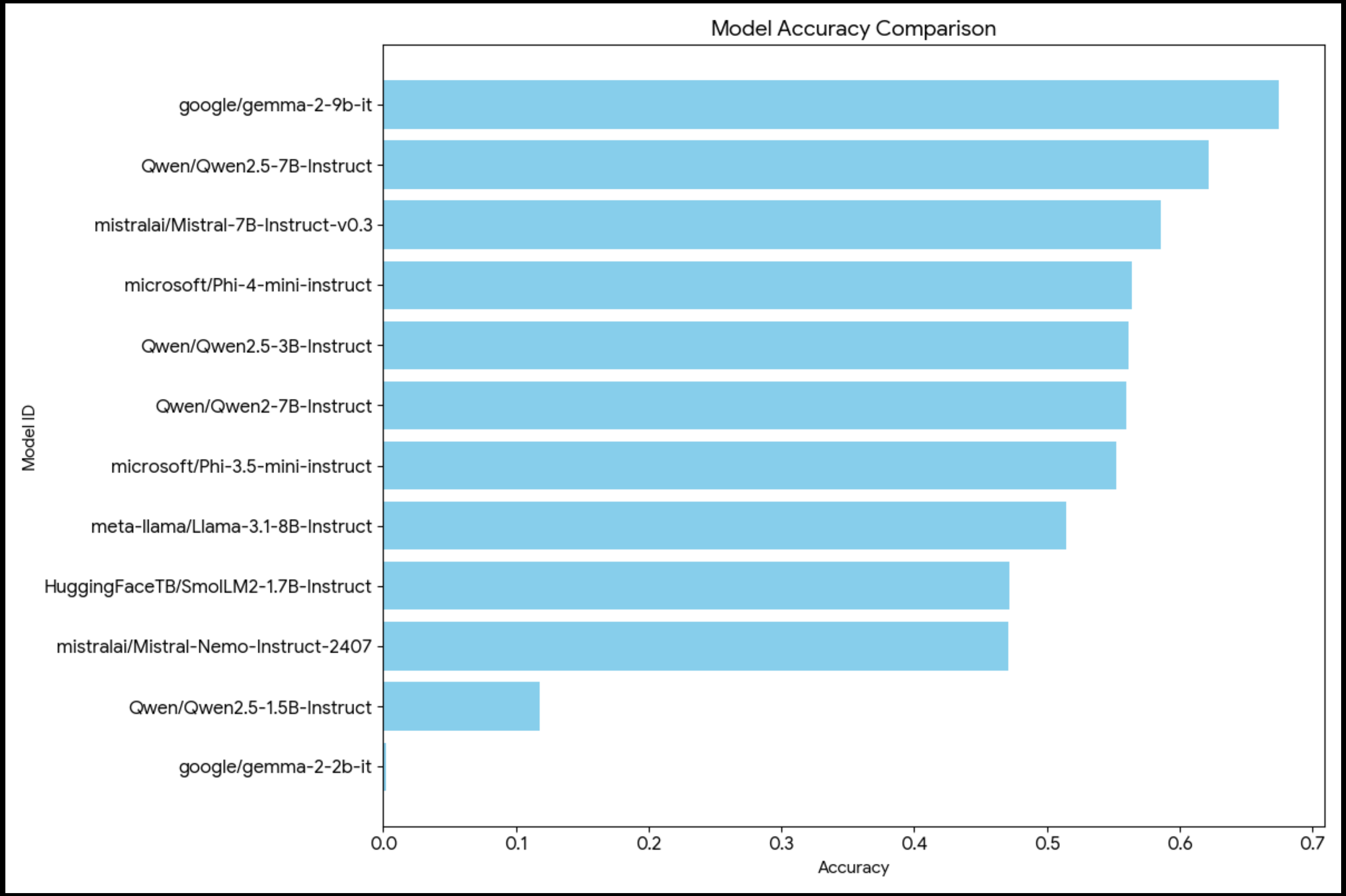
Zero Shot comparison via MCC



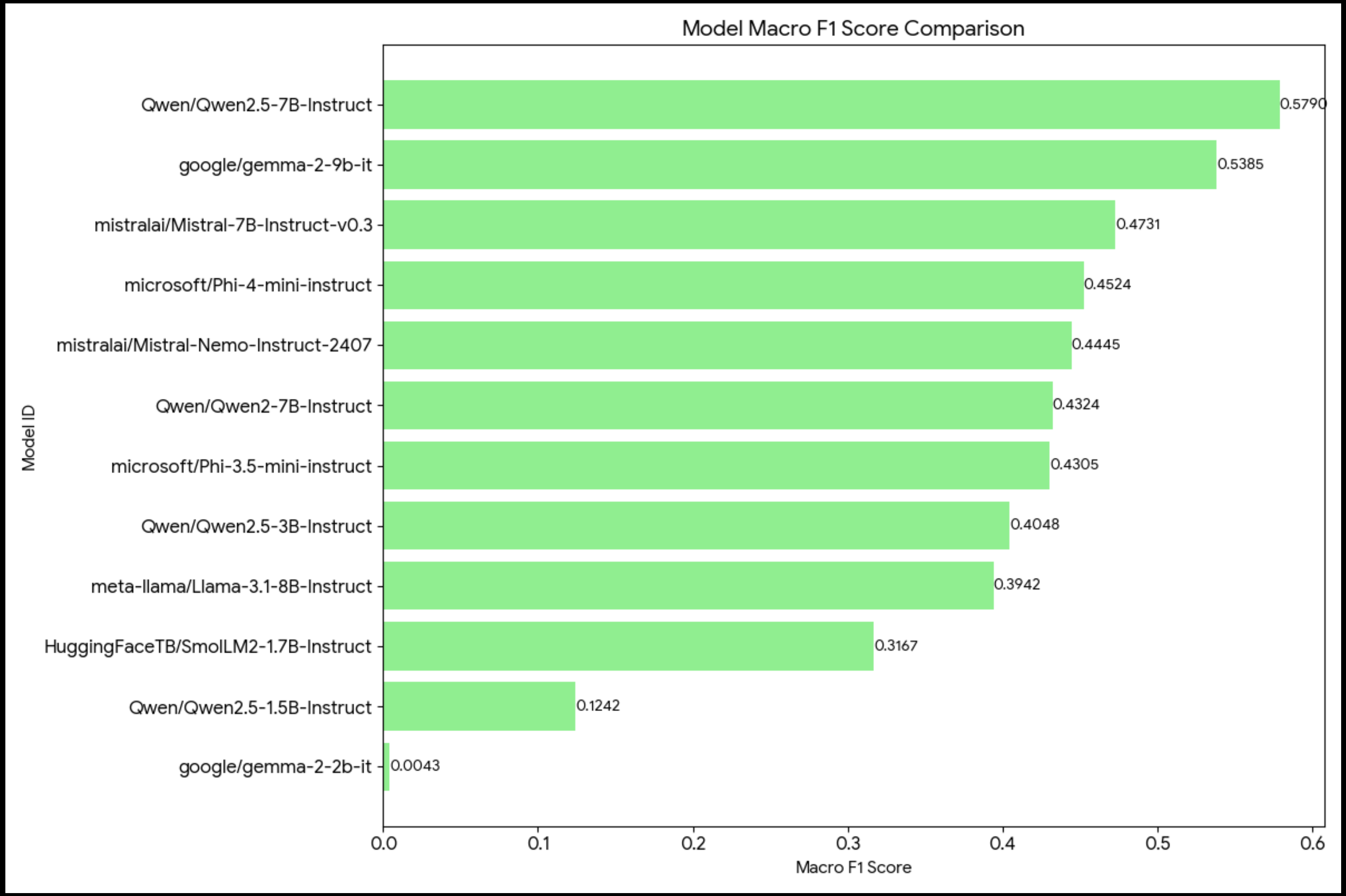
Zero Shot, Cohen's Kappa



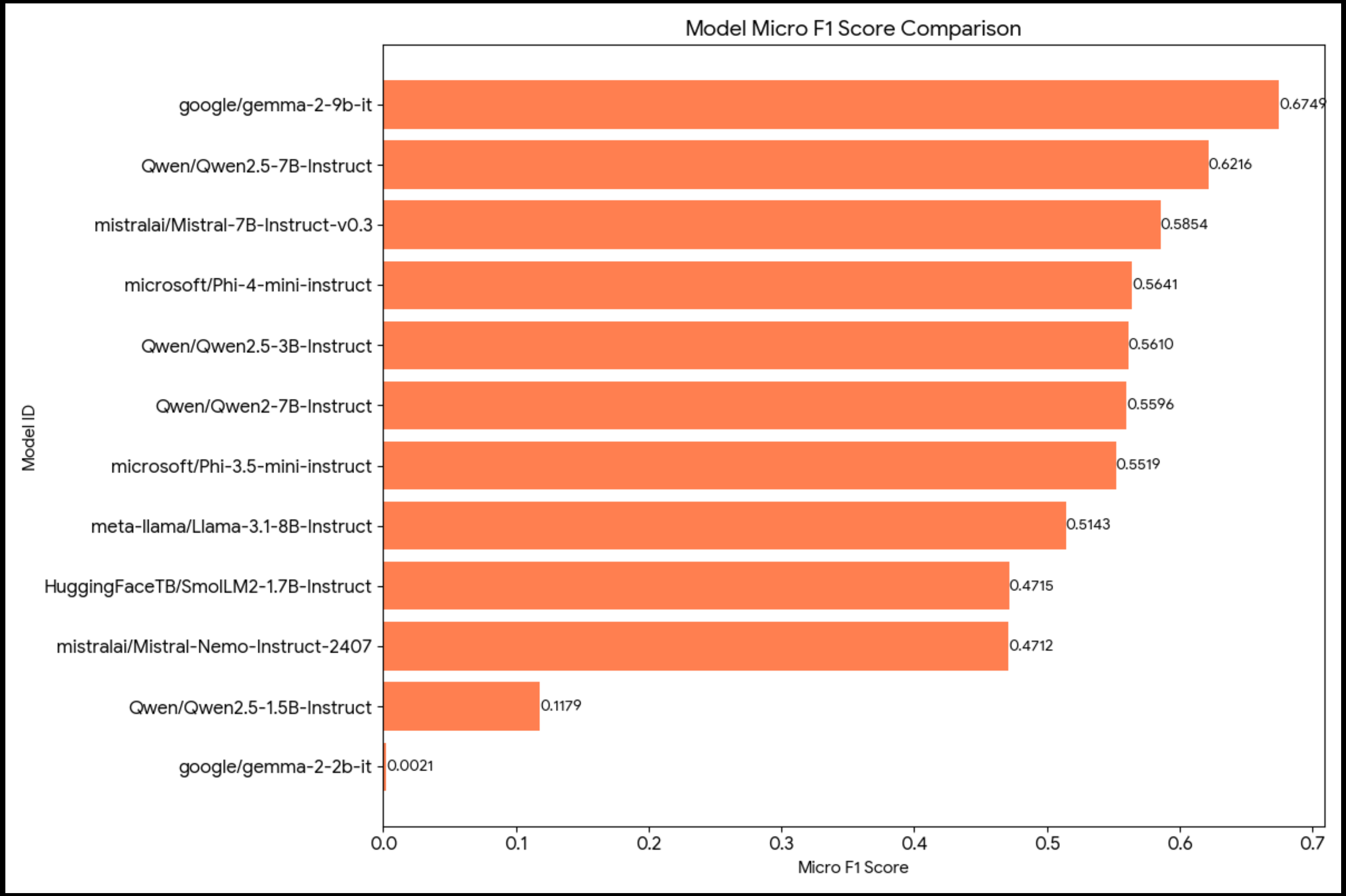
Few shot comparison via Accuracy



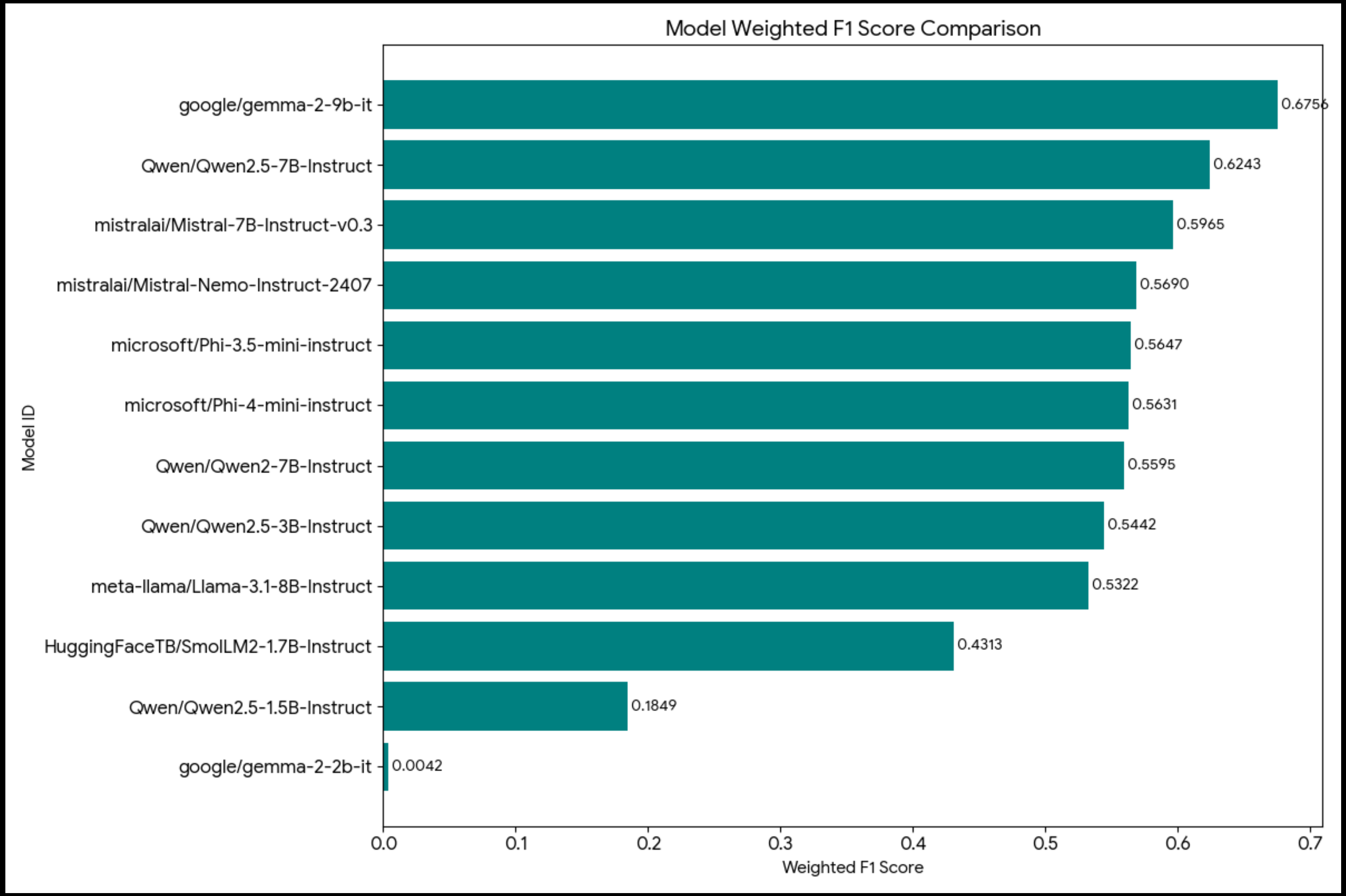
Few Shot Comparison via Macro F1



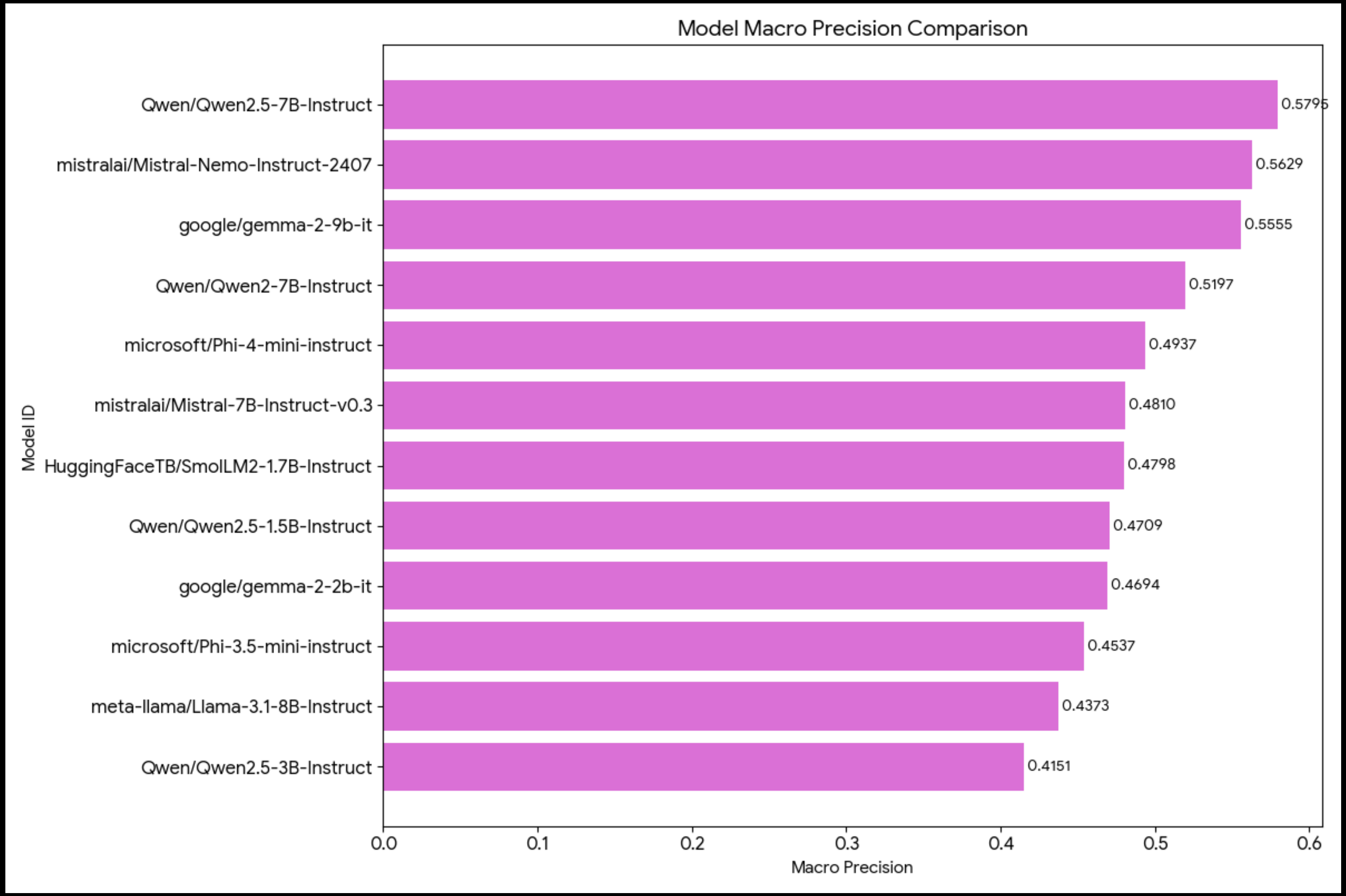
Few Shot Comparison via Micro F1



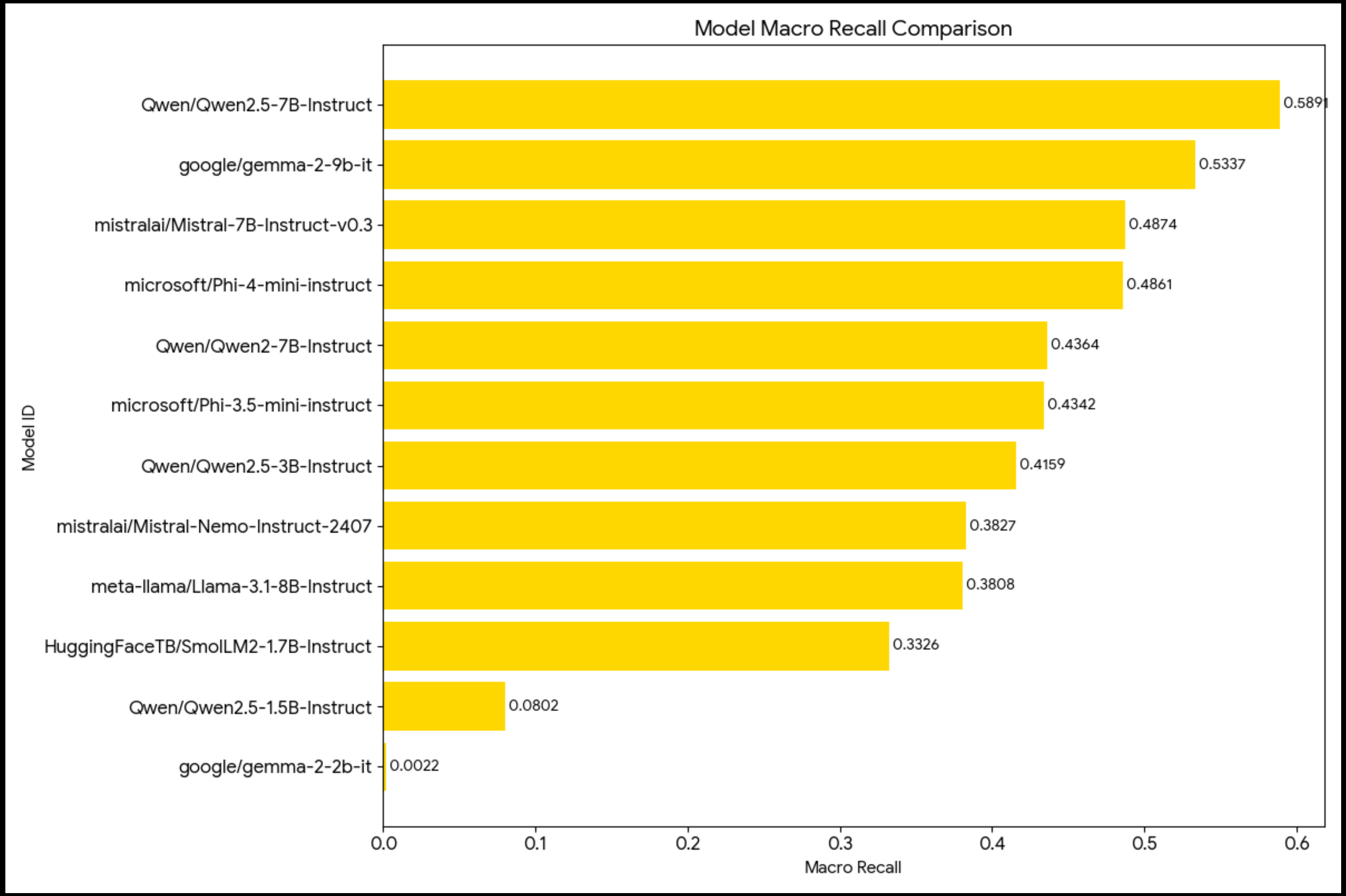
Few Shot Comparison via F1 Score



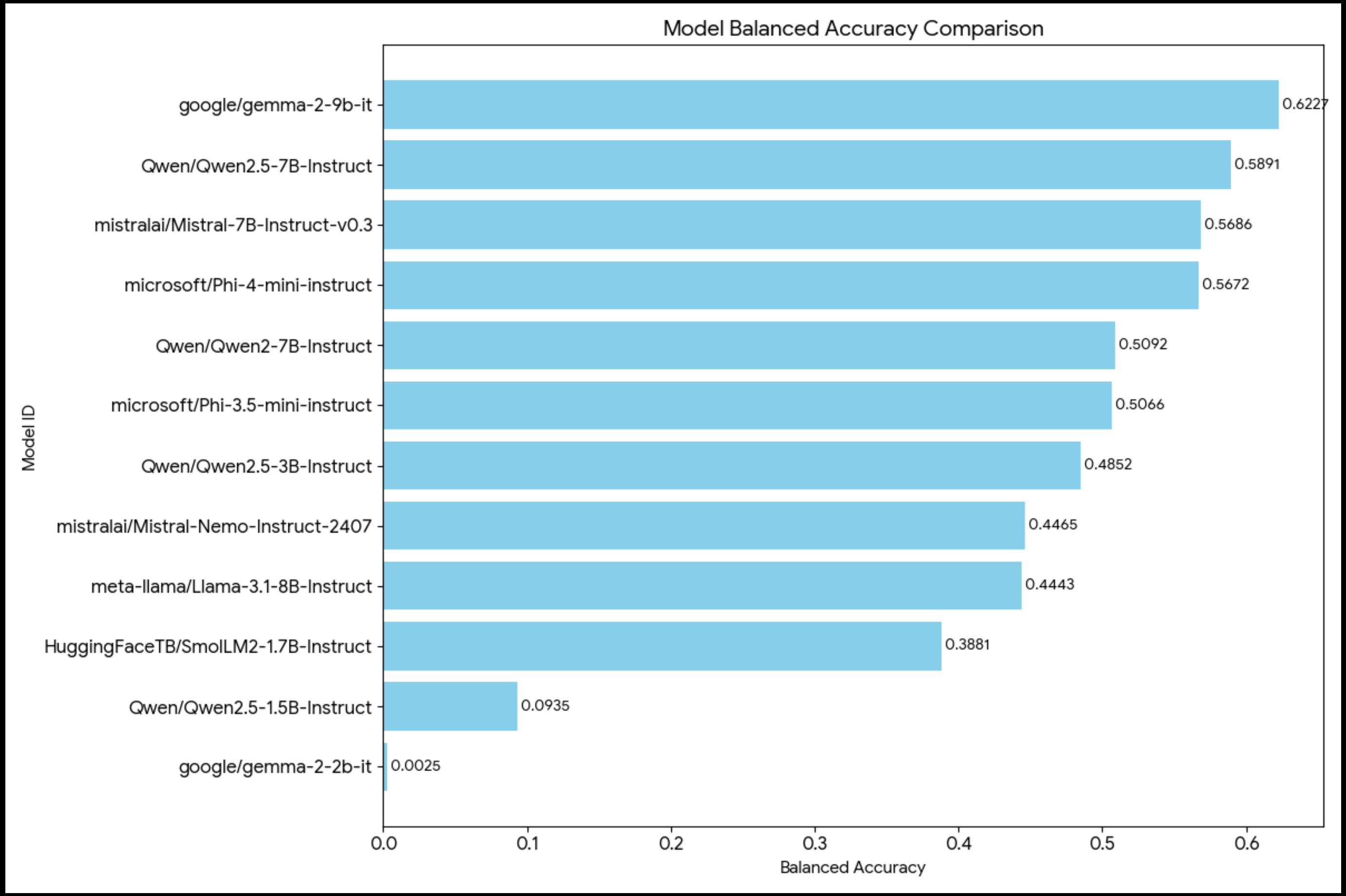
Few Shot Comparison via Macro Precision



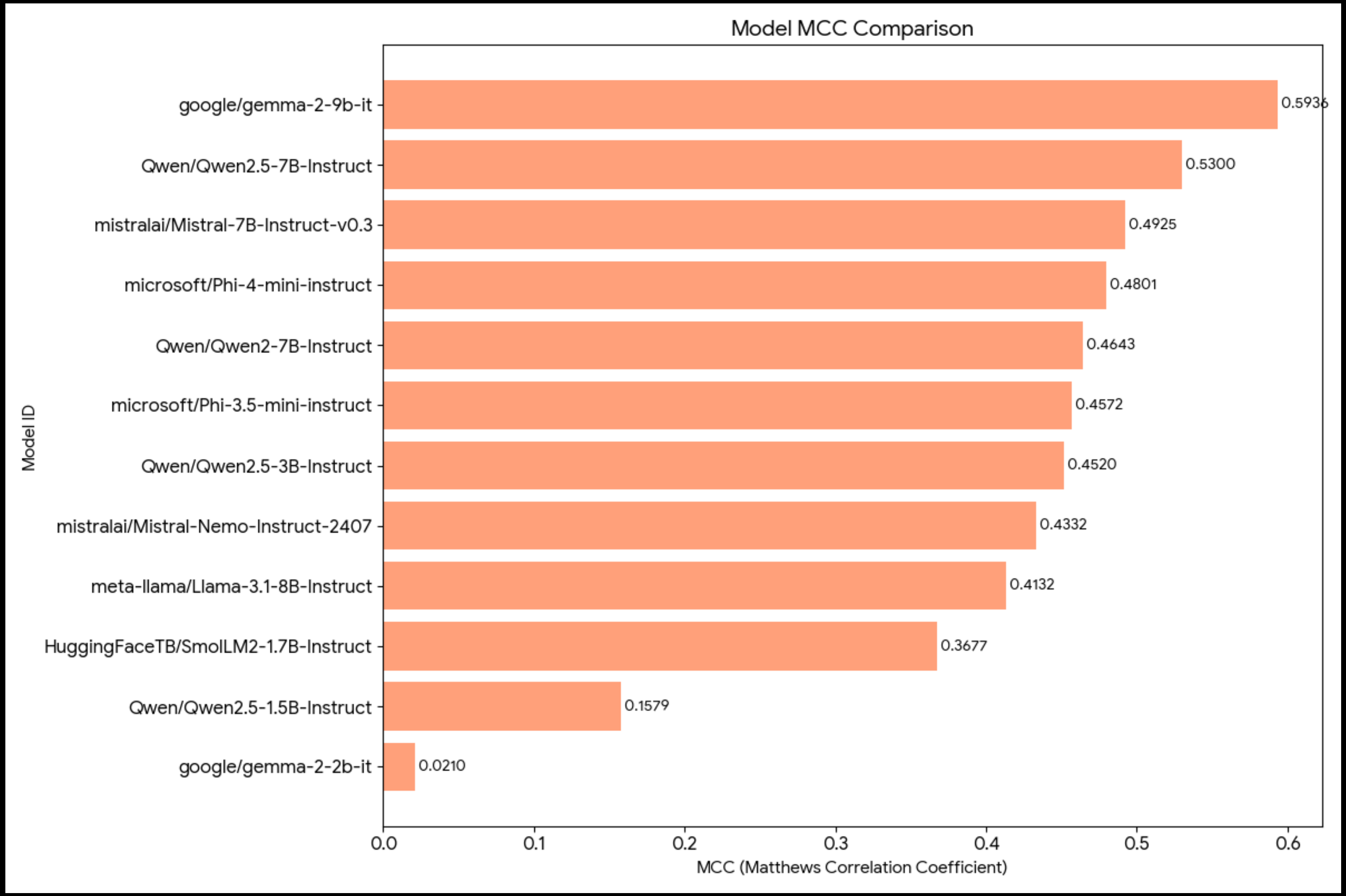
Few Shot Comparison via Macro Recall



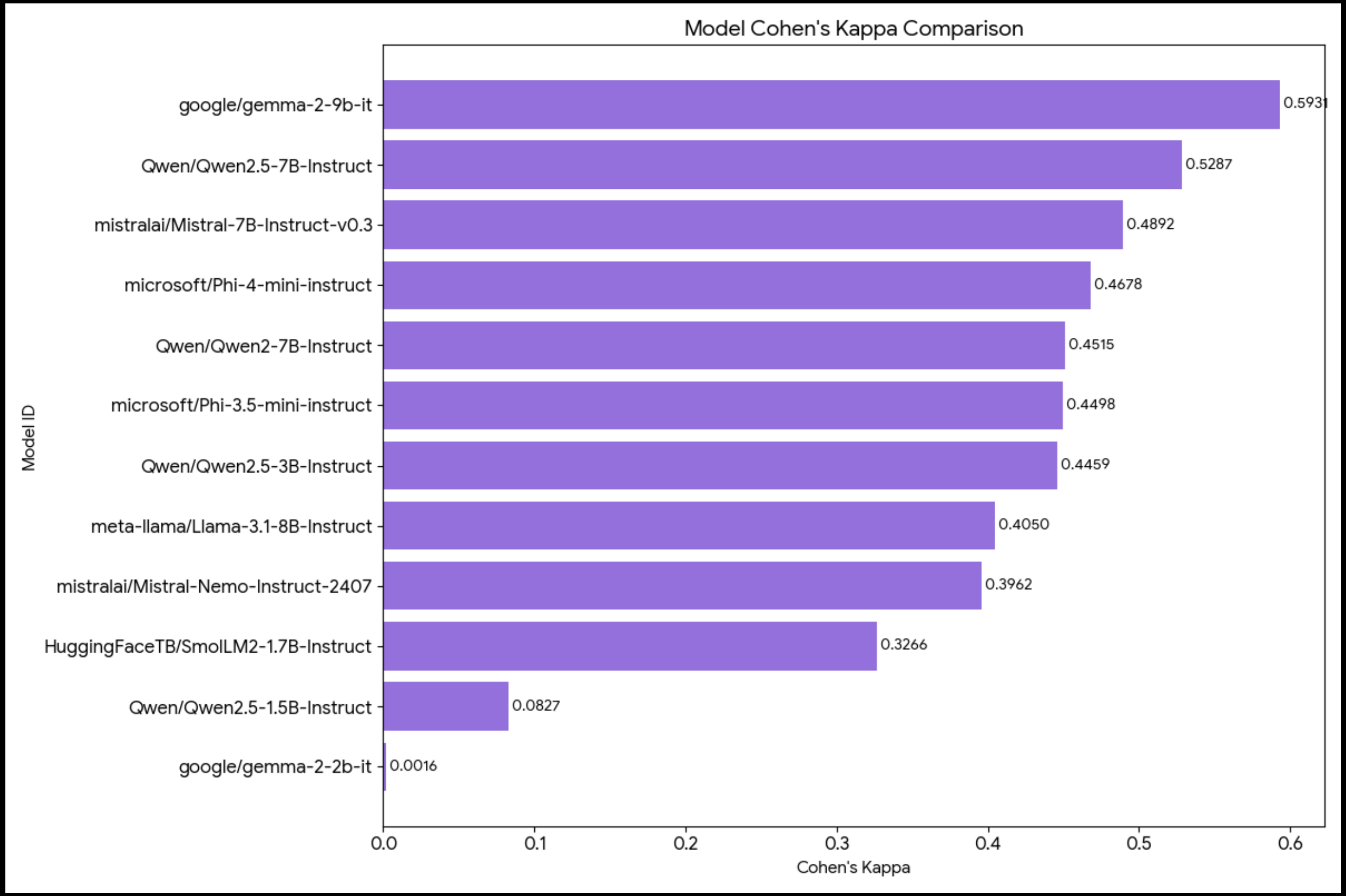
Few Shot Comparison via Balanced Accuracy



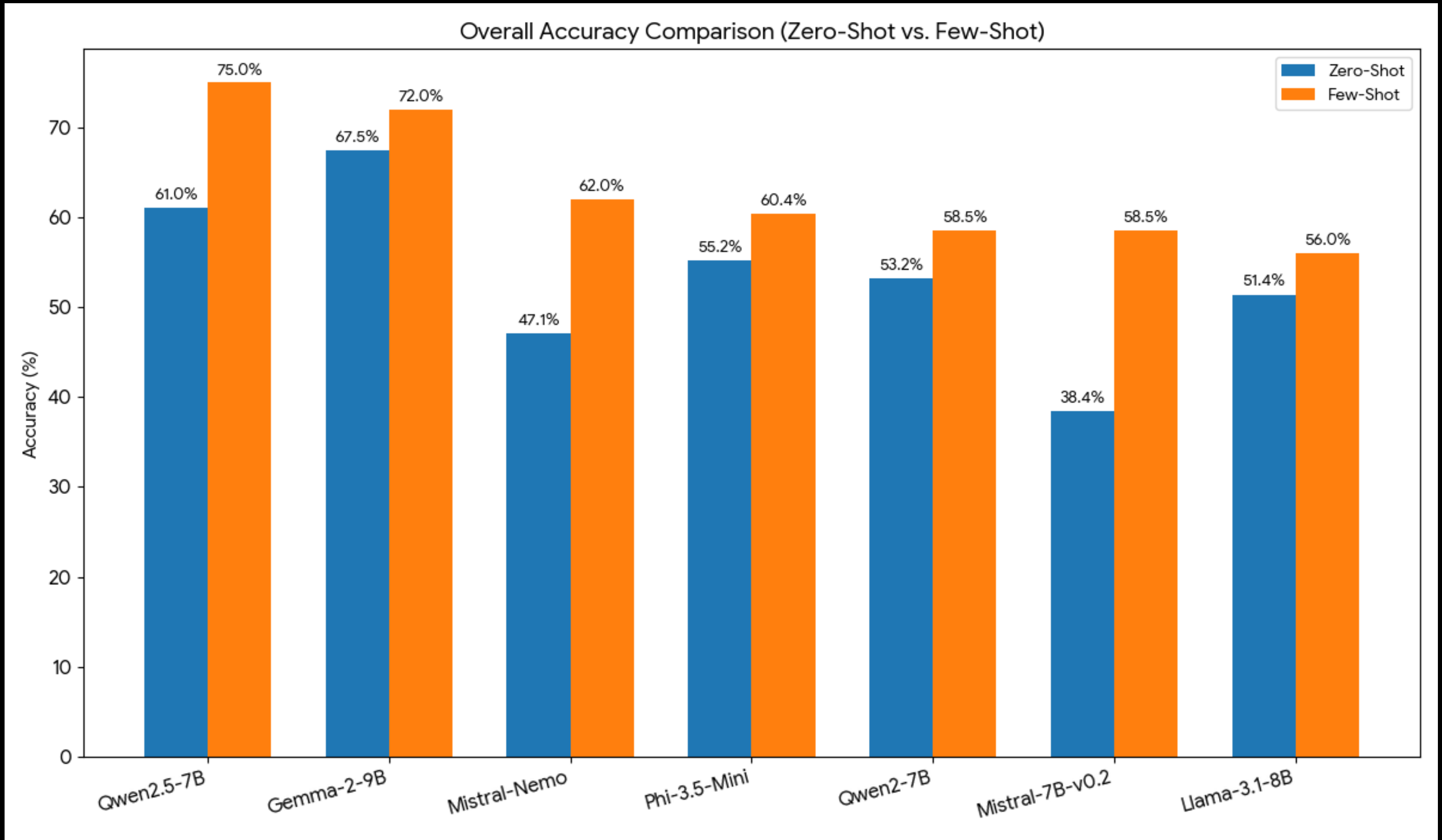
Few shot comparison via MCC



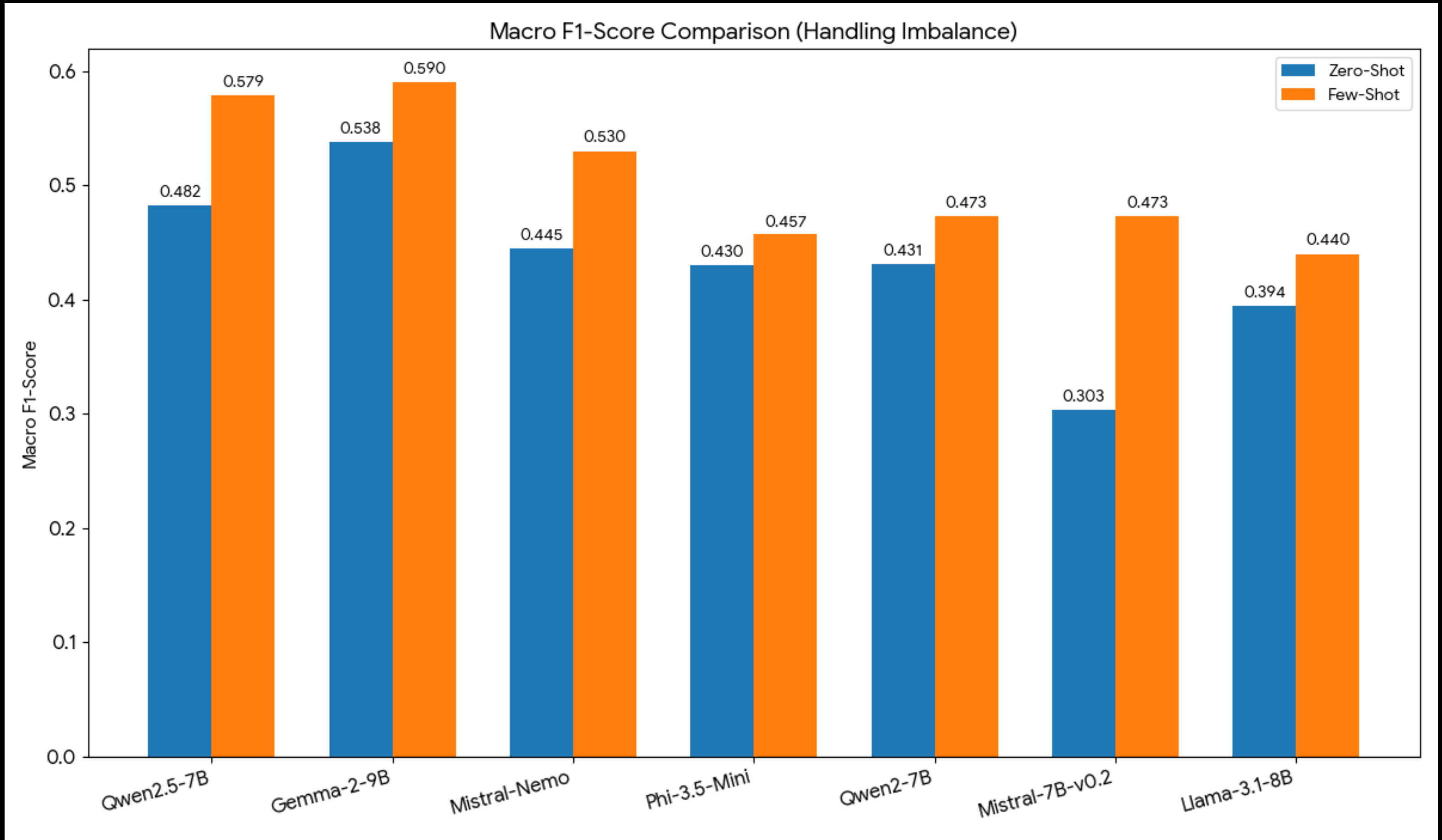
Few Shot Comparison via Cohen's Kappa



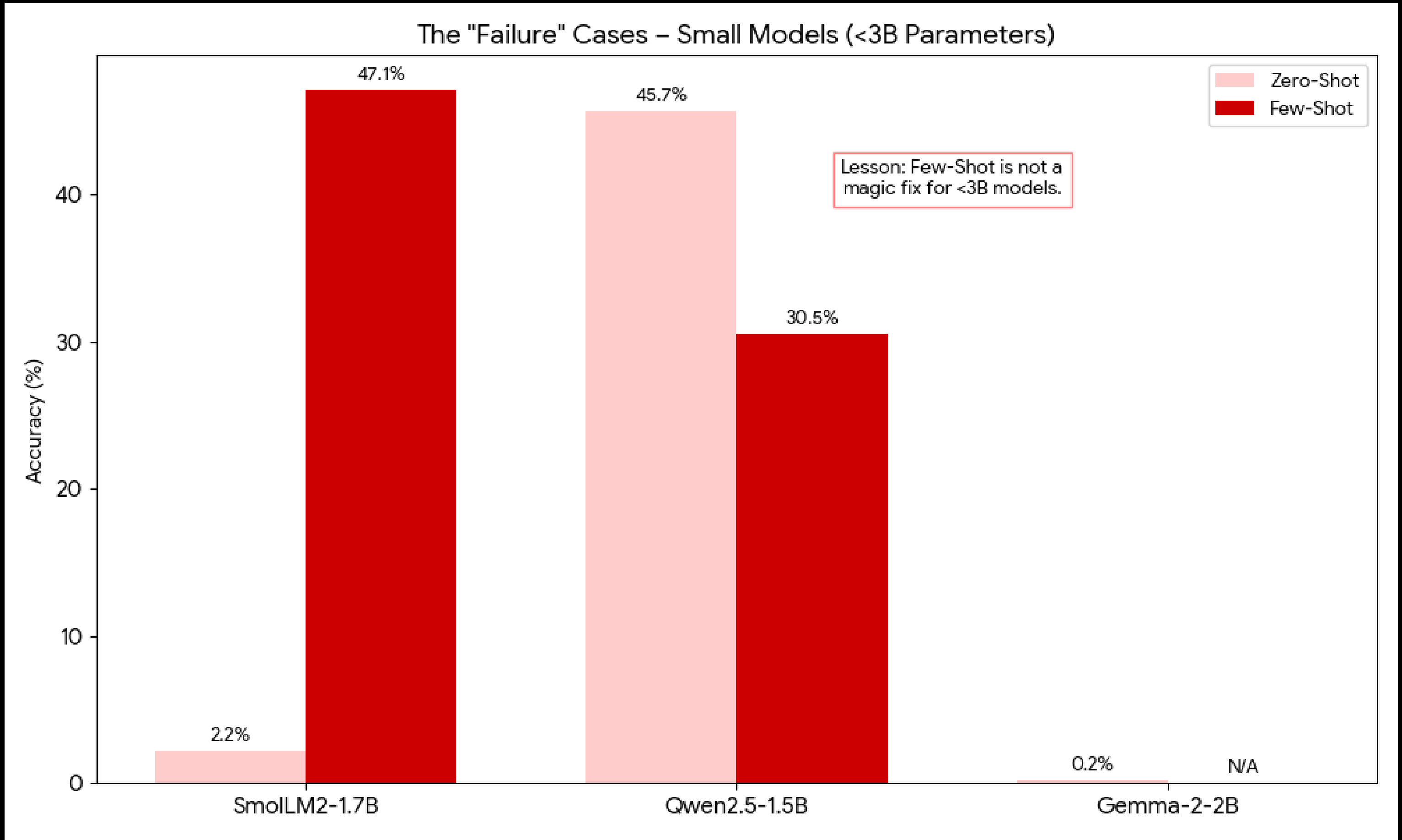
Zero vs Few Comparison (Accuracy)



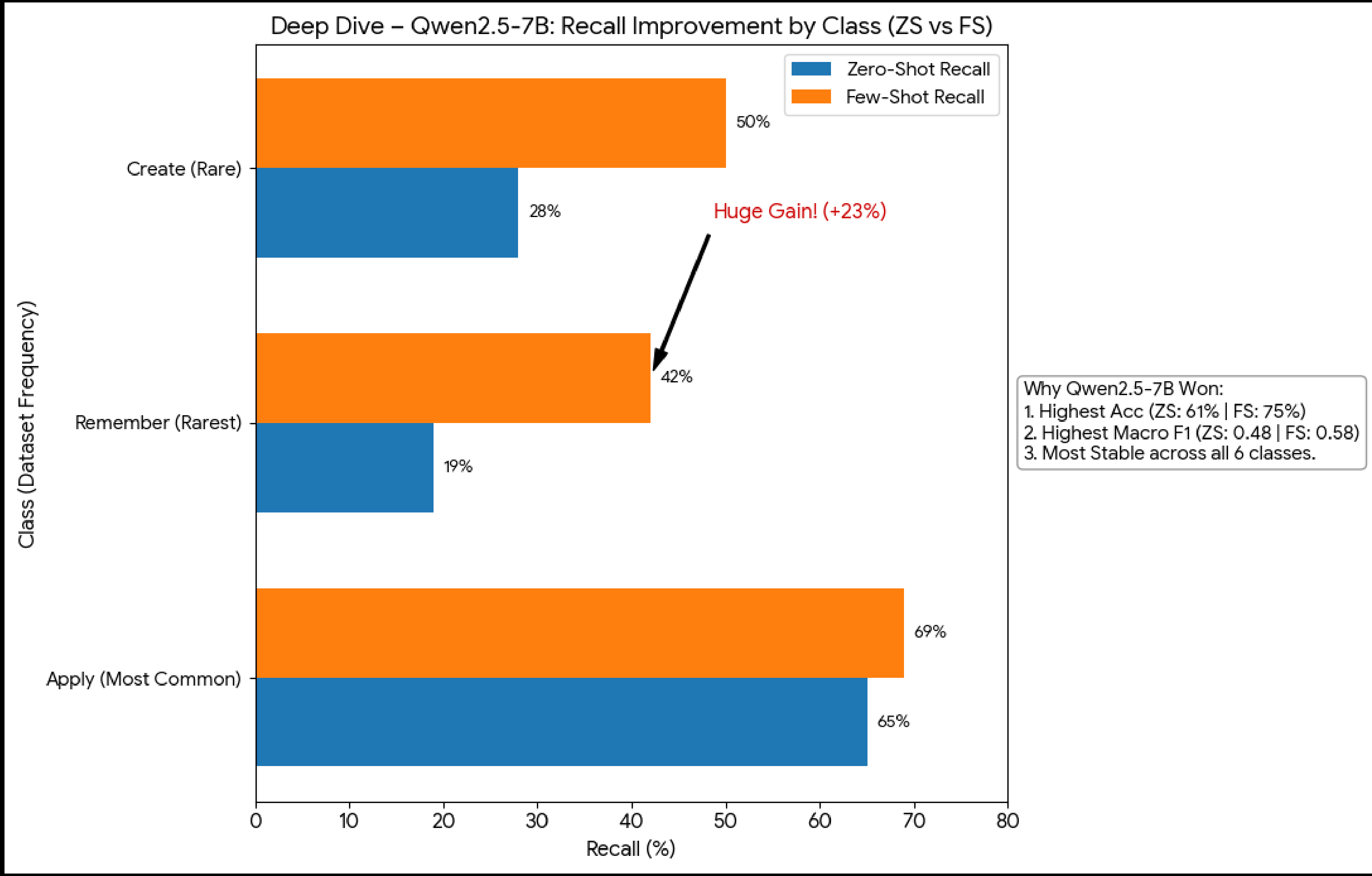
Zero vs Few Comparison (Macro F1 Score)



The “Failure” Cases



The Best Model Qwen2.5-7B



- Why Qwen2.5-7B Won:
- Highest Accuracy in both ZS (61%) and FS (75%).
- Highest Macro F1 in both ZS (0.48) and FS (0.58).
- Most stable performance across all 6 classes.
- Class-Level Improvement (Few-Shot vs Zero-Shot):
- "Apply" (Most Common): ZS Recall ~65% → FS Recall ~69%.
- "Remember" (Rarest): ZS Recall ~19% → FS Recall ~42%. (Huge Gain!)
- "Create": ZS Recall ~28% → FS Recall ~50%.
- Takeaway: Few-Shot prompting specifically helped the model find the rare classes it was ignoring in Zero-Shot.

ML RESEARCH • FINE-TUNING PIPELINE

Bloom's Taxonomy

LLM Fine-Tuning Pipeline

Teaching three 7–9B parameter models to classify academic learning outcomes into six cognitive levels using QLoRA.

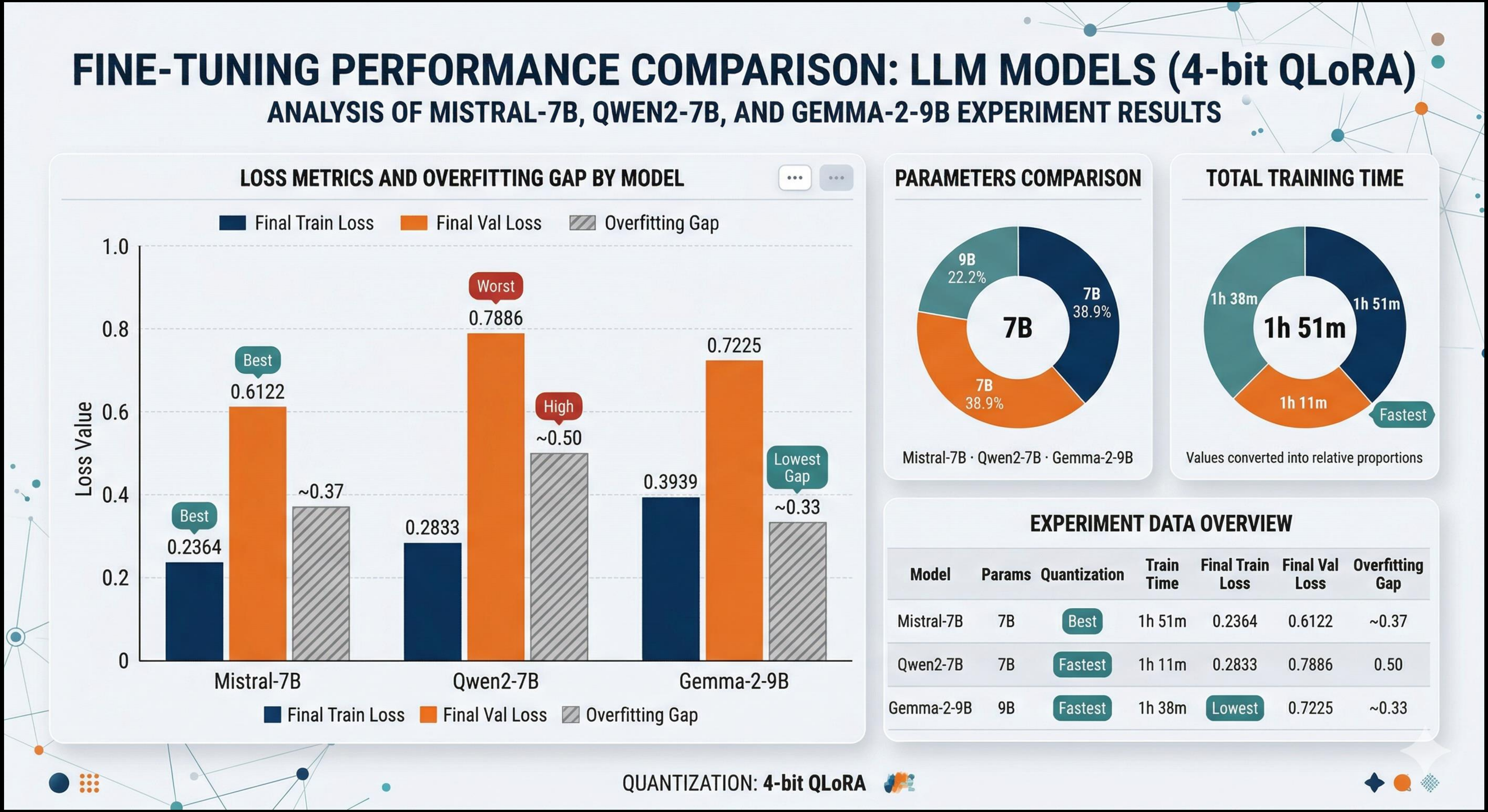
3	**21,379**	**18.77K**	**93.2%**
Models	Training samples	Clean rows	Best accuracy

NVIDIA RTX A6000 • 48GB VRAM • QLoRA 4-bit NF4

Based on Shot-experiment results, Three of the open-source models proved to be promising than the others, hence fine tuned

- Qwen 2.5 7B
- Gemma 2 9B
- Mistral 7B

Fine Tuning Comparison



Model	Params	Quantization	Train Time	Final Train Loss	Final Val Loss	Overfitting Gap
Mistral-7B	7B	Best	1h 51m	0.2364	0.6122	~0.37
Qwen2-7B	7B	Fastest	1h 11m	0.2833	0.7886	0.50
Gemma-2-9B	9B	Fastest	1h 38m	Lowest	0.7225	~0.33

THE TASK

Classifying what students actually learn

Every academic course lists dozens of learning outcomes —

“Define, explain, calculate, analyze, design...”

Each verb signals a different level of cognitive demand. Benjamin Bloom's taxonomy (1956, revised 2001) organizes these into six progressive tiers — a backbone of curriculum design.

THE QUESTION

Can a 7-billion parameter open model, fine-tuned on ~16K examples, reliably sort learning outcomes into Bloom's six levels?

Spoiler: two can. One spectacularly can't.

Higher-order thinking ↑

- **Create** — *design, compose, construct*
- **Evaluate** — *critique, judge, defend*
- **Analyze** — *compare, contrast, examine*
- **Apply** — *calculate, solve, demonstrate*
- **Understand** — *explain, summarize, classify*
- **Remember** — *define, list, identify*

Lower-order recall ↓

THE CONTENDERS

Three instruction-tuned open models

All fine-tuned identically: same data, same QLoRA config, same 3 epochs. Only the base weights differ.

01 Mistral-7B-Instruct

Mistral AI

7B parameters

Trainable	41.9M (0.57%)
Vocabulary	32,768
Runtime	1h 51m
Method	QLoRA 4-bit NF4

02 Qwen2-7B-Instruct

Alibaba Cloud

7B parameters

Trainable	40.4M (0.53%)
Vocabulary	151,643
Runtime	1h 38m
Method	QLoRA 4-bit NF4

03 Gemma-2-9B-IT

Google DeepMind

9B parameters

Trainable	54.0M (0.58%)
Vocabulary	256,000
Runtime	1h 11m
Method	QLoRA 4-bit NF4

THE DATASET

18,773 labelled learning outcomes

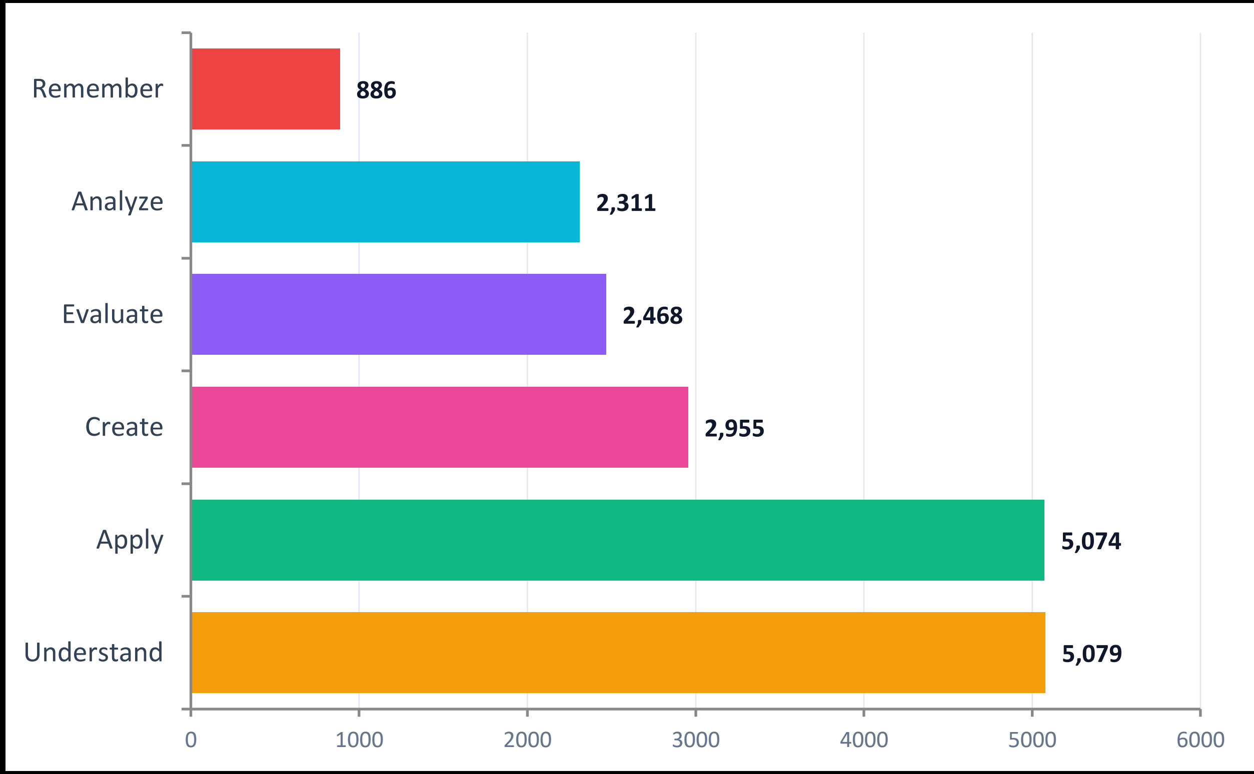
Streamed from sample_full.csv in 2,000-row chunks — a memory-safe pipeline that drops multi-labelled and stub rows.

PIPELINE FUNNEL

Raw rows	21,379
After chunked parse	18,773
Training split (85%)	15,957
Test split (15%)	2,816

2,607 rows dropped
(multi-label / unlabelled)

LABEL DISTRIBUTION



Class imbalance: Remember is ~5.7× rarer than Understand. Handled via stratified 85/15 split.

THE RECIPE

QLoRA: memory-efficient fine-tuning

4-bit NF4 quantised weights + low-rank adapters. Fits 9B models into 48 GB with room to train.

EPOCHS

3

Full passes over 15,957 rows

BATCH SIZE

4 × 4

Effective batch = 16 (grad accum)

LEARNING RATE

2e-4

Cosine schedule, 5% warmup

MAX SEQ. LEN

256

Prompts fit comfortably

LORA RANK

r = 16

α = 32, dropout 0.05

PRECISION

bf16

NF4 double-quant, bf16 compute

PROMPT TEMPLATE

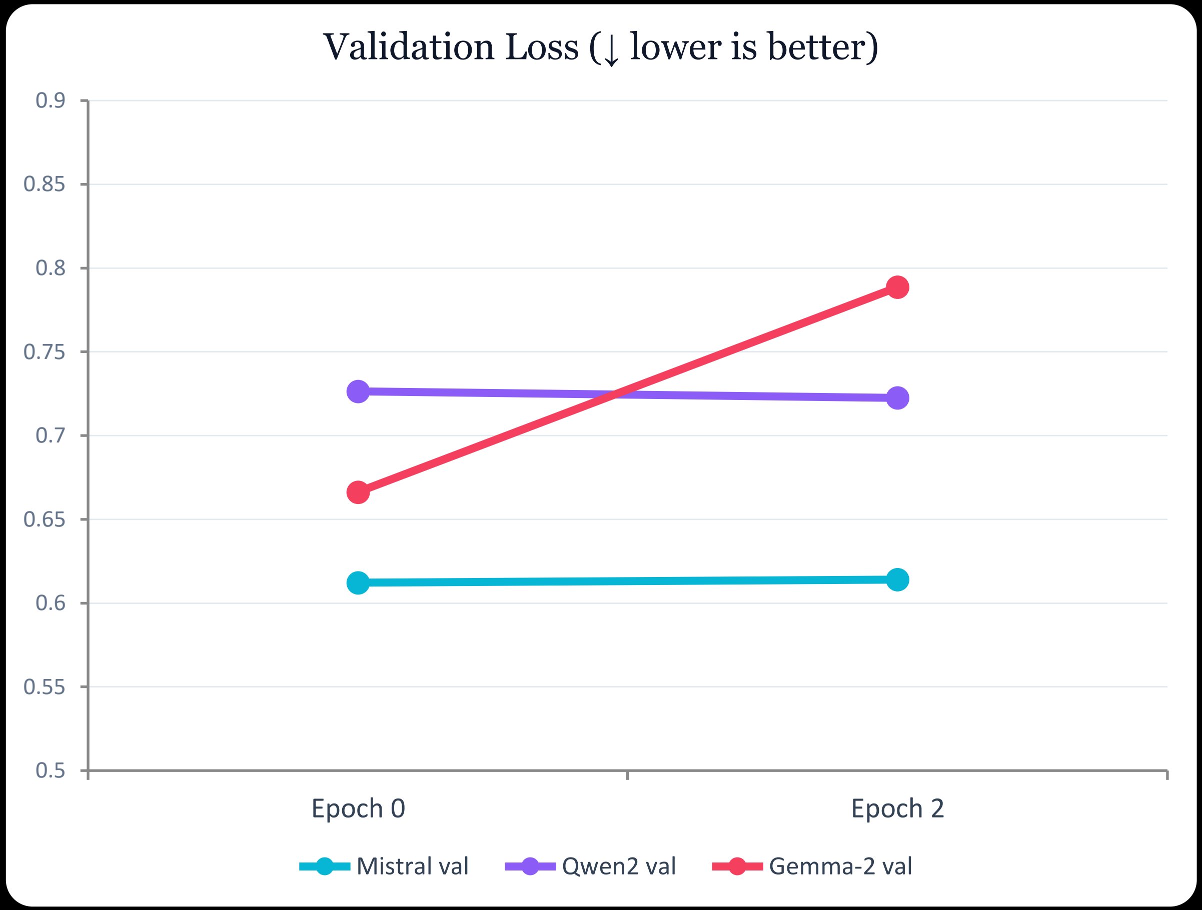
Instruction-style supervised fine-tuning

```
### System:
You are an expert educator. Classify the
following academic learning outcome into
exactly one of Bloom's Taxonomy cognitive
levels: Remember, Understand, Apply, Analyze,
Evaluate, or Create.

### Learning Outcome:
Analyze the health economic implications
of efficient perioperative practice.

### Bloom's Level:
Analyze
```

TRAINING DYNAMICS

How each model learned over time

Training and validation loss at each epoch. Lower train loss + higher val loss = classic overfitting signal.



HEADLINE RESULT

The Leaderboard

Evaluated on 2,816 held-out test samples. Weighted F1 is the primary metric.

1st — BEST

Mistral-7B-Instruct

93.22%

Accuracy

Weighted F1	Macro F1
0.9324	0.9206

2nd — STRONG

Qwen2-7B-Instruct

92.47%

Accuracy

Weighted F1	Macro F1
0.9250	0.9149

3rd — FAILED

Gemma-2-9B-IT

20.24%

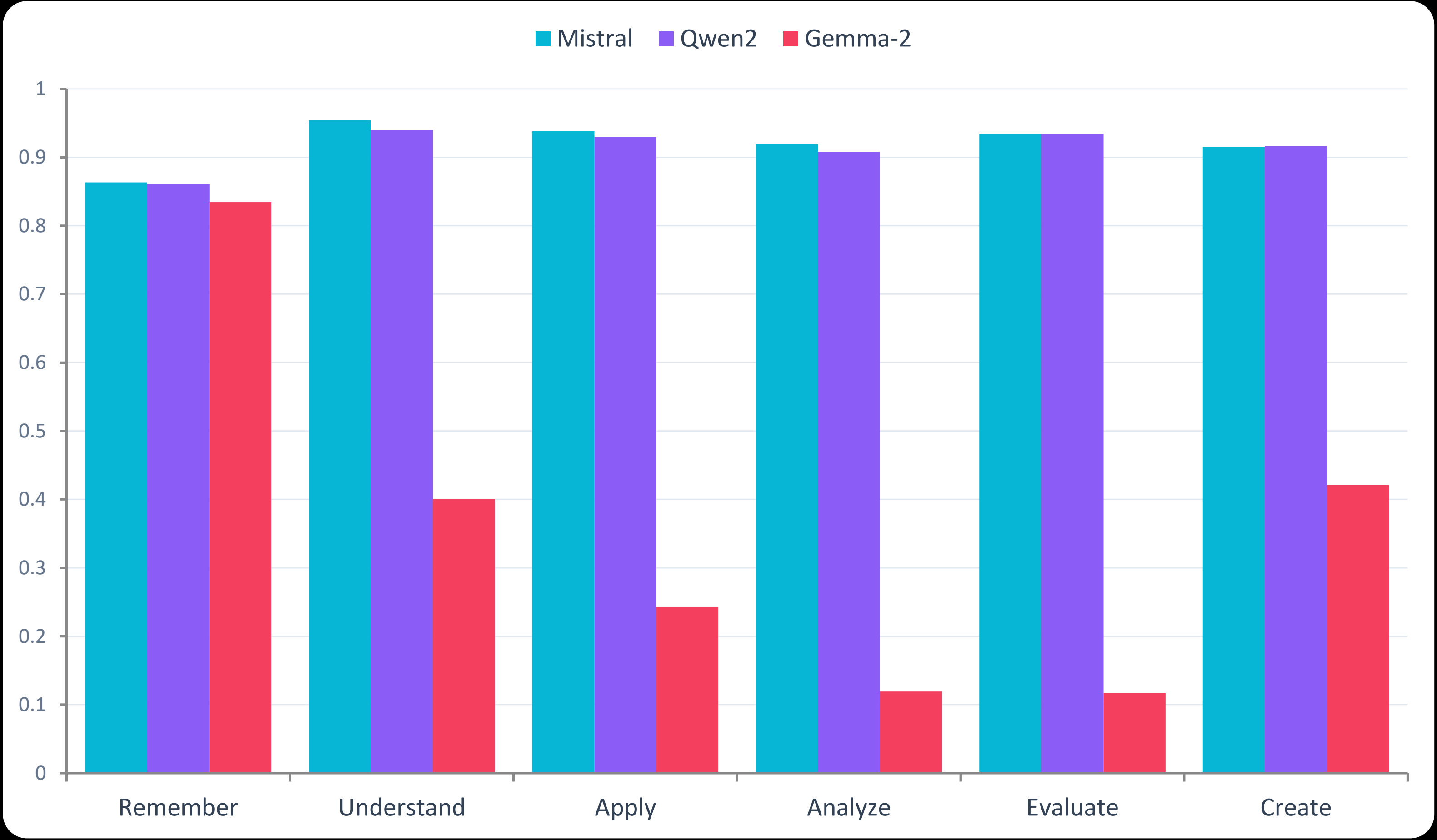
Accuracy

Weighted F1	Macro F1
0.3097	0.3050

PER-CLASS F1

How each model handles each Bloom level

Gemma's collapse is uniform across higher-order levels — precision is strong, but recall falls off a cliff.



WINNER

Mistral dominates across the board

Best F1 on 4 of 6 classes. Most balanced recall-precision profile.

CLOSE

Qwen2 matches on Evaluate & Create

Essentially tied with Mistral on the hardest creative levels.

BROKEN

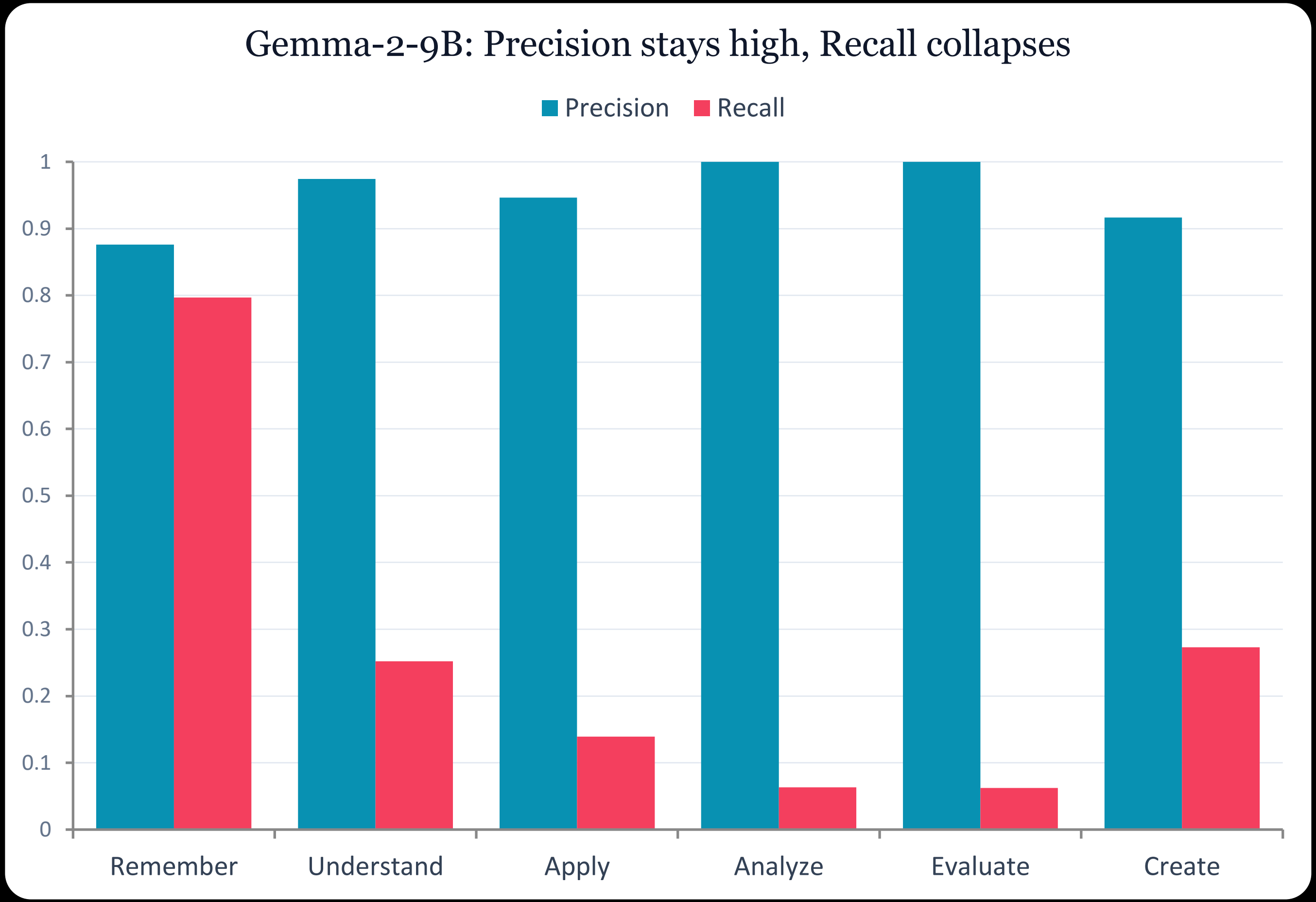
Gemma collapses above Remember

F1 drops from 0.83 → 0.12 on Analyze & Evaluate. See next slide.

CASE STUDY

The Gemma paradox — precise yet wrong

Its training loss dropped to 0.28 (lowest of the three). Yet accuracy stalled at 20%. What happened?



1

Precision → ~1.0

When Gemma *does* predict a class, it's almost always right. It isn't confused about meaning — it's silent on most inputs.

2

Recall → ~0.06

Only ~6% of Analyze/Evaluate outcomes get labelled at all. The model abstains or produces tokens that don't match any Bloom label.

3

Likely cause

Gemma-2's safety/instruction format differs from Mistral & Qwen. Without a chat template, outputs drift and the parser captures 'Unknown'.

4

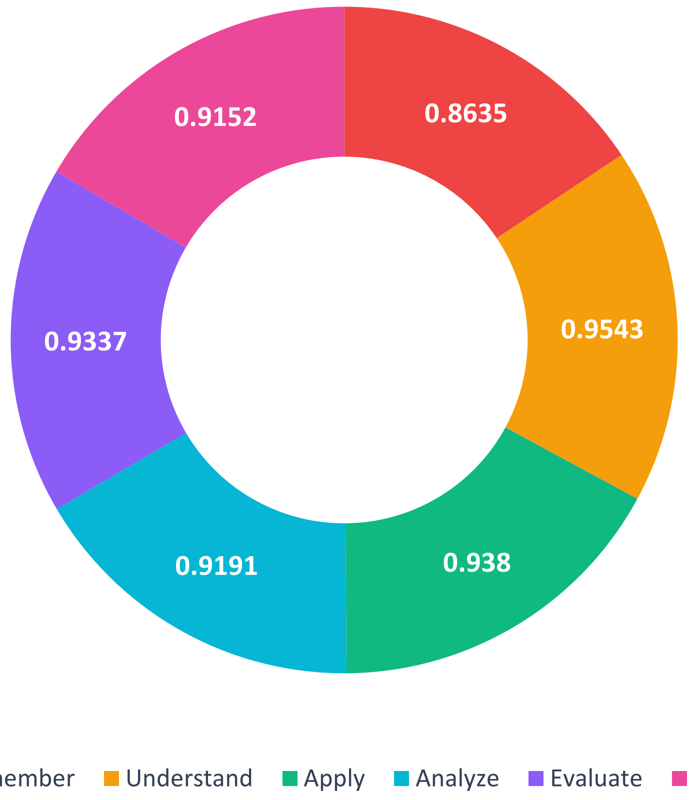
The fix

Apply Gemma's official chat template, or retrain with `attn_implementation="eager"` AND a matching prompt structure.

WINNER SPOTLIGHT

Mistral-7B — full per-class breakdown

F1 score per Bloom level



Class	Precision	Recall	F1	Support
Remember	0.85	0.88	0.86	133
Understand	0.96	0.94	0.95	762
Apply	0.94	0.93	0.94	761
Analyze	0.92	0.92	0.92	347
Evaluate	0.95	0.91	0.93	370
Create	0.88	0.95	0.92	443

Macro F1
0.9206

Weighted F1
0.9324

Accuracy
93.22%

IN THE WILD

Live predictions from the best model

Six unseen learning outcomes. Mistral classifies each in under 100 ms on an A6000.

Input		Prediction
"Define the key terms used in thermodynamics."	→	**Understand**
"Explain the principles of natural selection."	→	**Understand**
"Calculate the acceleration of a falling object using Newton's 2nd law."	→	**Apply**
"Compare and contrast supervised and unsupervised ML."	→	**Evaluate**
"Critique a peer's experimental design for potential confounds."	→	**Evaluate**
"Design an algorithm to solve the travelling salesman problem."	→	**Create**

```
$ classify_learning_outcome("Design a curriculum unit on climate change.")  →  Create
```

TAKEAWAYS

What this experiment proved

01

QLoRA works — at scale.

Fine-tuning <1% of parameters lifted instruction-tuned models to 93% accuracy on a 6-way classification task.

02

Bigger ≠ better.

The 9B Gemma failed while both 7B models excelled. Architecture, chat templates, and tokenizer alignment matter more than raw parameter count.

03

Class imbalance was survivable.

Remember class had only 886 rows (<5% of data) yet still achieved F1 ≥ 0.86 in both winning models. Stratified splitting + QLoRA handled it.

04

Always inspect recall, not just loss.

Gemma's training loss was the lowest of the three. A loss curve alone would have declared it the winner. Class-level recall told the real story.

Mistral-7B-Instruct-v0.3 • 93.22% accuracy • 0.9324 weighted F1 • ~1h 51m training on a single A6000

Based on Shot-experiment results, Three of the open-source models proved to be promising than the others, hence fine tuned

- Qwen 2.5 7B
- Gemma 2 9B
- Mistral 7B

Fine Tuning Comparison